DC Electrical Machines

Electrical Machine

- Faraday’s law is the basis of generator action (Art. 1.5 and 1.7)

- Ampere’s force law is the basis of motor action (Art. 1.6)

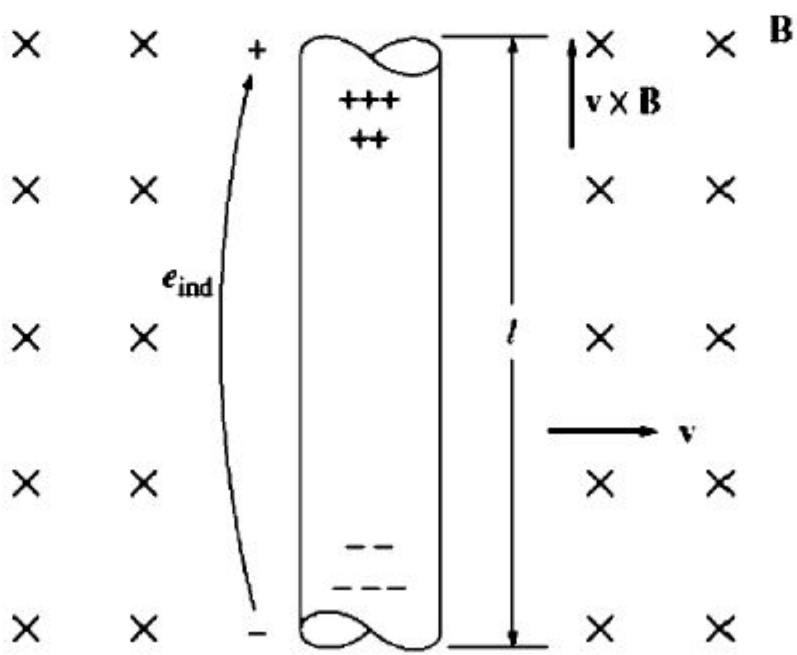
Faraday's Law (1)

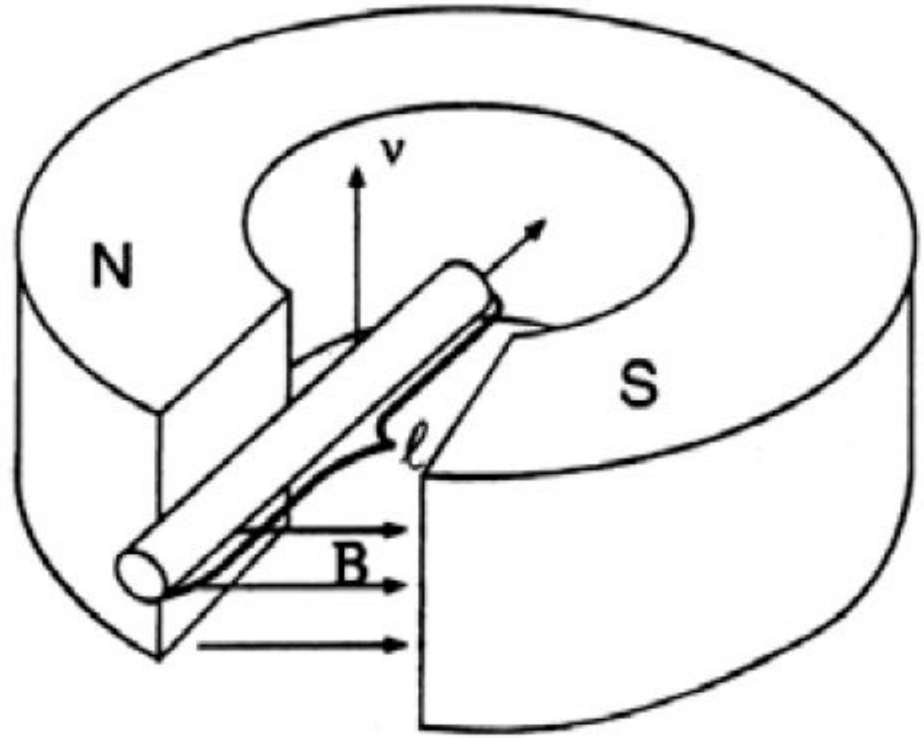
The law presents in two different forms:

- A moving conductor cutting flux lines of a constant magnetic field has a voltage induced in it.

- A changing magnetic flux inside a loop of conducting material will induce a voltage in the loop.

Faraday’s Law (2) **Art. 1.7**



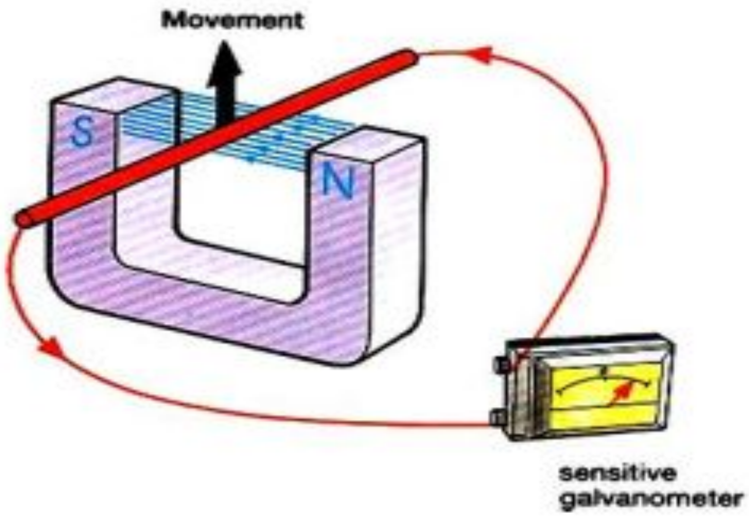


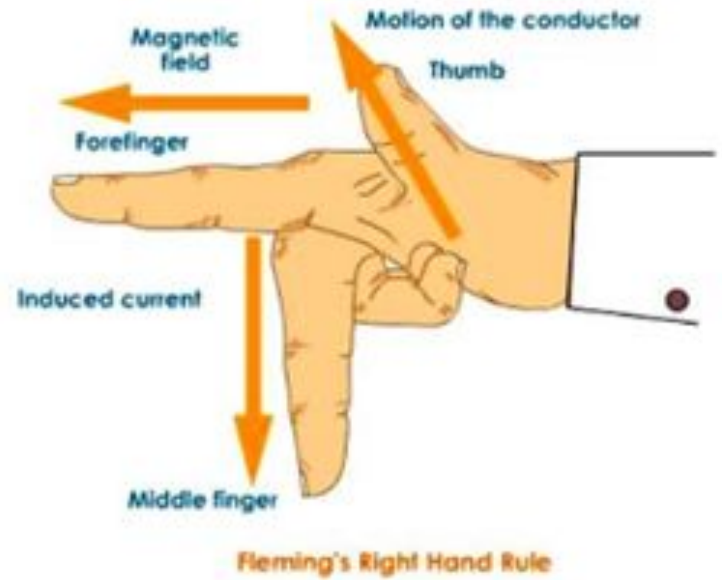
$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

Direction of Induced emf

The direction of induced voltage in a conductor in magnetic field is determined by Fleming's right hand rule:

If Forefinger points direction of flux from North to South and Thumb points direction of motion of the conductor, the Middle finger will point the direction of the induced voltage.

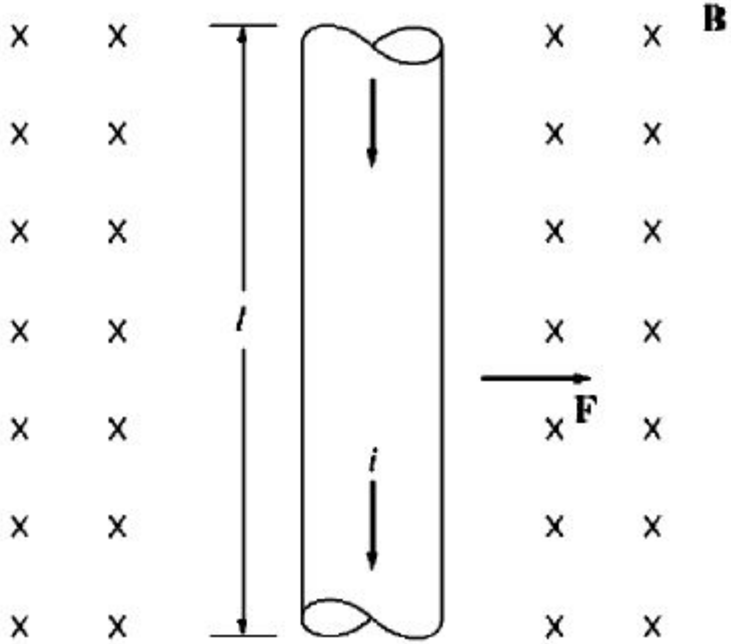


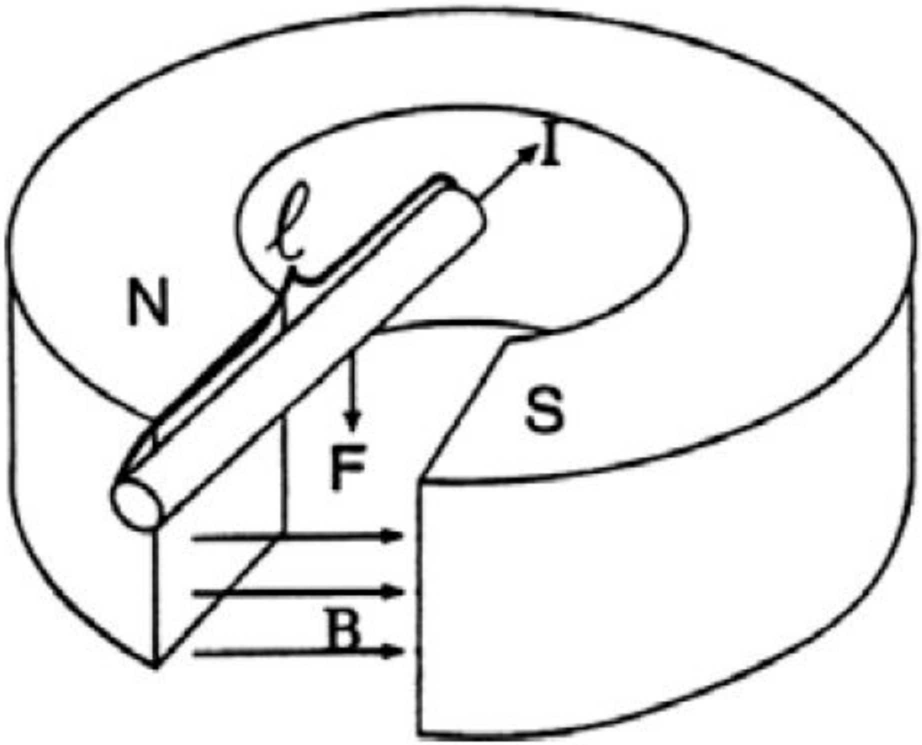


Ampere’s Force Law (1)

- A force is generated on a current carrying conductor located in a magnetic field.

Ampere's Law (2) **Art. 1.6**



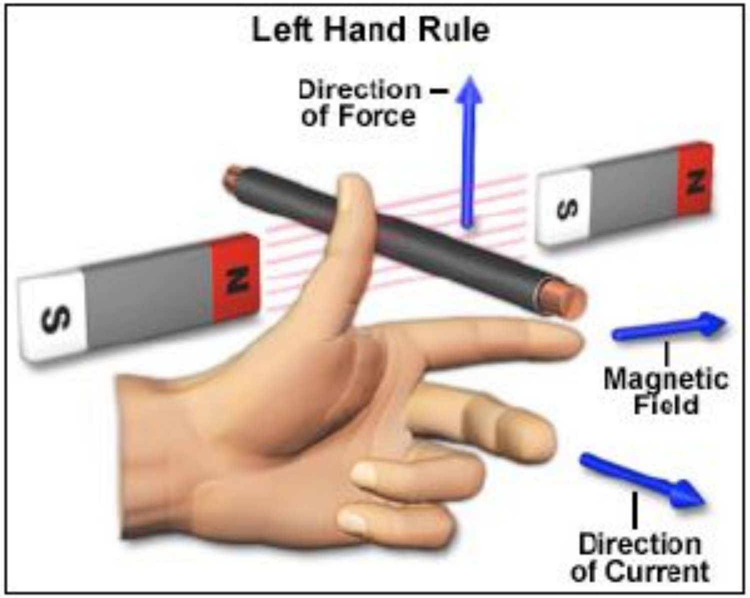


$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

Direction of Force

The direction of the force on a conductor in magnetic field is determined by the Fleming's left hand rule:

If Forefinger points the direction of flux from North to South and Middle finger points the direction of current in the conductor, the Thumb will point the direction of force on the conductor.



Loading Effect-Lenz's Law

Lenz's law:

An induced electromotive force (emf) always gives rise to a current whose magnetic field opposes the change in original magnetic flux.

Lenz's law explains that

- An increased force is required to drive the generator as its load increases.
- An increased current supply to the motor is required as its load increases.

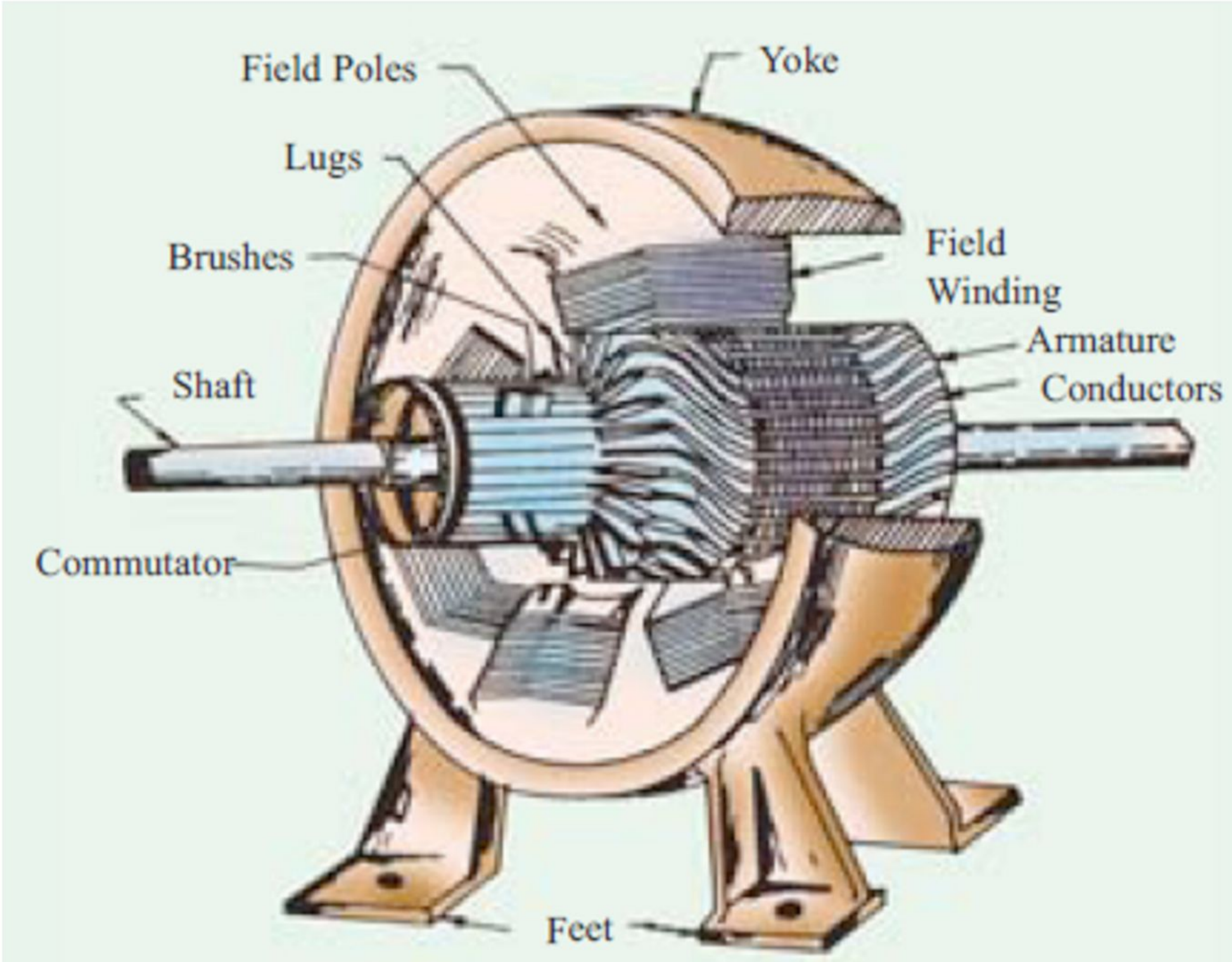
Direct current (dc) Machine

Application of dc Machine

DC motors are used in a wide variety of industrial drives, such as-
Robots, machine tools, petrochemical, pulp, paper and steel mills, oil drilling rigs, and mining. They are also used extensively in automotive systems and railroads.

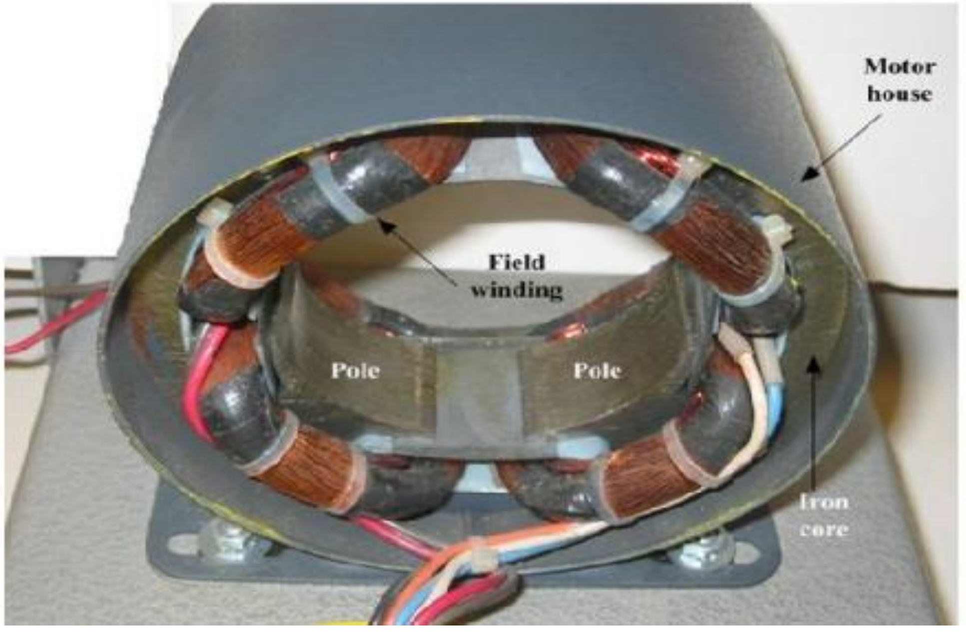
Like ac machines, dc machines have ac voltages and currents within them.

Practical dc Machine (2)



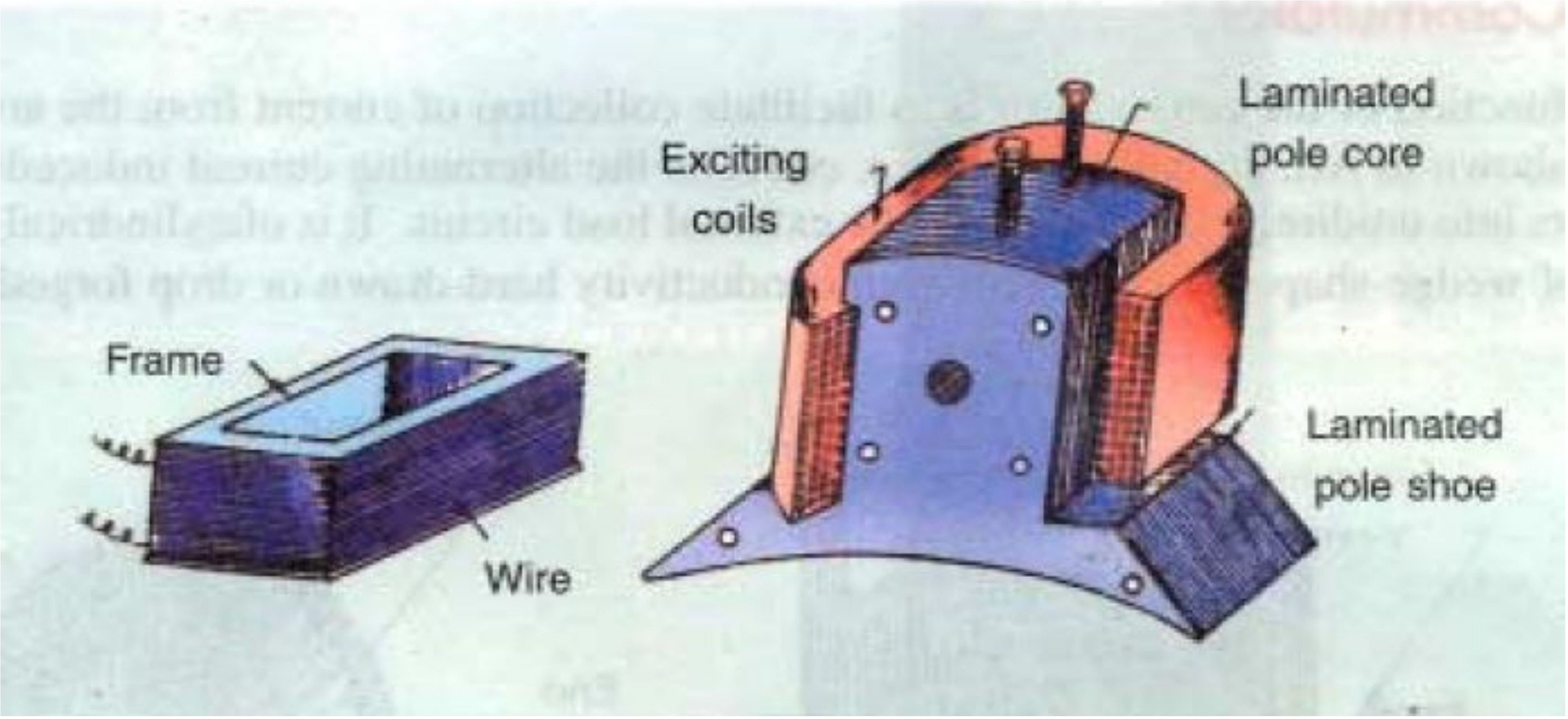
Different Parts of a dc Machine (1)

Yoke: The outer frame of a dc machine and made up of cast iron or steel. It provides mechanical strength to the whole assembly as well as carries the magnetic flux produced by the field winding.



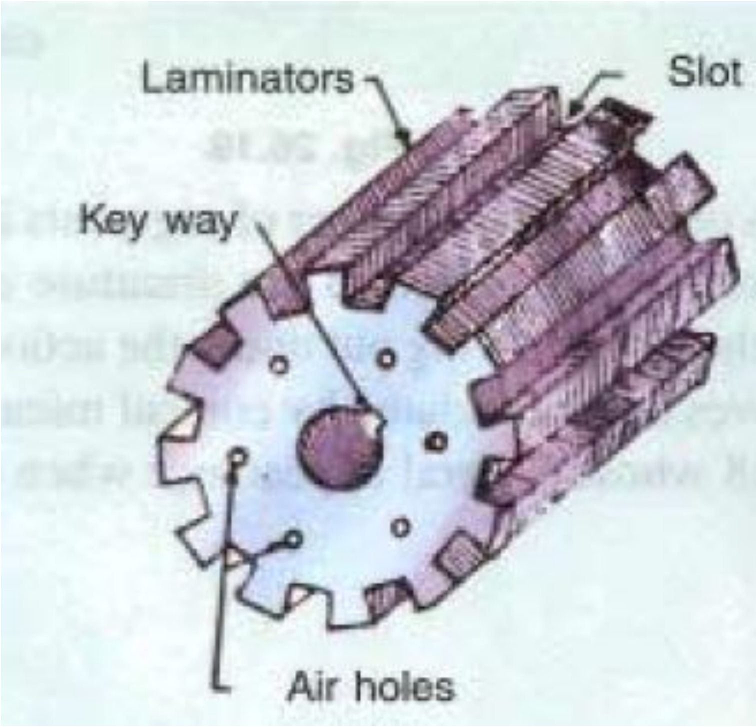
Different Parts of a dc Machine (2)

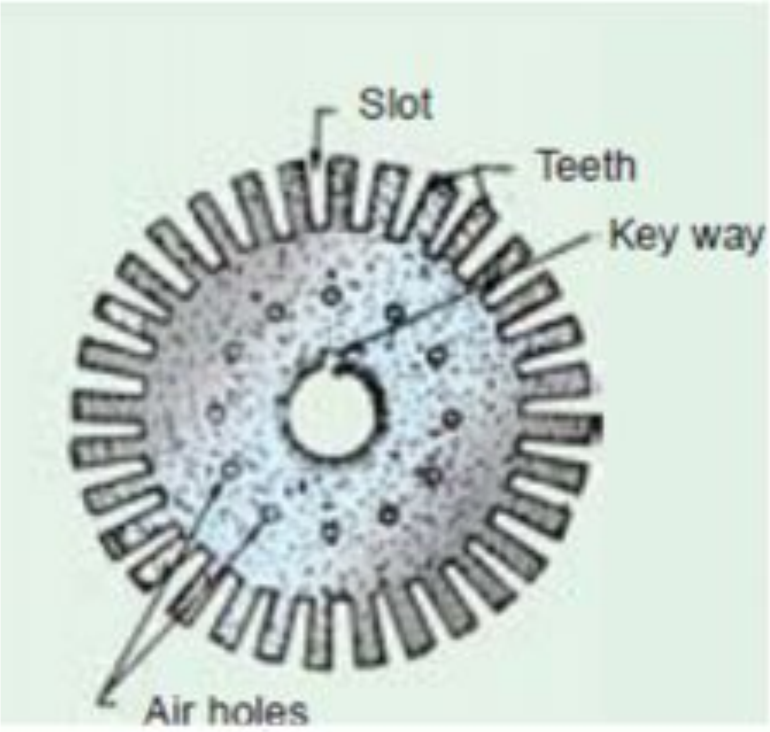
Pole: In old machines pole core are solid piece of cast iron or cast steel. In new construction, laminated steel are used for pole core construction.



Different Parts of a dc Machine (3)

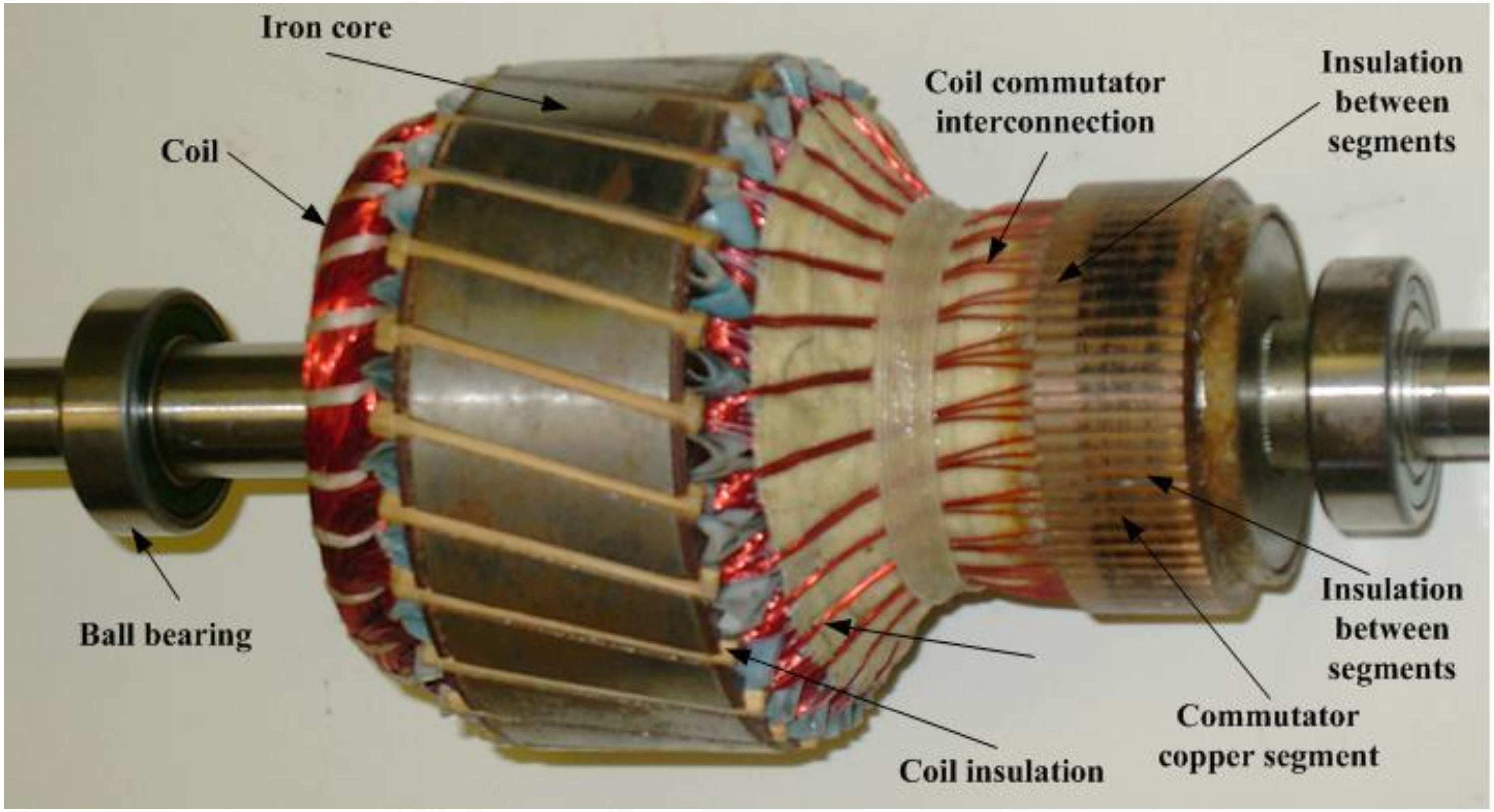
Armature: The moving part of the dc generator is called the armature. It consists of a shaft upon which all parts are mounted. Armature is made of laminated sheet steel and it houses the conductor or coils where emf (voltage) is induced.





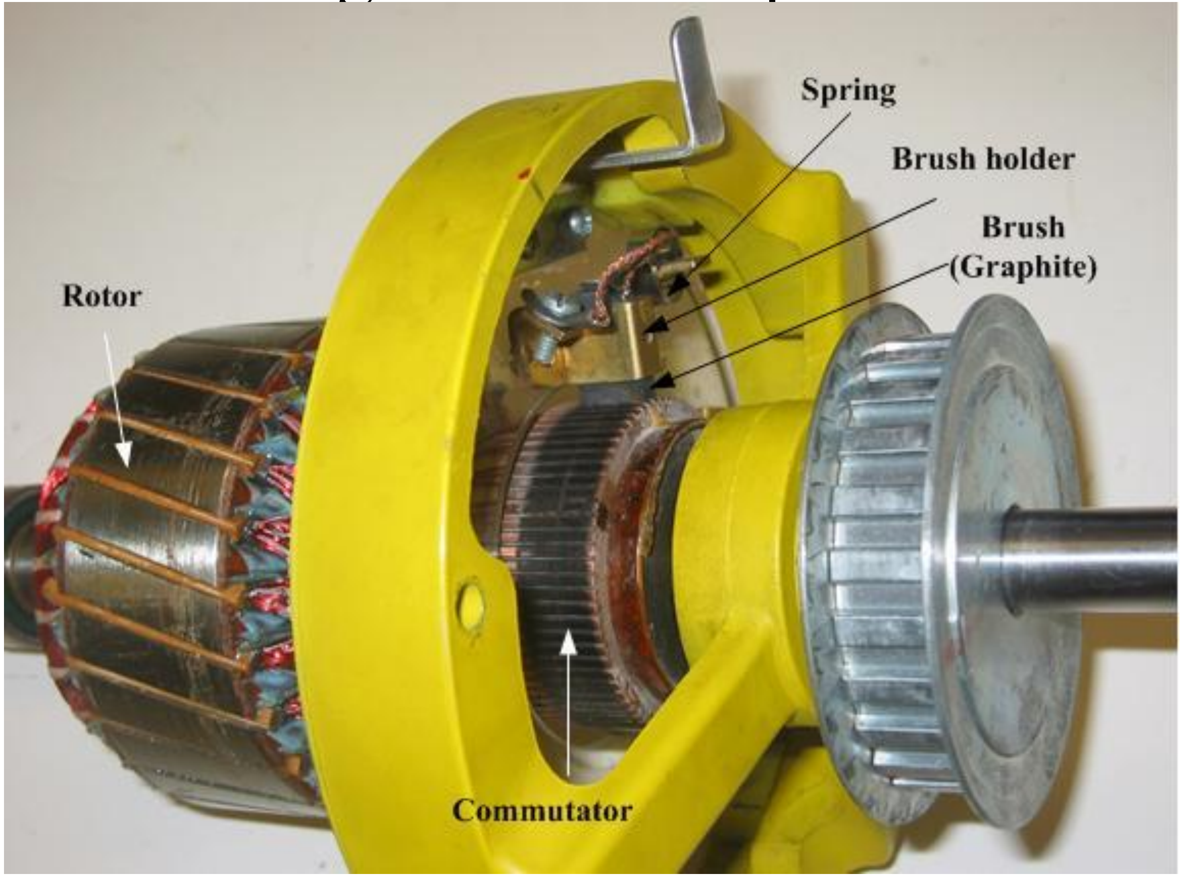
Different Parts of a dc Machine (4)

Commutator: Its function is to facilitate collection of current from the armature conductor and converts it from alternating into unidirectional in the external circuit.



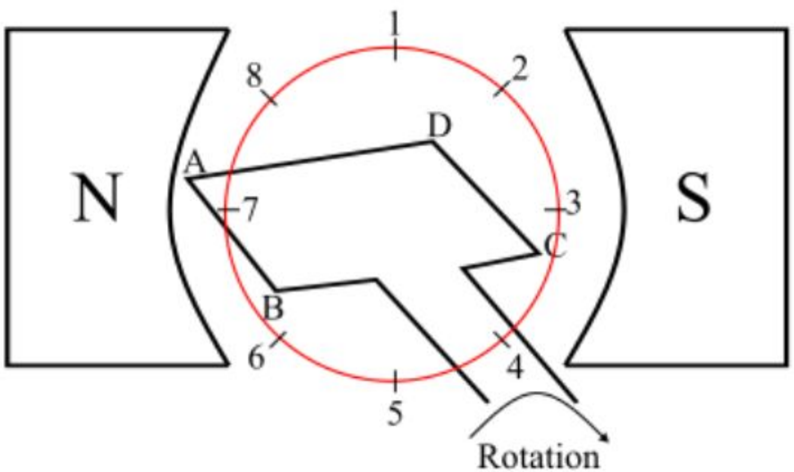
Different Parts of a dc Machine (5)

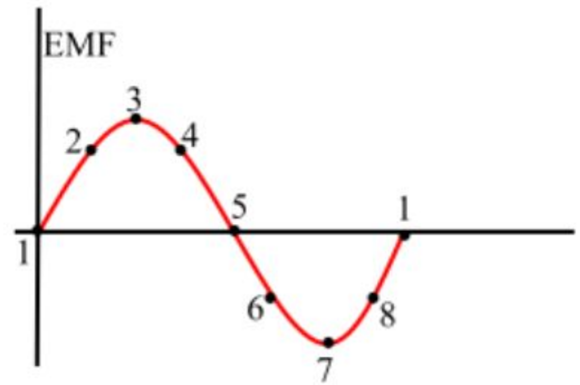
Brushes: They are usually made of carbon or graphite; they collect current from commutator and supply it to the external circuit and placed in the magnetic neutral plane.



Working Principle

Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.





Working Principle

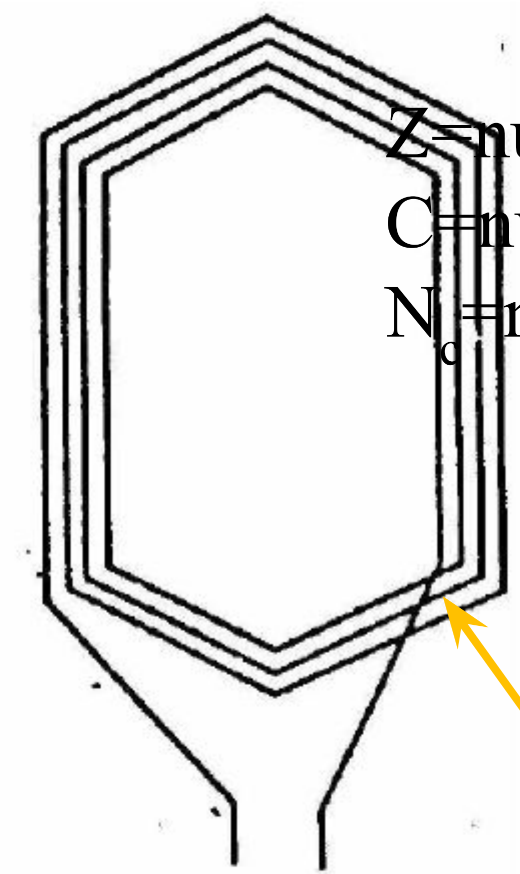
The EMF induced in a DC generator can be explained as follows

- When the loop is in position-1, the generated EMF is zero because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a small EMF is generated.
- When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated EMF is maximum.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a reduced EMF is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

It is clear that the generated EMF in the loop is alternating one.

Armature Winding in real Machines (1)

Art. 26.08 to 26.30 BL Theraja

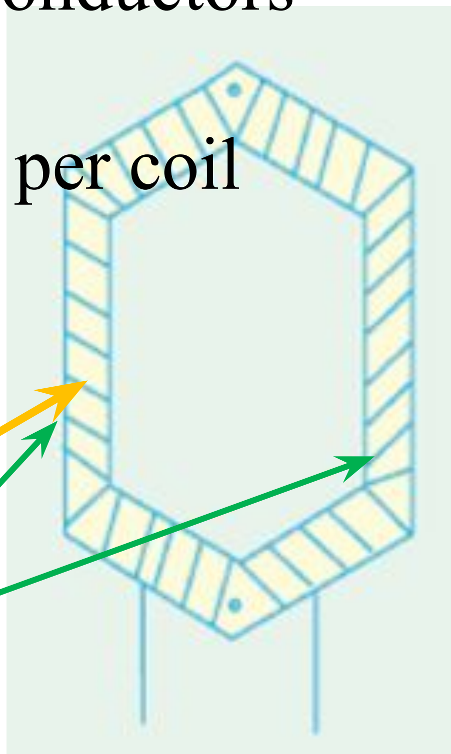


Z=number of total conductors

C=number of coils

N_c=number of turns per coil

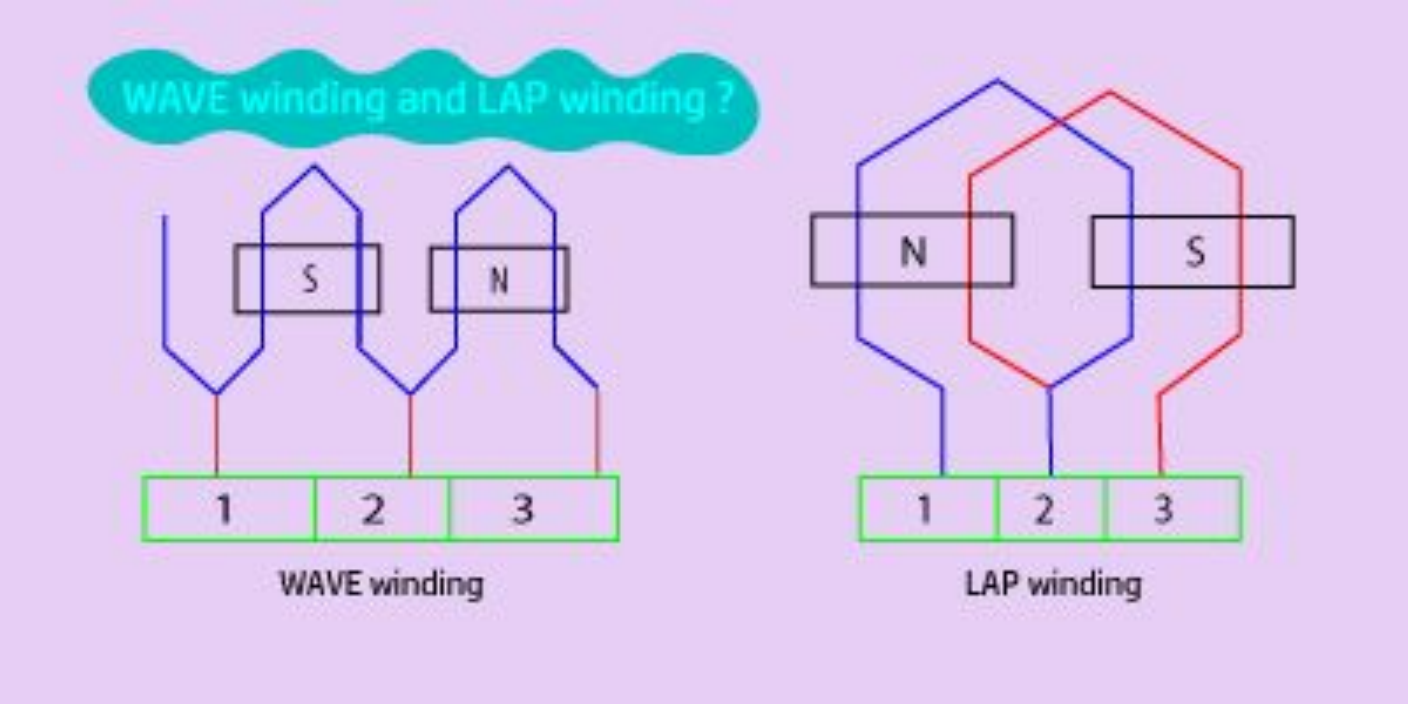
$$Z = 2CN_c$$



Coil Coil sides

Armature Winding in real Machines (2)

Two types of armature winding are generally used to place the conductors on the armature namely; Lap and wave winding.



Features of Lap winding

1) The total brush number is equal to the number of poles.

2) Number of parallel paths is equal to the number of poles.

3) The emf between +ve and –ve brushes is equal to the emf in any of the parallel paths. Therefore, generated emf =(average induced emf/conductor)xZ/P.

4) If I_A is the armature current of the machine, then current in a parallel path is I_A/P.

Features of Wave winding

1) The total brush number is only two.

2) Number of parallel paths is also two.

3) The emf between +ve and –ve brushes is equal to the emf in any of the parallel paths. Therefore, generated emf =(average induced emf/conductor)xZ/2.

4) If I_A is the armature current of the machine, then current in a parallel path is $I_A/2$.

Comparison of Lap & Wave windings

- ✔For a given number of poles and conductors, wave winding gives more emf than lap winding.
- ✔Equalizer connection is not necessary in wave winding, but it must in lap winding.
- ✔Lap wound is suitable for low voltage high current machines.

Equalizer connection

There are many parallel paths in lap-wound machine. The reluctance of each path may be different which causes the voltage in each parallel path to differ, and the unequal voltages in turn cause flow of a circulating current through the windings and brushes. To reduce the circulating current, points on the winding, which should be at the same potential, are brought to the same potential by connecting them with a conductor. These connections, called equalizer connections, confine the circulating current to the winding, thus reducing the sparking at the brushes.

A wave winding requires no equalizer connections. This is because each path has conductors in series under all poles of the machine. Any difference in the magnetic flux from the poles will produce a different voltage in the conductors, but both paths will be equally affected and the total induced emf of each path will always be the same.

Generated emf of a real Generator

(Art. 8.5)

Voltage induced in a single conductor:

$$e_{in} = vBl = r\omega Bl$$

Voltage of a parallel path:

$$E_A = \frac{Zr\omega Bl}{a} = \left(\frac{ZP}{2\pi a}\right)\frac{2\pi rBl}{P}\omega$$

$$= \left(\frac{ZP}{2\pi a}\right)\phi\omega = K\phi\omega$$

$$\boxed{\omega = 2\pi\frac{n}{60}}$$

φ=Flux/pole in Weber

Z=Total number of armature conductor

P=No. of poles

n=Armature rotation in revolution per minute (rpm)

a=No. of parallel paths

Induced Torque of a real Motor

(Art. 8.5)

Torque induced in a single conductor: $$\tau_{cond} = rilB$$

Induced torque on a motor:

$$\tau_{ind} = ZrilB = \frac{ZrI_A lB}{a} = \left(\frac{ZP}{2\pi a}\right)\frac{2\pi rlB}{P} I_A$$

$$= \left(\frac{ZP}{2\pi a}\right)\phi I_A = K\phi I_A$$

φ=Flux/pole in Weber

Z=Total number of armature conductor

P=No. of poles

I_A=Armature (Motor) current in A

a=No. of parallel paths

Types of dc Generators (1)

(Art. 9.11~9.17)

Generators are classified according to the way in which their field circuits are excited : (a) Separately excited generators and (b) Self-excited generators.

(a)Separately excited generators: The field coils are energized from an independent external source of dc current to produce magnetic flux.

(b)Self-excited generators: The field coils are energized by the current produced by the generators themselves. There are three types of self-excited generators named according to the manner in which their field coils are connected to the armature.

Types of Self-excited Generators

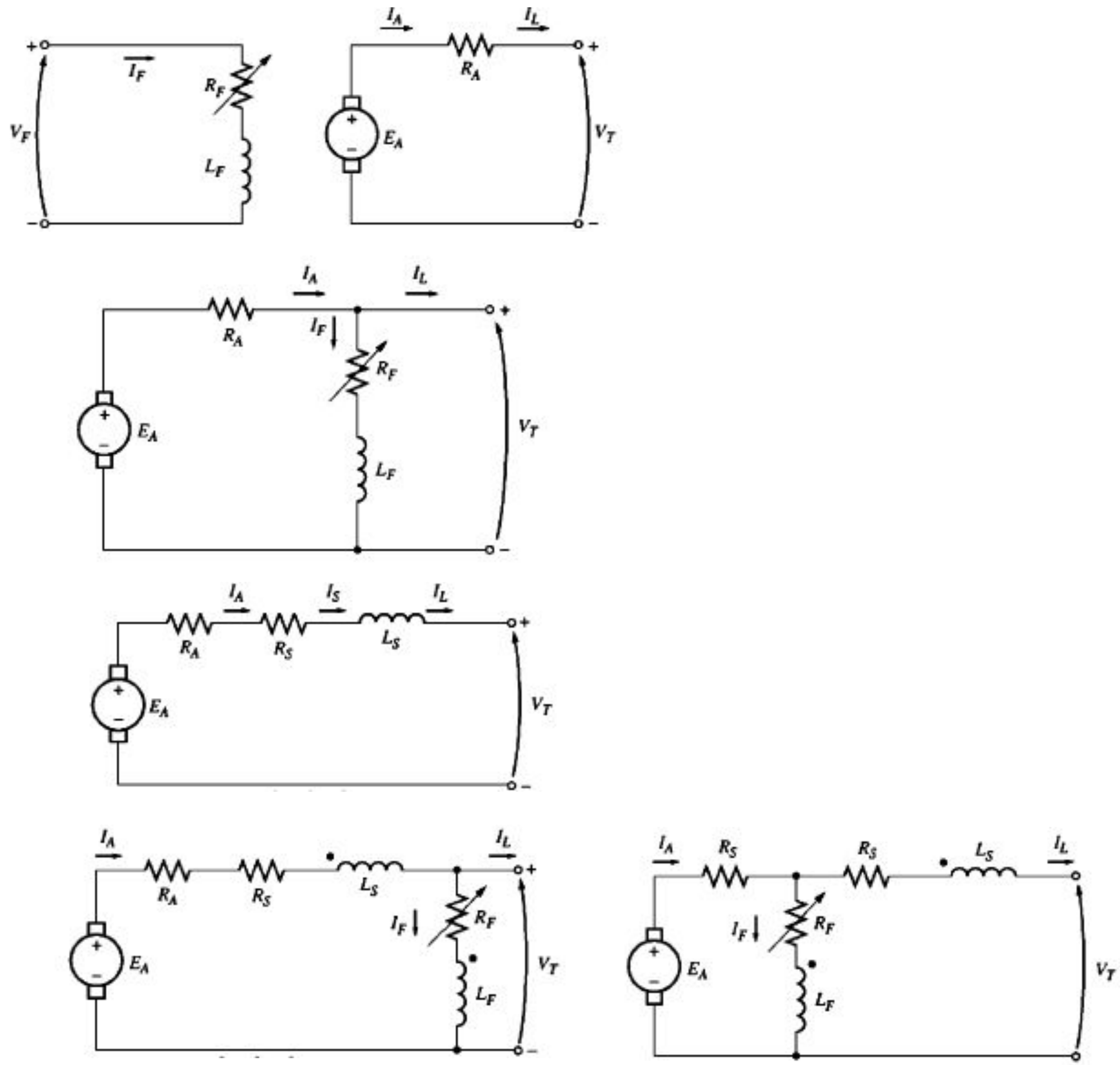
(i) Shunt generator: Field coil/circuit is connected in parallel with the armature circuit. Full voltage of the generator applies across both of the circuits.

(ii) Series generator: Field coil/circuit is connected in series with the armature circuit. These generators have very rare use.

(iii) Compound wound: Uses both a series and a shunt field windings; and can be either short-shunt or long-shunt. When the series field aids the shunt field, generator is said to be commutatively compounded. However, if the series field opposes the shunt field, the generator is said to be differentially compounded.

Noted that series field windings uses very few turns and has low resistance; the shunt field always has a very high resistance.

Types of dc Generators (2)



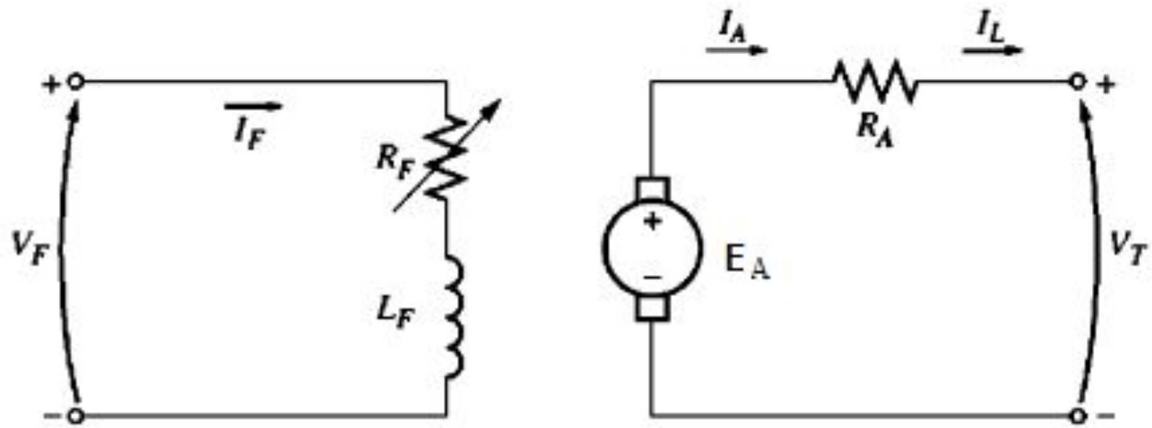
DC Generator

The various types of dc generators differ in their terminal (voltage-current) characteristics.

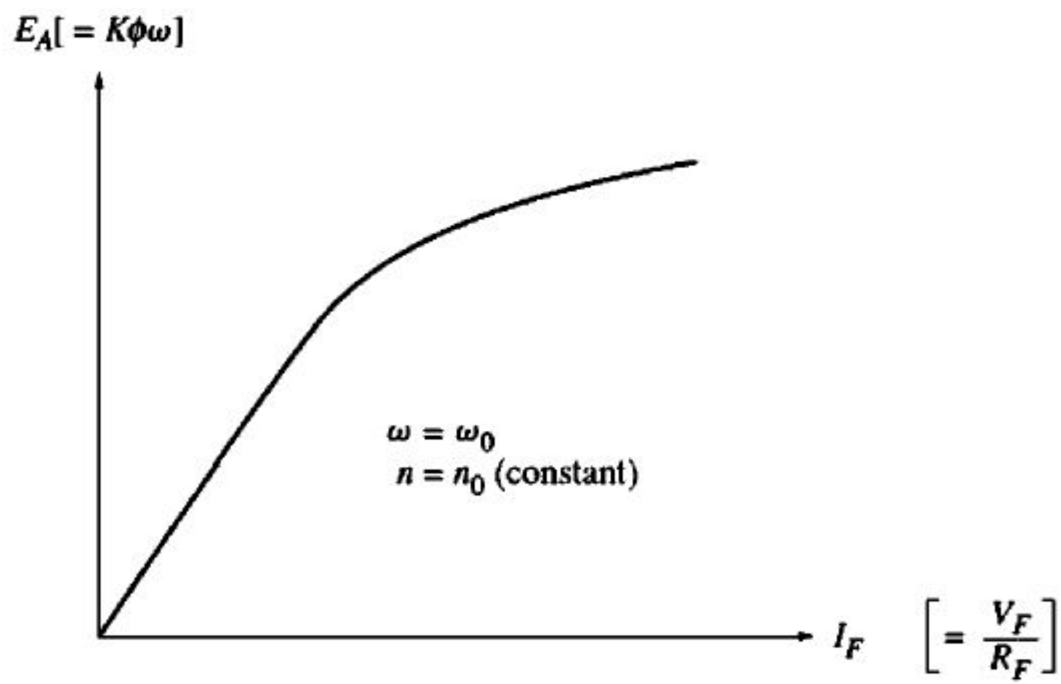
DC generators are compared by their voltages, power ratings, efficiencies, and voltage regulations.

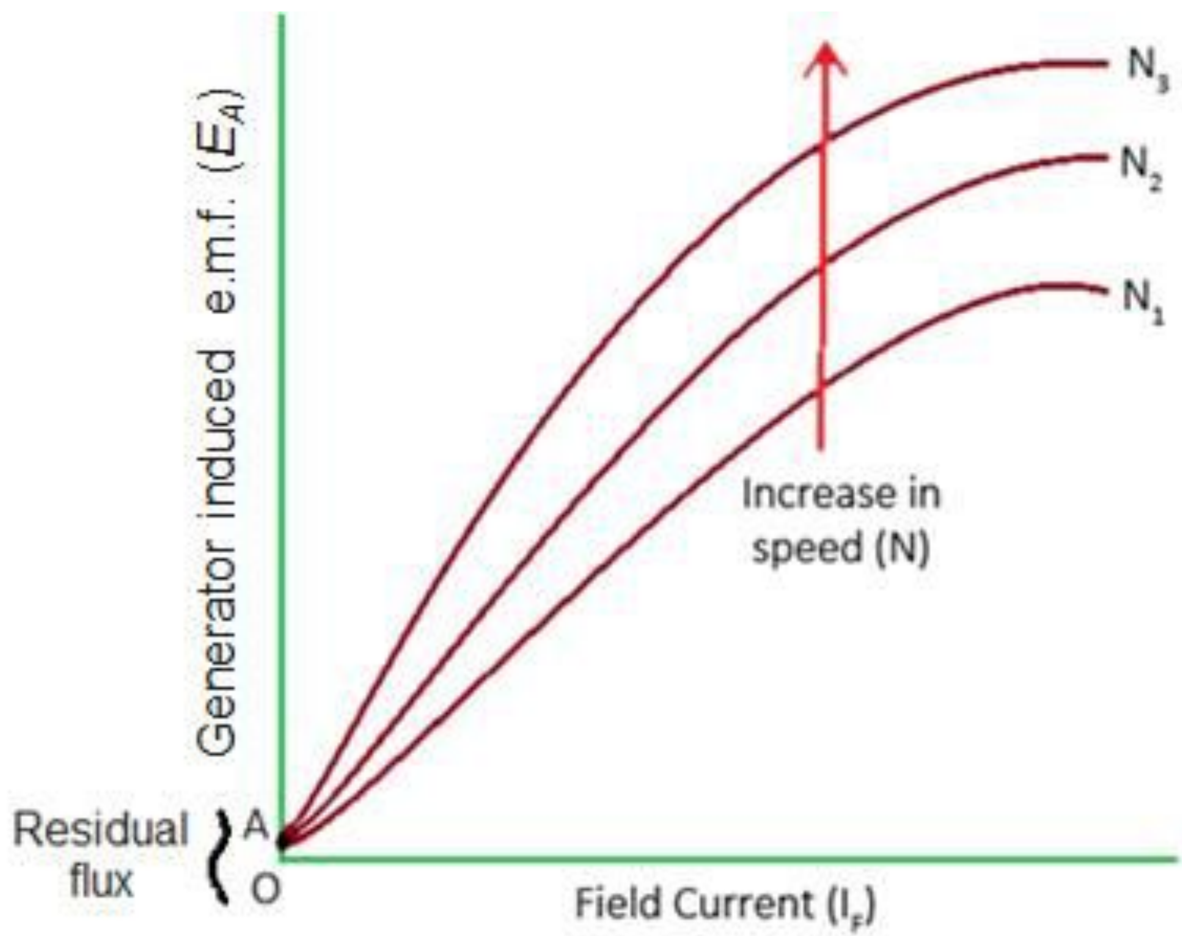
Separately Excited Generator(1)

(Art. 9.12)



Open Circuit Characteristic (OCC)

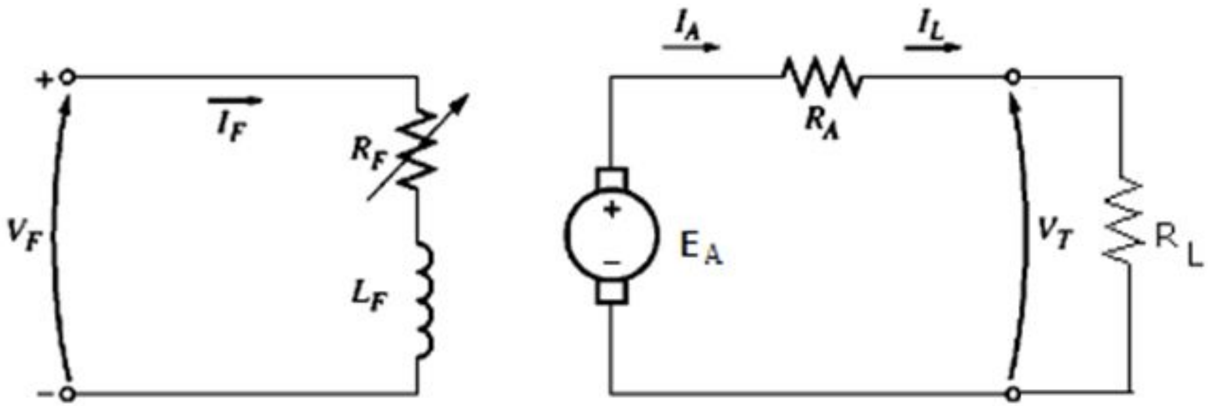




Open Circuit Characteristic (O.C.C.)

Separately Excited Generator(2)

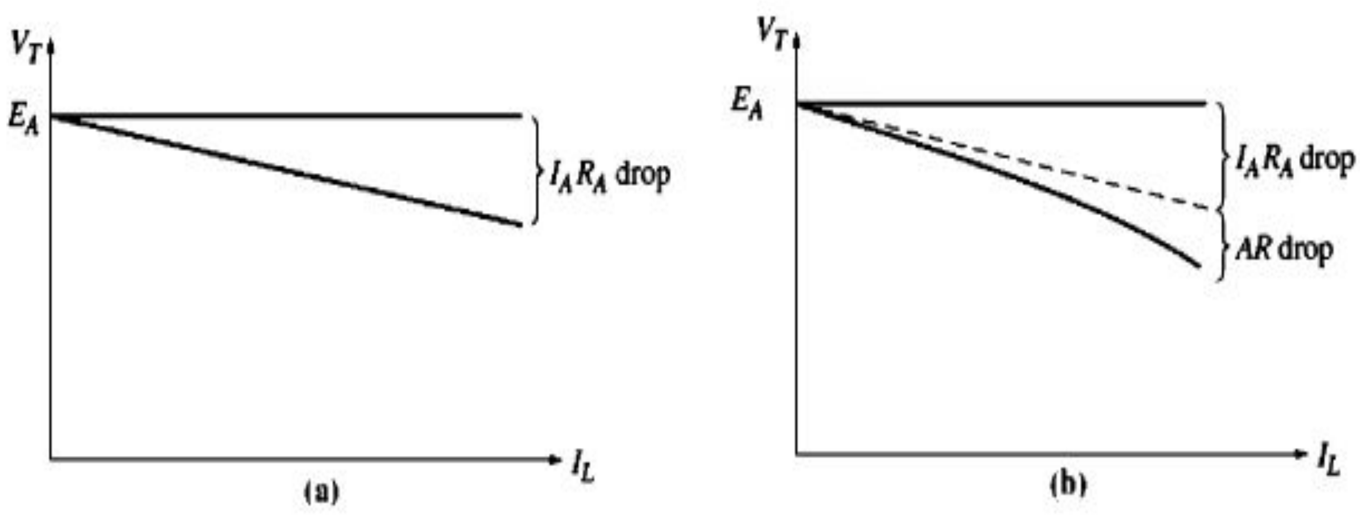
Terminal Characteristic **(Art. 9.12)**



$$I_F = \frac{V_F}{R_F}$$

$$I_L = I_A = \frac{V_T}{R_L}$$

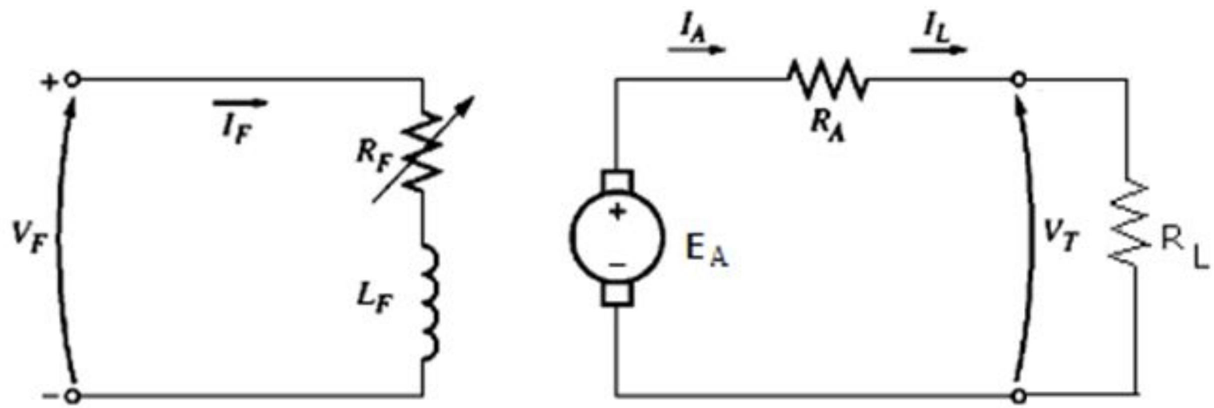
$$V_T = E_A - I_A R_A$$



Separately Excited Generator (3)

Control of terminal voltage

(Art. 9.12)

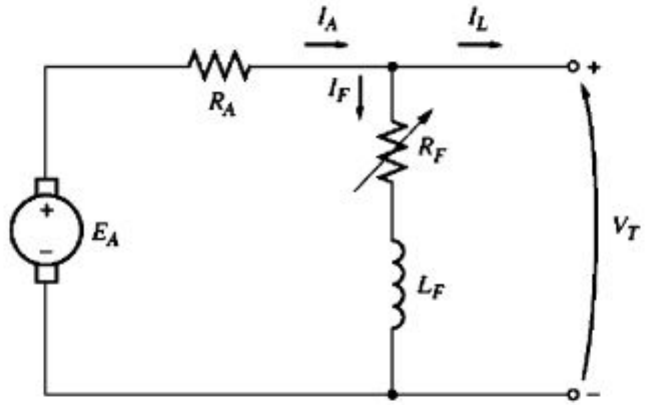


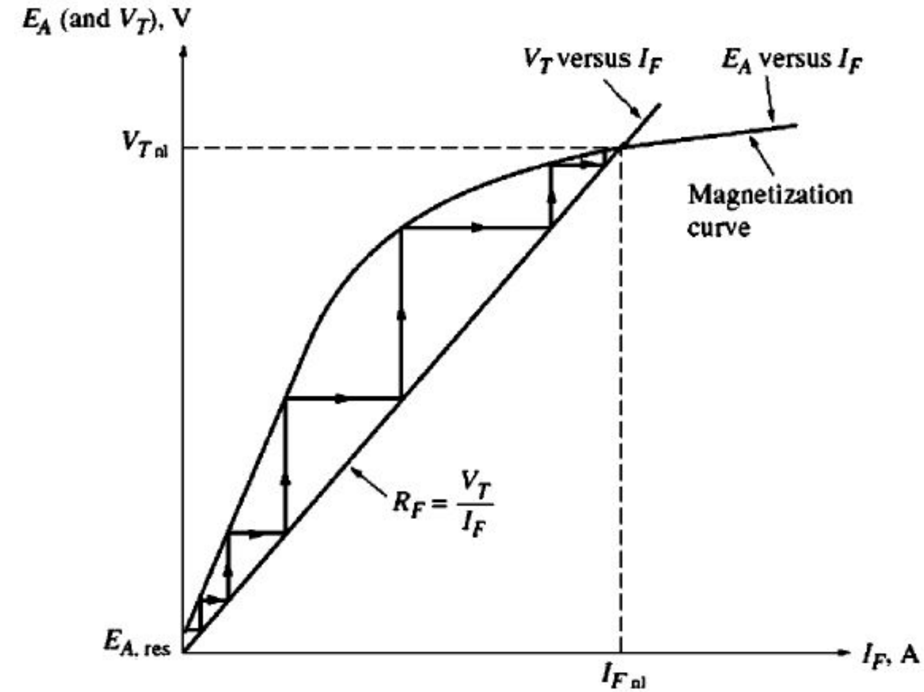
Terminal voltage can be controlled in two ways:

- *Change the speed:* If $\omega\uparrow$, then $E_A\uparrow = K\varphi\omega$, *so* $V_T\uparrow = E_A - I_A R_A$
- *Change the field current (widely used):* If $R_F\downarrow$, $I_F\uparrow = V_F/R_F$. *Therefore, flux* $\varphi\uparrow$.

$$E_A\uparrow = K\varphi\omega, \text{ so } V_T\uparrow = E_A - I_A R_A.$$

Shunt Generator(1) **(Art. 9.13)**

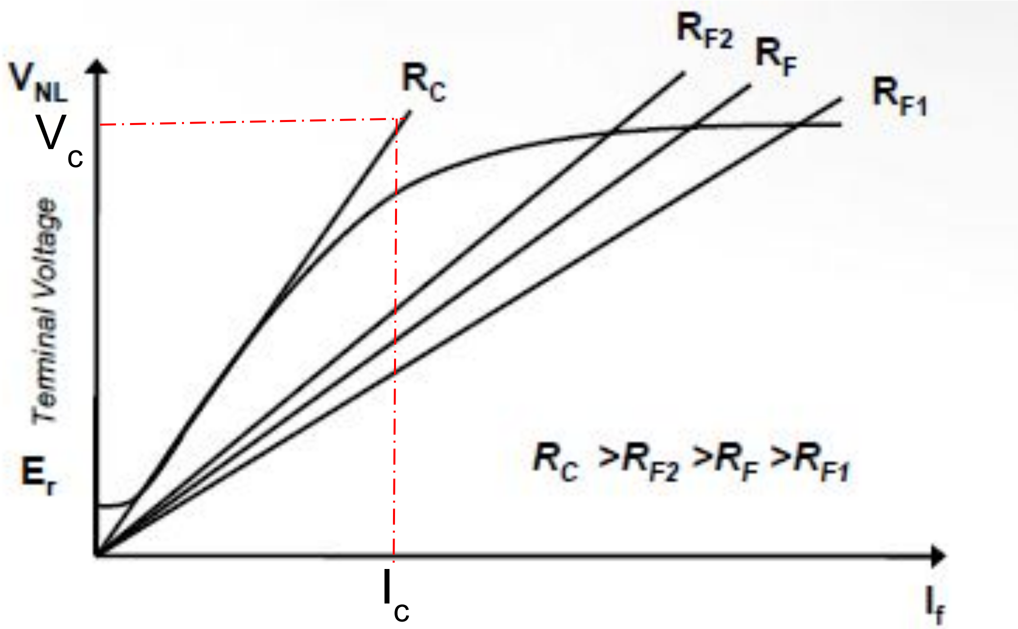




Conditions for voltage build-up in a shunt generator:

- There must be some residual magnetism in the generator pole.
- Field connection and armature rotation should be such that the induced voltage due to residual magnetism must produce flux in the same direction as the residual flux.
- Shunt field resistance should be less than the critical resistance.

Shunt Generator(2) <u>**(Art. 9.13)**</u>



To determine the critical resistance R_c, draw a tangent to the lower part of the OCC. If V_c is the voltage corresponding to the field current I_c, then the critical resistance R_c is given by:

$$\mathbf{R_c = (V_c / I_c)}$$

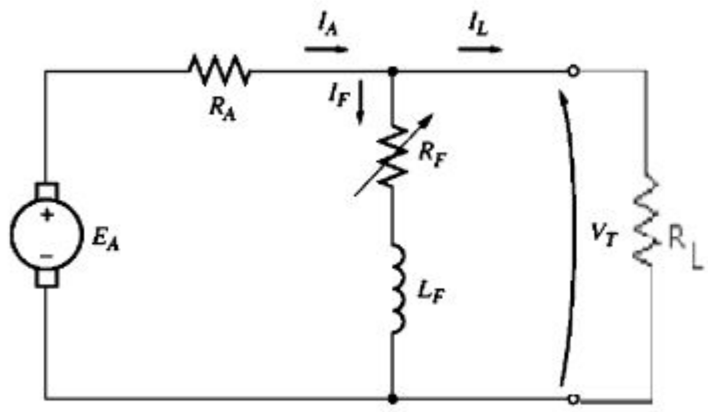
Lower value of the field circuit resistance will cause the shunt generator to build-up faster to a higher voltage; the generator will not build-up if the field resistance is greater or equal to the critical resistance.

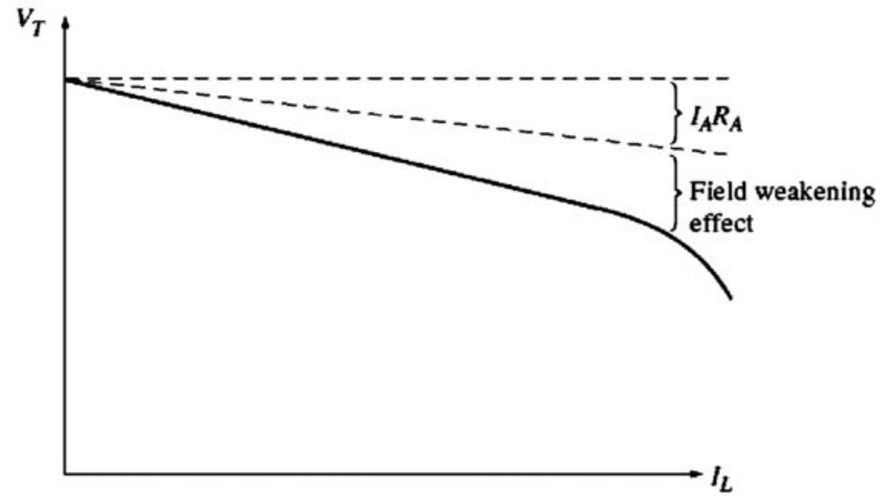
Shunt Generator (3) **(Art. 9.13)**

Load/External Characteristic

$$I_F = \frac{V_T}{R_F}$$

$$I_L = \frac{V_T}{R_L}$$



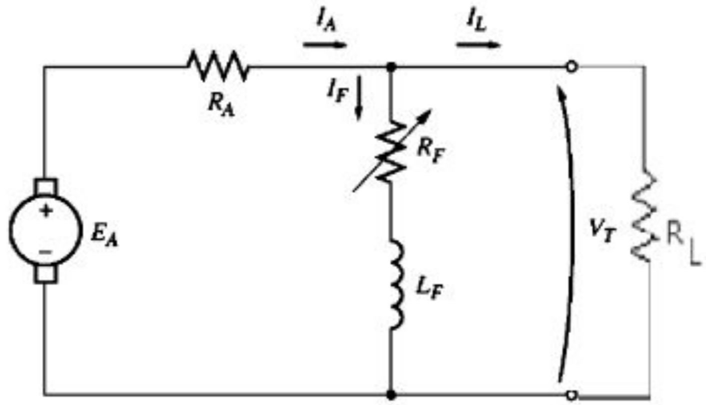


$$I_A = I_L + I_F;\ V_T = E_A - I_A R_A$$

In a shunt generator, the rate of decrease in output voltage is higher than that of separately excited generator, because increase in load current increases the armature resistance drop as well as decreases the field flux.

Shunt Generator (4) **(Art. 9.13)**

Control of terminal voltage



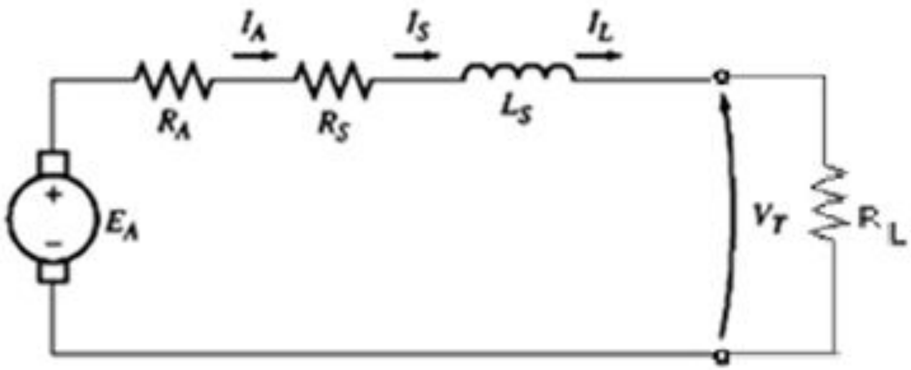
Terminal voltage can be controlled in two ways:

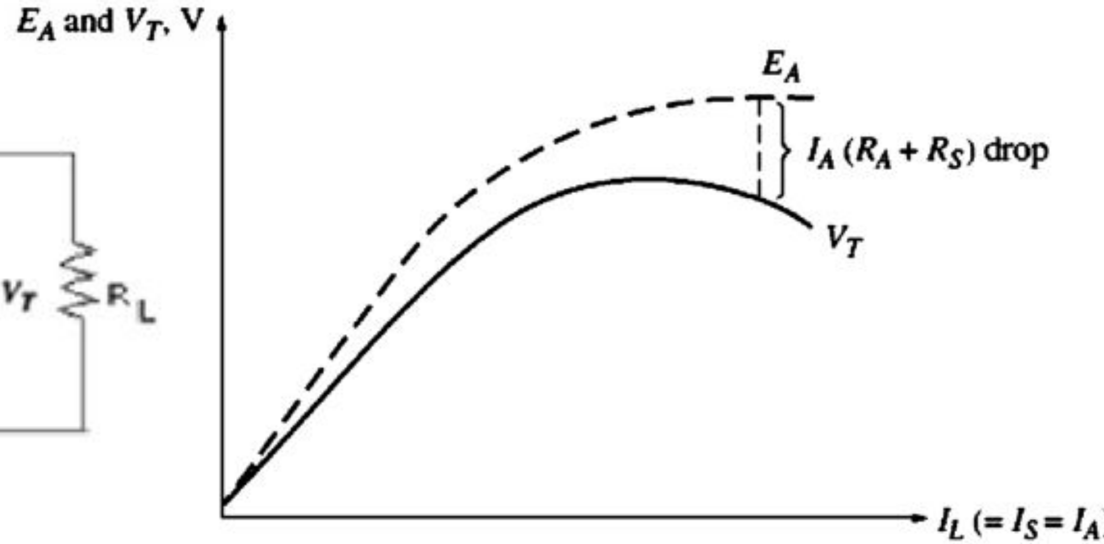
- *Change the speed:* If $\omega \uparrow$, then $E_A \uparrow = K\varphi\omega$, *so* $V_T \uparrow = E_A - I_A R_A$

- *Change the field resistance(widely used):* If $R_F \downarrow$, $I_F \uparrow = V_T/R_F$. *Therefore, flux* $\varphi \uparrow$.

$$E_A \uparrow = K\varphi\omega, \text{ so } V_T \uparrow = E_A - I_A R_A.$$

Series Generator **(Art. 9.14)**





$$I_L = \frac{V_T}{R_L}, \quad V_T = E_A - I_L(R_A + R_S)$$

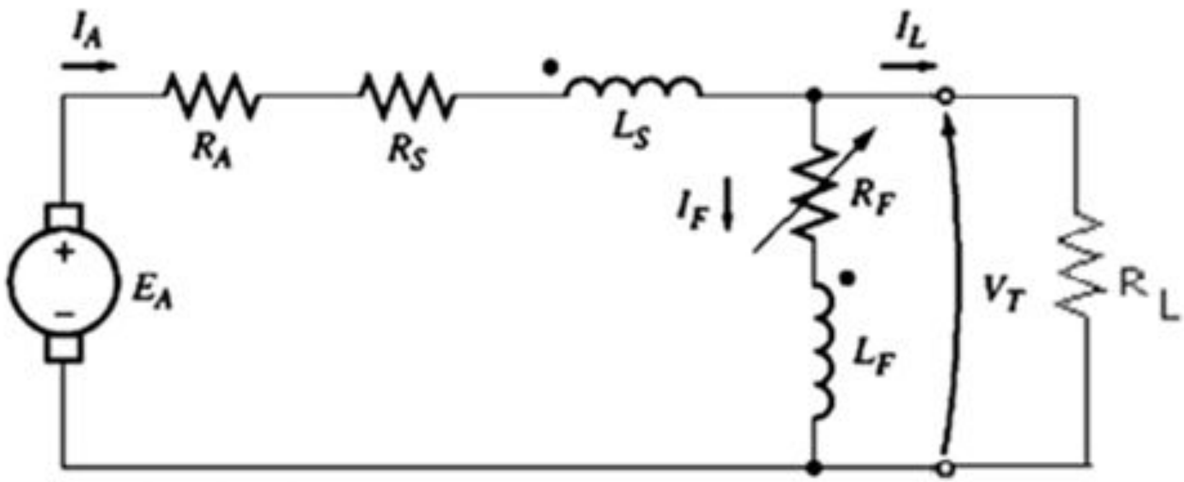
$$E_A = K\phi\omega \quad \Rightarrow E_A \propto \phi \propto I_L$$

The output voltage of a series generator varies with load current. This is undesirable in most applications. This is why this type of generator is rarely used.

Cumulative Compound Generator (1)

Long-shunt Connection

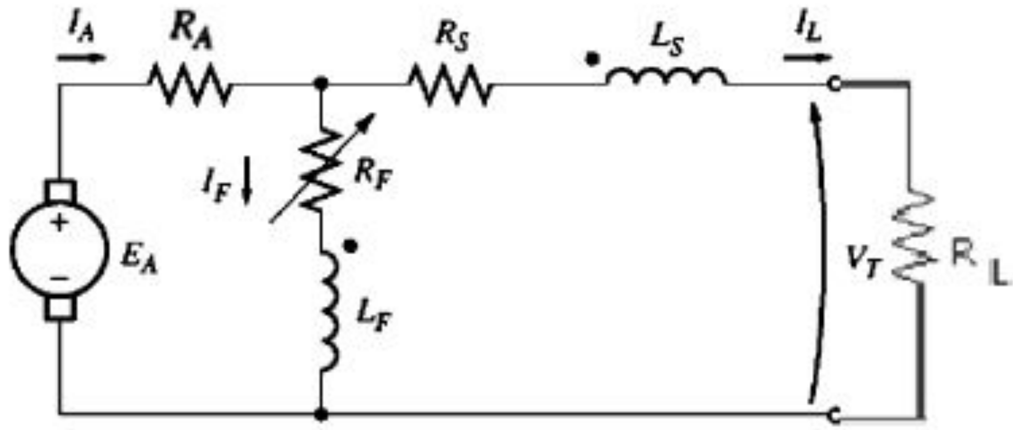
(Art. 9.15)



$$I_F = \frac{V_T}{R_F}, I_L = \frac{V_T}{R_L} \qquad \text{and } I_A = I_L + I_F$$

$$\therefore V_T = E_A - I_A(R_A + R_s)$$

Cumulative Compound Generator (2)
Short-shunt Connection



$$I_F = \frac{V_T + I_L R_S}{R_F}, I_L = \frac{V_T}{R_L} \quad \text{and } I_A = I_L + I_F$$

$$\therefore V_T = E_A - I_A R_A - I_L R_s$$

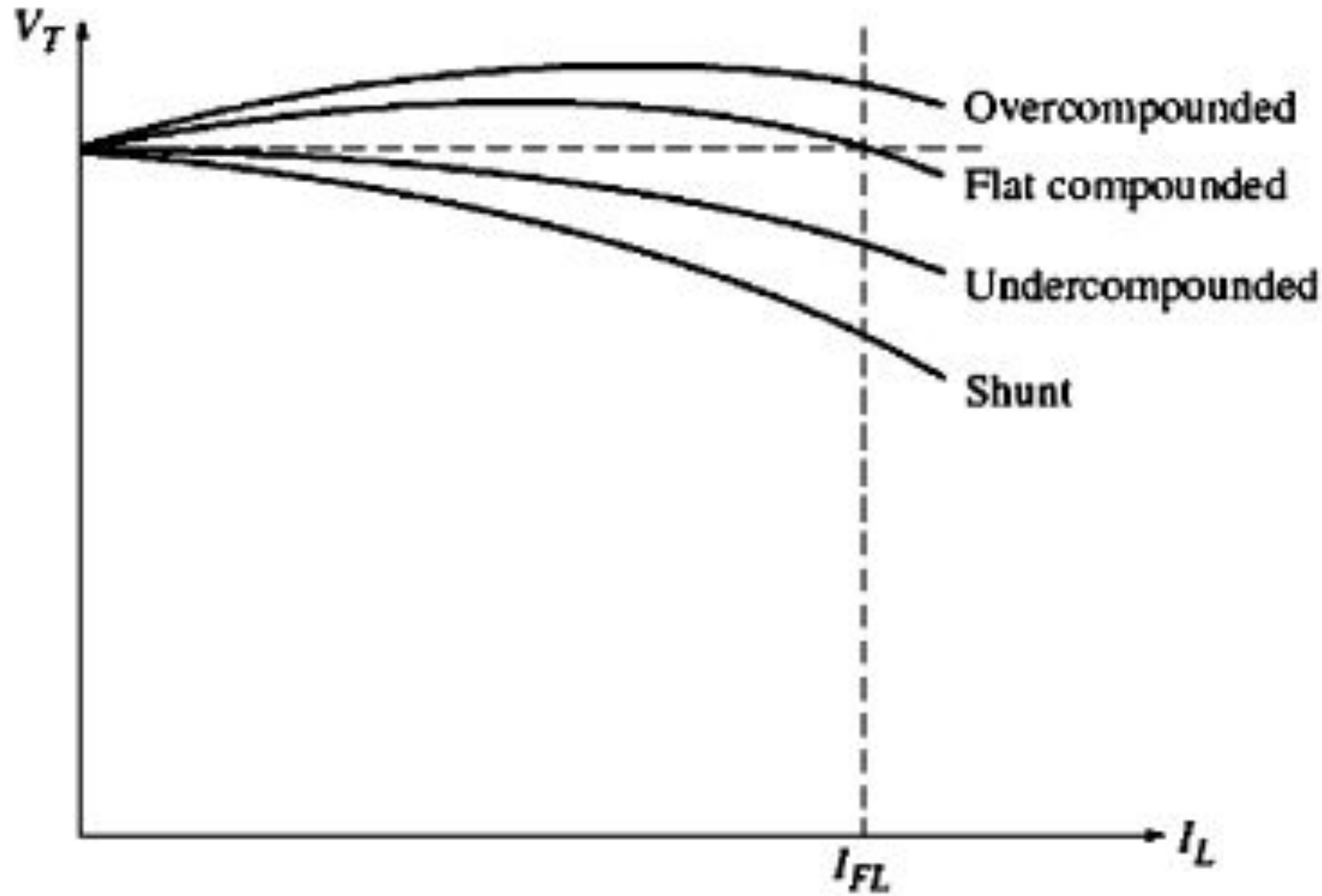
Cumulative Compound Generator (3) Terminal Characteristic

If I_A increases

1) $I_A(R_A+R_S)$ drop increases; the output voltage V_T decreases
2) Series field flux as well as the induced voltage E_A increases; the output voltage V_T increases

Which of the above effect predominates in a machine completely depends on how many series turns are placed on the poles of the machine.

Cumulative Compound Generator (4)

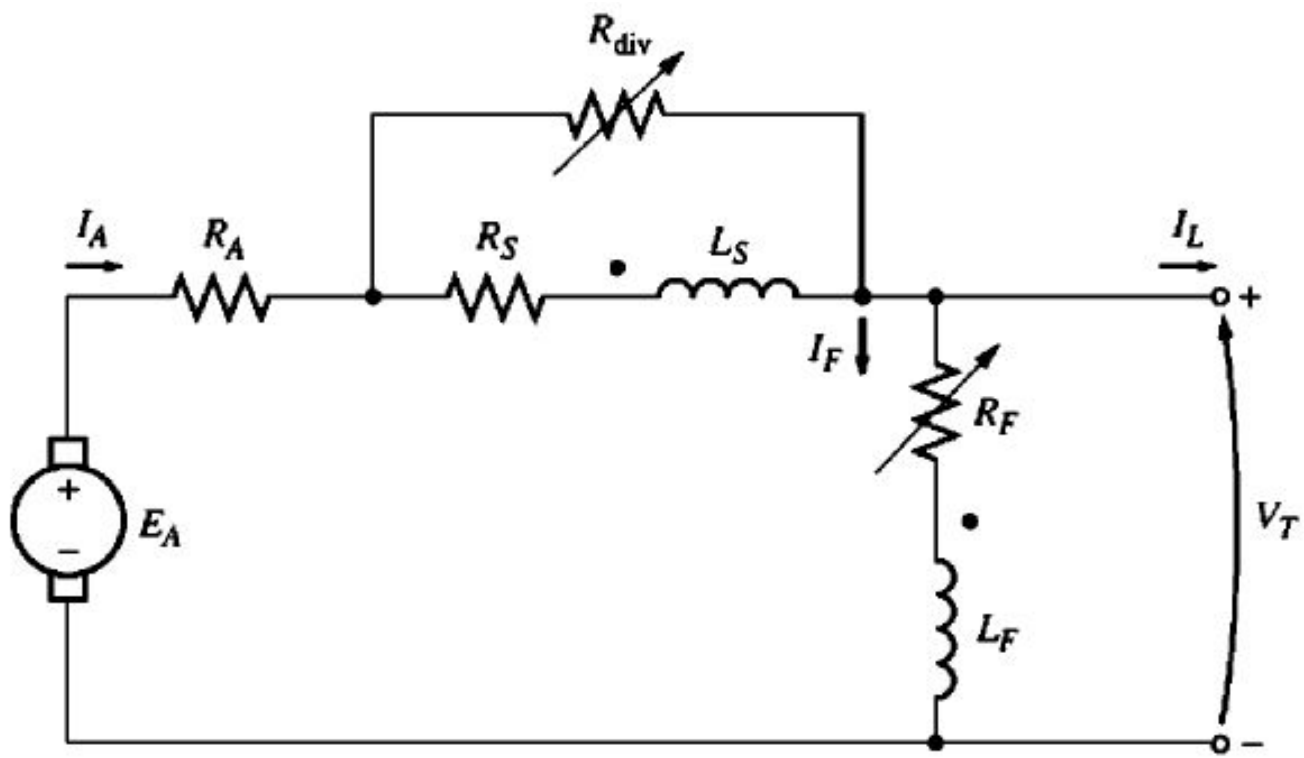


Cumulative Compound Generator (5) Control of terminal voltage

Terminal voltage can be controlled in two ways:

- *Change the speed:* An increase in ω causes $E_A = K\varphi\omega$ to increase; increasing the terminal voltage V_T.
- *Change the field resistance(widely used):* A decrease in R_F causes I_F to increase, which increases the flux φ, and $E_A = K\varphi\omega$ increases. Finally, an increase in E_A increases V_T

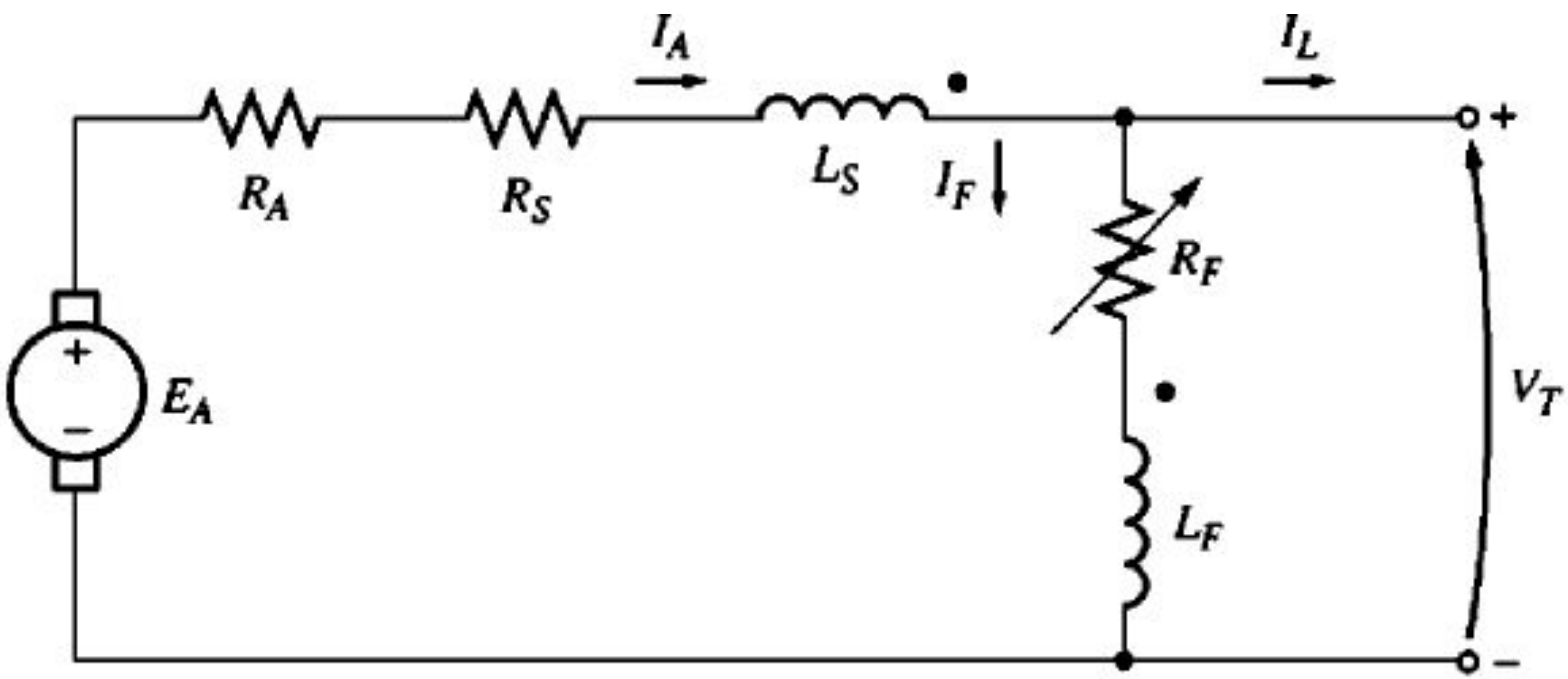
Cumulative Compound Generator (6)



Compound Generator with diverter resistor

Differentially Compound Generator (1)
Long Shunt

(Art. 9.16)



$$\mathscr{F}_{\text{net}} = N_F I_F - N_{\text{SE}} I_A - \mathscr{F}_{\text{AR}}$$

$$I_A = I_L + I_F \qquad I_F = \frac{V_T}{R_F}$$

$$V_T = E_A - I_A(R_A + R_S)$$

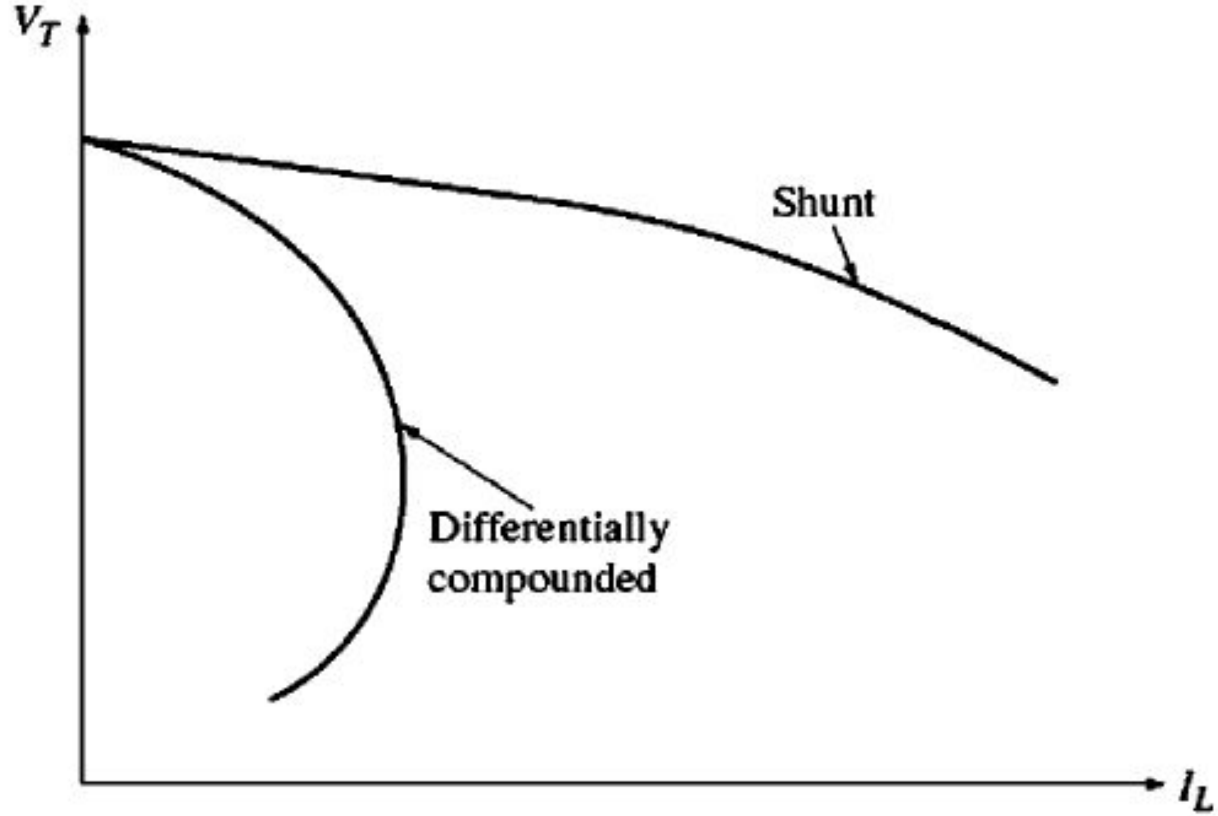
Differentially Compound Generator (2) Terminal Characteristic

If I_A increases

1) $I_A(R_A+R_S)$ drop increases; the output voltage V_T decreases
2) Total machine flux as well as the induced voltage E_A decreases; the output voltage V_T also decreases

Since both effects tend to decrease V_T, the terminal voltage drops drastically in comparison to other generators.

Differentially Compound Generator (3)
Terminal Characteristic



Causes of Voltage Drops in DC Generator

The major causes of voltage drop in a dc generator are:

1) Armature resistance drop
2) Brush contact drop
3) Armature reaction voltage drop
4) Voltage drop due to flux weakening in shunt generator.

Voltage Regulation of a DC Generator

It can be seen from the external characteristics that the terminal voltage of a dc generator falls slightly as the load current increases. *Voltage regulation* is defined as the percentage change in terminal voltage when full load is removed from the generator terminals. It is expressed by the following equation:

$$\text{Voltage Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{E_A - V_T(\mathit{at}\ \text{Full Load})}{V_T(\mathit{at}\ \text{Full Load})} \times 100\%$$

Problems

A dc shunt generator has an induced voltage on open circuit of 127 Volts. When the machine is on load, the terminal voltage is 120 Volts. Find the load current if the field circuit resistance is 15 ohms and the armature resistance is 0.02 ohm. Ignore armature reaction loss and brush drop.

$$I_f = \frac{V_t}{R_f}, \quad I_L = \frac{V_t}{R_L}, \quad I_a = I_L + I_f \text{ and } V_t = E_g - I_a R_a$$

$$I_a = \frac{E_g - V_t}{R_a} = \frac{127 - 120}{0.02} = 350A \quad \text{and } \mathrm{I_f} = \frac{120}{15} = 8A$$

$$\therefore I_L = I_a - I_f = 350 - 8 = 342A$$

Problems

A separately excited dc generator, when running at 1000 rpm supplied 200 A at 125 V. What will be the load current when the speed drops to 800 rpm if I_f is unchanged? Given that the armature resistance is 0.04 ohms and brush drop is 2V.

$$V_t = E_A - I_a R_a$$

$$E_A = \mathrm{K}'\phi n$$

$$At\, 1000 \text{ rpm}, E_A = V_t + I_a R_a + 2 = 125 + 200 \times 0.04 + 2 = 135$$

$$\frac{E_{A1}}{E_{A2}} = \frac{\mathrm{K}'\phi n_1}{\mathrm{K}'\phi n_2} \Rightarrow E_{A2} = E_{A1}\frac{n_2}{n_1} = 135\frac{800}{1000} = 108V$$

$$I_a R_a = E_A - V_t - 2 = 108 - I_a R_L - 2$$

$$\therefore I_a = 106/(0.625 + .04) = 159.4$$

Problems

A 4-pole lap wound dc shunt generator has a useful flux per pole of 0.07 Wb. The armature winding consists of 220 turns each of 0.004 ohm resistance. Calculate the terminal voltage when running at 900 rpm if the armature current is 50 A.

$$E_A = \frac{ZP}{2\pi a}\phi\omega$$

Here P=a=4, φ=0.07 Wb, N=900 rpm and z=220x2=440

$$E_A = \frac{0.07 \times 4 \times 900 \times 440}{60 \times 4} = 462V$$

Armature resistance/path=(220*0.004)/4=0.22 ohm. Therefore R_a=0.22/4 =0.055 ohm.

$$\therefore V_t = E_g - I_a R_a = 462 - 50 \times 0.055 = 459.25V$$

Problems

A 100-kW, 250-V DC shunt generator has an armature resistance of 0.05 Ω and field circuit resistance of 60 Ω. With the generator operating at rated voltage, determine the induced voltage at (a) full load, and (b) half-full load.

(a) At full load,

$V_t=E_g-I_aR_a$

$I_f=250/60=4.17$ A

$I_{L\_FL}=100{,}000/250=400$ A

$I_a=I_{L\_FL}+I_f=400+4.17=404.17$ A

$E_g=V_t+I_aR_a=250+404.17*0.05=270.2$ V

(b) At half load,

$I_f=250/60=4.17$ A

$I_{L\_HL}=50{,}000/250=200$ A

$I_a=I_{L\_HL}+I_f=200+4.17=204.17$ A

$E_g=V_t+I_aR_a=250+204.17*0.05=260.2$ V

Problems

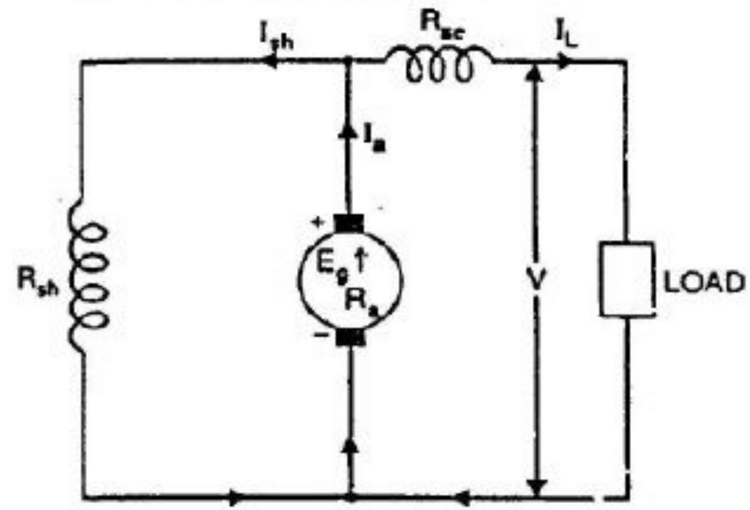
A 48-KW 240-V *short shunt compound* generator has the following parameters:

$R_a = 0.03\ \Omega$; $R_{se} = 0.015\ \Omega$; $R_f = 120.0\ \Omega$;

Take 2 V as the total brush contact drop.

(i) Calculate the induced voltage at full load.

(ii) Calculate the voltage regulation at full load.



Solution:

Problems

$$(i)\, I_L = \frac{48\times 1000}{240} = 200\,A;$$

$$Field\ circuit\ voltage: V_f = V_t + I_L R_{se} = 240 + 200\times 0.015 = 243V$$

$$\therefore Field\ current: I_f = \frac{V_f}{R_f} = \frac{243}{120} = 2.025\,A$$

$$I_a = I_L + I_f = 200 + 2.025 = 202.025\,A$$

$$E_a = V_t + I_L R_{se} + I_a R_a + V_{BD} = 243 + 202.025\times 0.03 + 2 = 251.0607V$$

$$(ii)\, Voltage\ regulation = \frac{E_a - V_t}{V_t}\times 100 = \frac{251.0607 - 240}{240}\times 100 = 4.61\%$$

Problems

The OCC for a certain self-exited, 125V, 50kW, 1750rpm shunt generator is shown in the following. The no load voltage with the rheostat shorted is 156V. Determine (a) The field circuit resistance; (b) Field-rheostat setting that will provide a no load voltage of 140V; (c) Armature voltage if the rheostat is set to 14.23Ω; (d) Field-rheostat setting that will cause critical resistance; (e) Armature voltage at 80% rated speed, and the rheostat shorted; (f) Rheostat setting required to obtain a no load armature voltage of 140V at 1750rpm if the shunt field is separately excited from a 120V dc source. **(Hubert: Example 12.1, pp 479)**

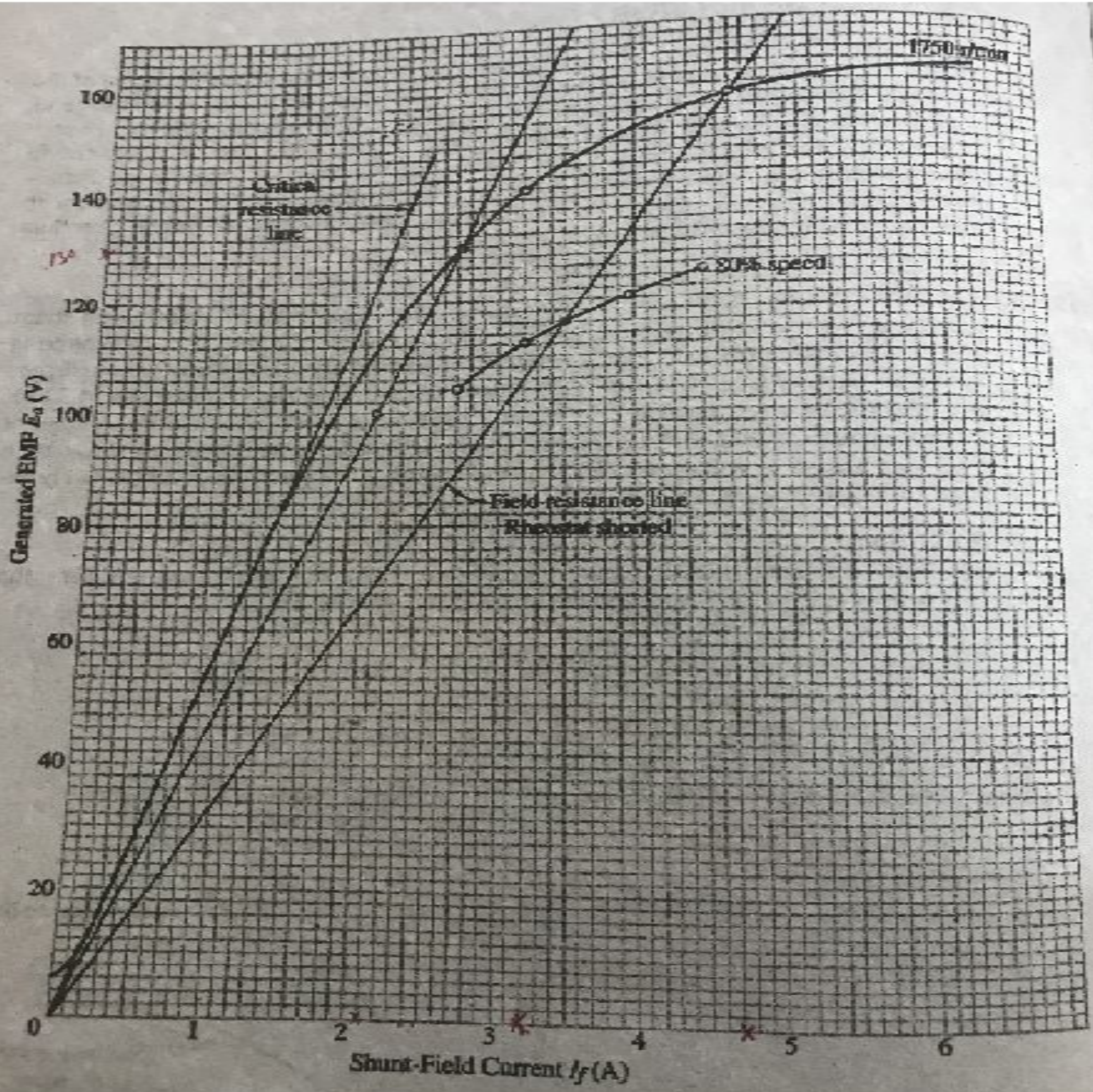


FIGURE 12.4
Magnetization curves and field-resistance lines for Example 12.1.

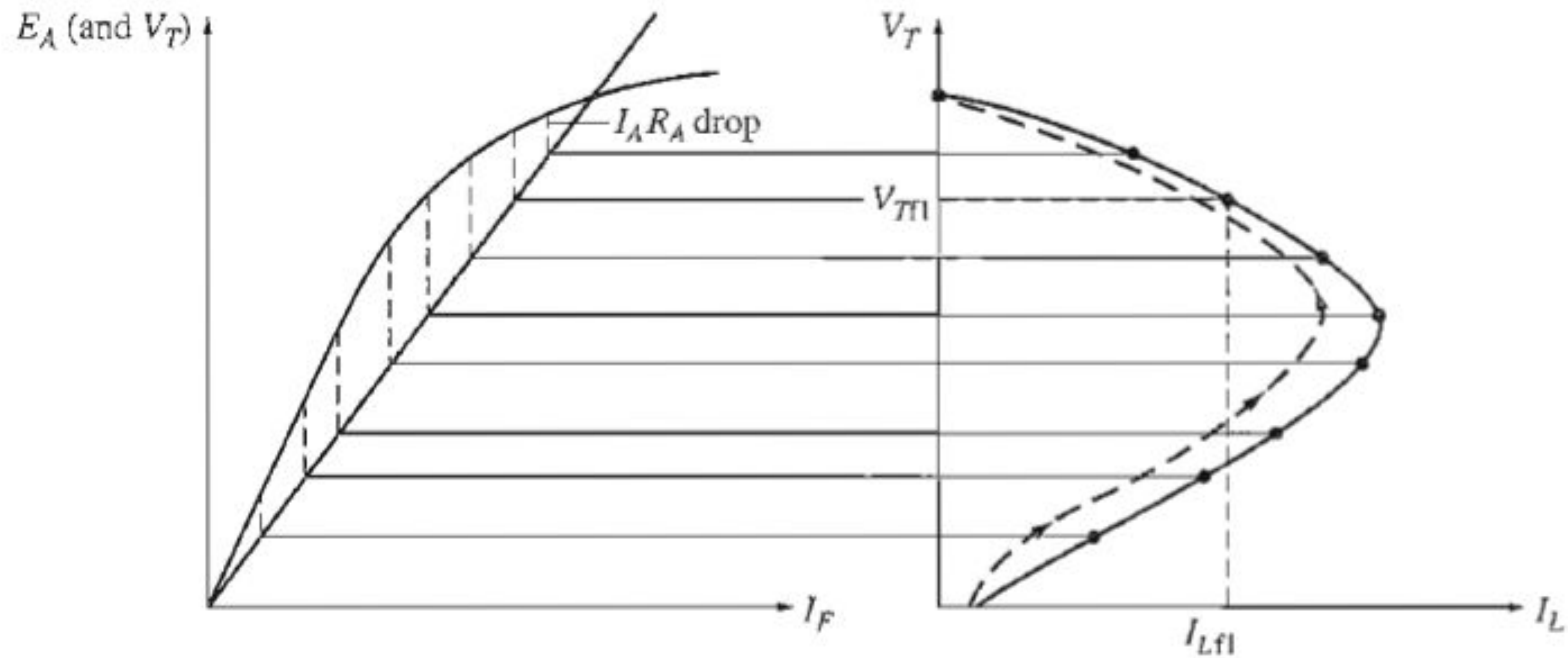
Problems

A 4-pole, long shunt, compound generator supplies 100 A at a terminal voltage of 500V. If armature resistance is 0.02 ohm, series field resistance is 0.04 ohm and the shunt field resistance 100 ohm, find the generated voltage. Assume drop per brush is 1 Volts and neglect armature reaction. (Ans: 508.3 V)

Home Work

A 20 kW compound generator works on full load with a terminal voltage of 250V. The armature, series and shunt winding resistances are, 0.05 ohm, 0.025 ohm, and 100 ohm, respectively. Calculate the generated voltage assuming the generator is connected as short-shunt. (Ans: 256.13 V)

Terminal Characteristic of Shunt Generator



1) For a fixed I_F , read I_A R_A (difference between OCC and Field-resistance line; find I_A.
2) Calculate I_L from $I_L=I_A-I_F$.
3) For a fixed I_F , field-resistance line with gives the terminal voltage V_T.
4) At $I_L=0$, the intersection of field-resistance line and OCC gives V_T.
5) The minimum I_L is obtained from V_{res}/R_A.

HW

The magnetization characteristic of a dc generator is given as follows:

I_F(A):	0	0.4	0.8	2	5	6	8	10	12	14
E_A(V)	20	22.5	25	82.5	200	232.5	275	300	320	335

The shunt field resistance is 24.5 Ω. What will be the no load voltage if the generator is used as a shunt generator? If the generator is used as separately excited with its field supplied from a 120V dc source, what will be its no load voltage?

An 8-pole wave wound dc shunt generator is supplying a load of 12.5Ω resistor at 250V. The voltage regulation is found to be 2.016%. If the generator is used as separately excited with its field connected to a 250V dc source, what will be its voltage regulation? Assume the armature resistance of the machine is 0.24Ω.

DC Motor

Application of DC Motor

Direct current motors are seldom used in ordinary industrial applications because all electric utilities supply AC. However, for special applications such as in steel mills, mines, cars, and electric trains, it is sometimes advantageous to use DC motors. The reason is that the torque-speed characteristics of DC motors can be varied over a wide range while retaining high efficiency.

Working Principle of dc motor

After the field coil is excited if a dc current is passed through the armature conductors from a dc source connected across the armature terminal, the armature experiences a torque and armature starts rotating. The direction of rotation can be obtained from the Fleming's left hand rule. The rotation of the armature also induces emf in the armature coils which is known as back emf (counter emf). This back emf is less than the supply voltage thus the current flow is in the reverse direction to that of generator case. The back emf serves to limit the armature current to the value just sufficient to take care of the developed mechanical power to drive the load.

Classification of dc motors

1. Separately excited dc motor
2. Shunt dc motors
3. Series dc motors
4. Compound dc motors
5. Permanent magnet dc motor

Analysis of dc motors

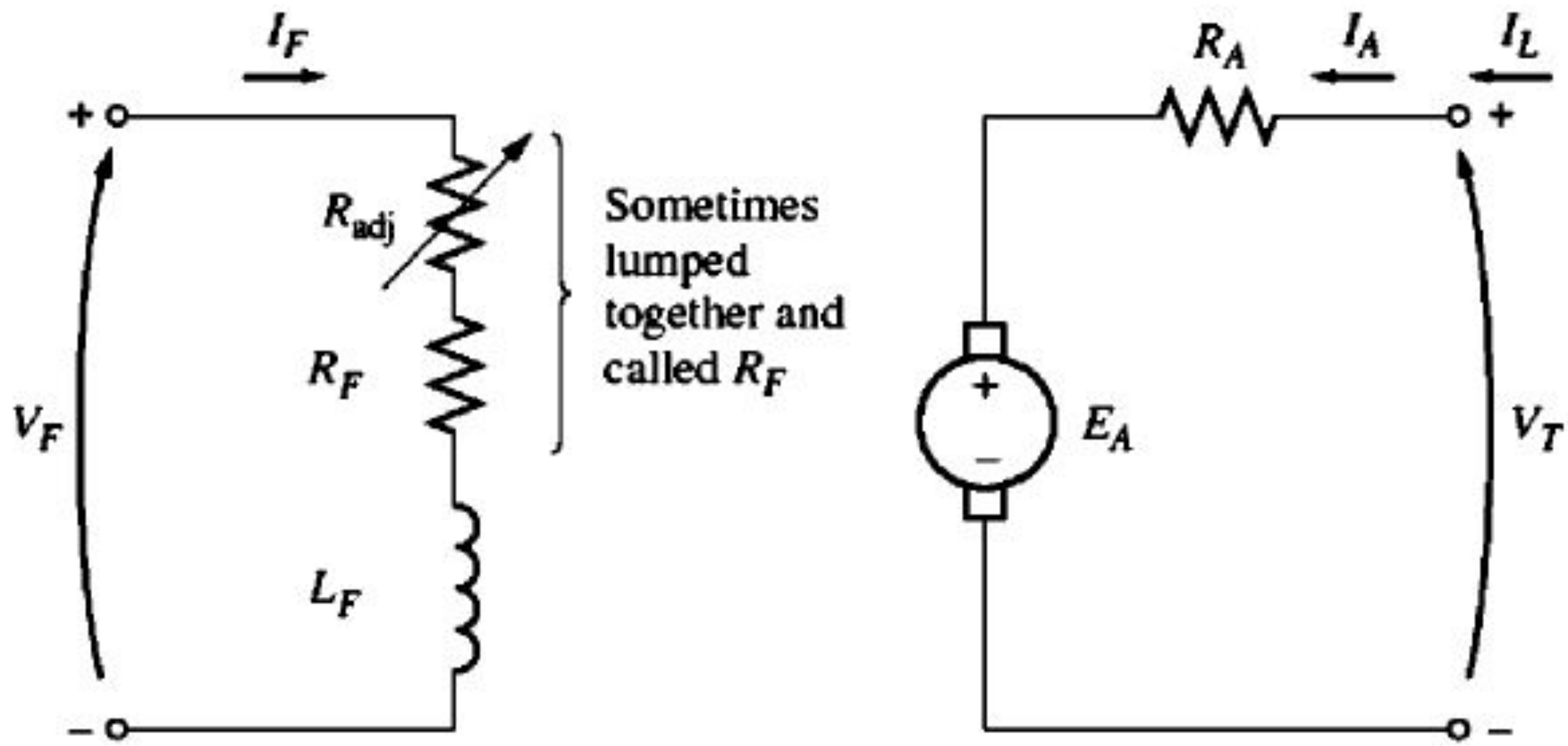
$$\boxed{\tau_{\text{ind}} = K\phi I_A} \quad \boxed{E_A = K\phi\omega}$$

E_A in the case of motor is known as back emf or counter emf.

The above two equations, KVL and magnetization curve are the basis of dc motors analyses.

For a motor, the output quantities are shaft torque and speed, the terminal characteristic of a motor is thus a plot of its output torque versus speed.

Separately excited motors

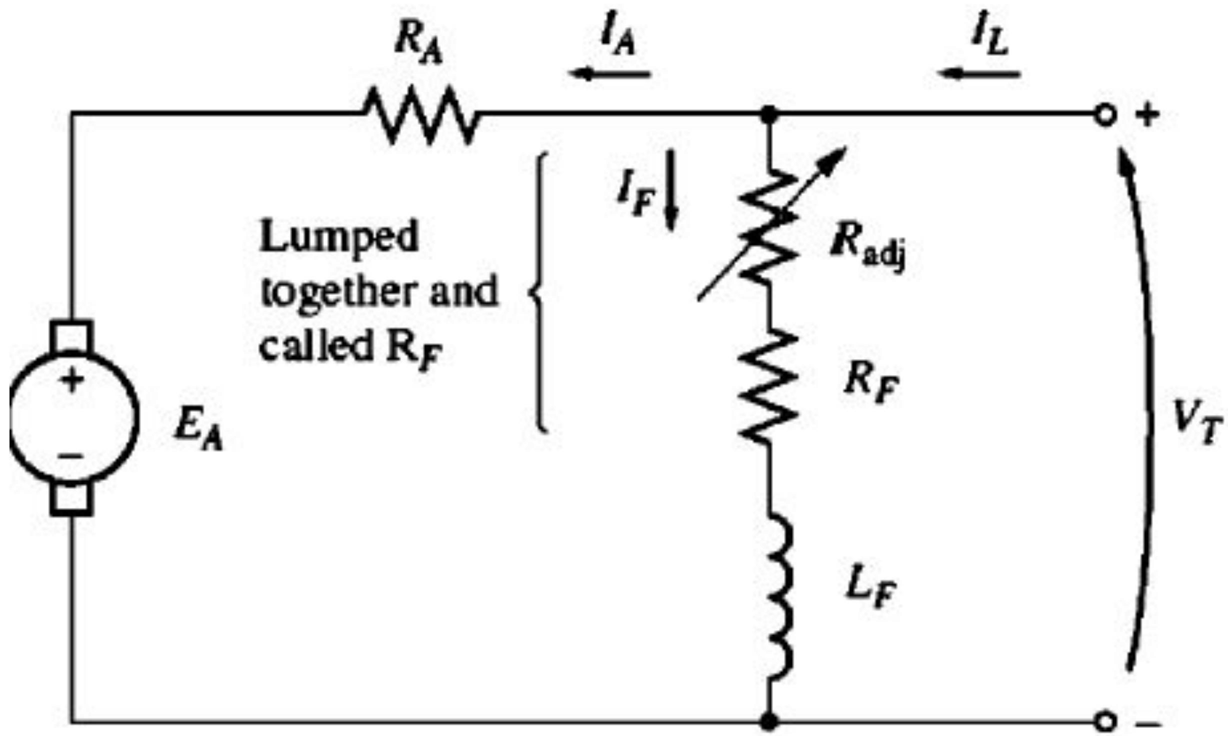


$$I_F = \frac{V_F}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A$$

Shunt motors



$$I_F = \frac{V_T}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A + I_F$$

Loading effect on Shunt motors

In motors, there are two torques (load torque and induced torque). At running condition, these two torques are equal except the losses.

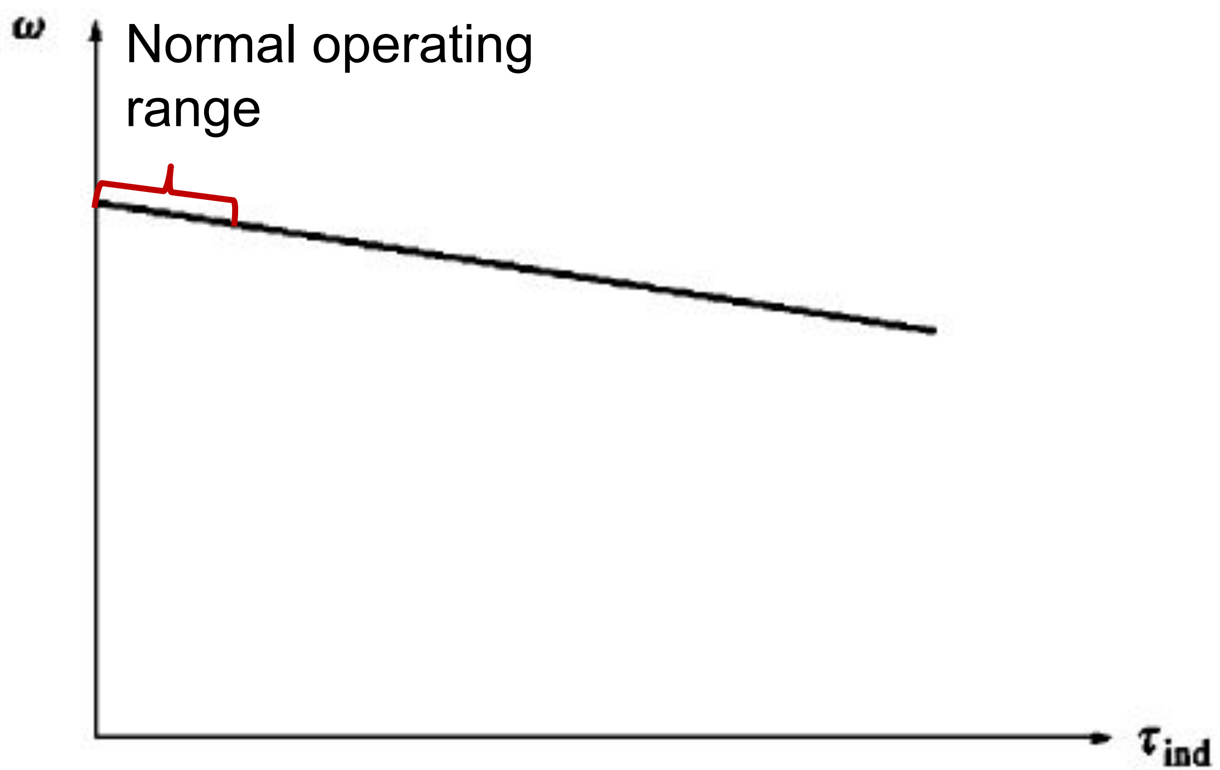
$$\tau_{load} = \tau_{ind} - losses \approx \tau_{ind}$$

If load torque exceed the induced torque, motor will start to slow down. Thus $E_A = K\varphi\omega$ decreases, so $I_A = (V_T - E_A)/R_A$ increases. The increase in I_A will eventually increase the induced torque. Finally induced torque will balance the load torque at a lower motor speed.

If induced torque exceeds the load torque, motor will start to accelerate.

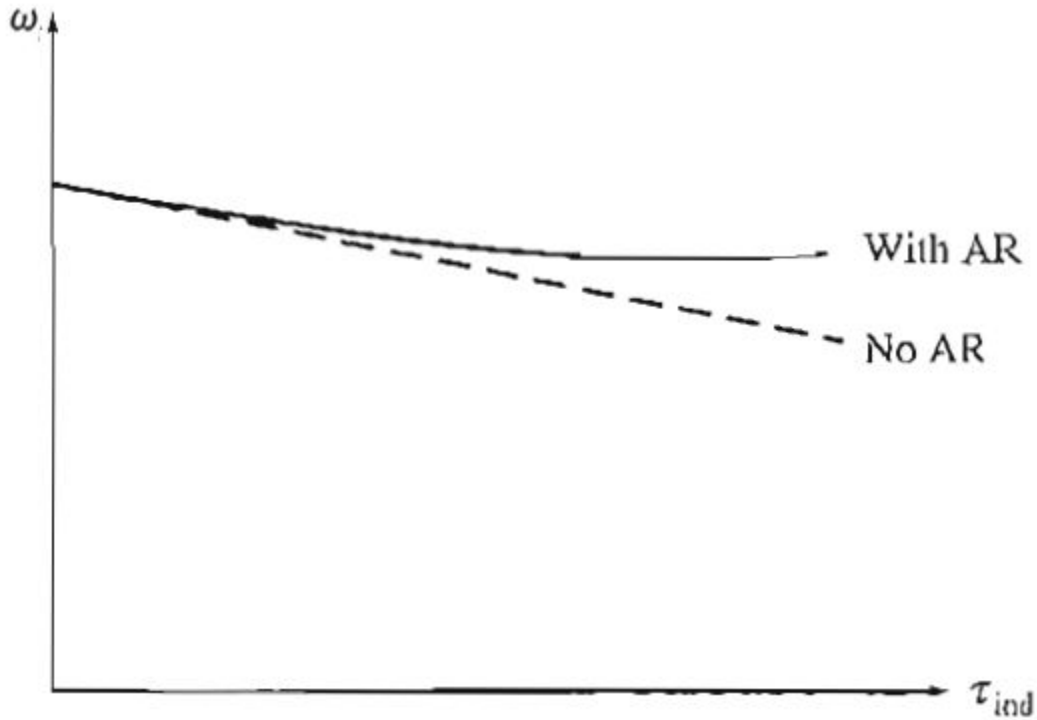
Terminal Characteristics of Shunt motors (1)

$$V_T = E_A + I_A R_A = K\phi\omega + \frac{\tau_{ind}}{K\phi} R_A \quad \Rightarrow \quad \omega = \frac{V_T}{K\phi} - \frac{\tau_{ind}}{(K\phi)^2} R_A$$



Terminal Characteristics of Shunt motors (2)

$$V_T = E_A + I_A R_A = K\phi\omega + \frac{\tau_{ind}}{K\phi} R_A \quad \Rightarrow \quad \omega = \frac{V_T}{K\phi} - \frac{\tau_{ind}}{(K\phi)^2} R_A$$

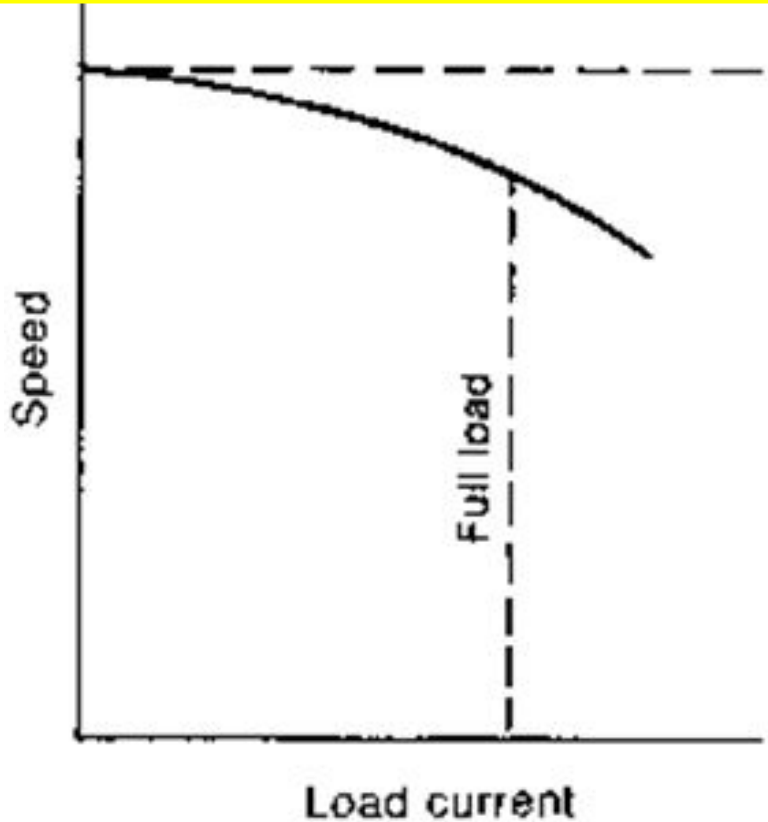


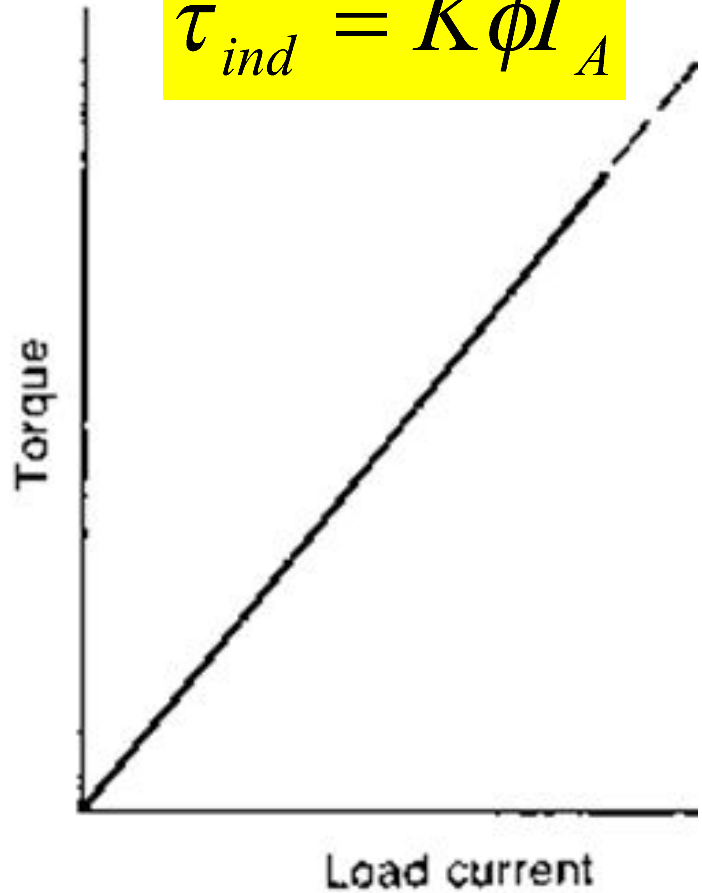
Terminal Characteristics of Shunt motors (3)

$$V_T = E_A + I_A R_A = K\phi\omega + I_A R_A$$

$$\Rightarrow \ \omega = \frac{V_T - I_A R_A}{K\phi}$$

$$\tau_{ind} = K\phi I_A$$





Speed control of Shunt motors (1)

<u>Most common methods are</u>:

1) Field resistance/field flux control

2) Terminal voltage control

<u>Less common method</u>:

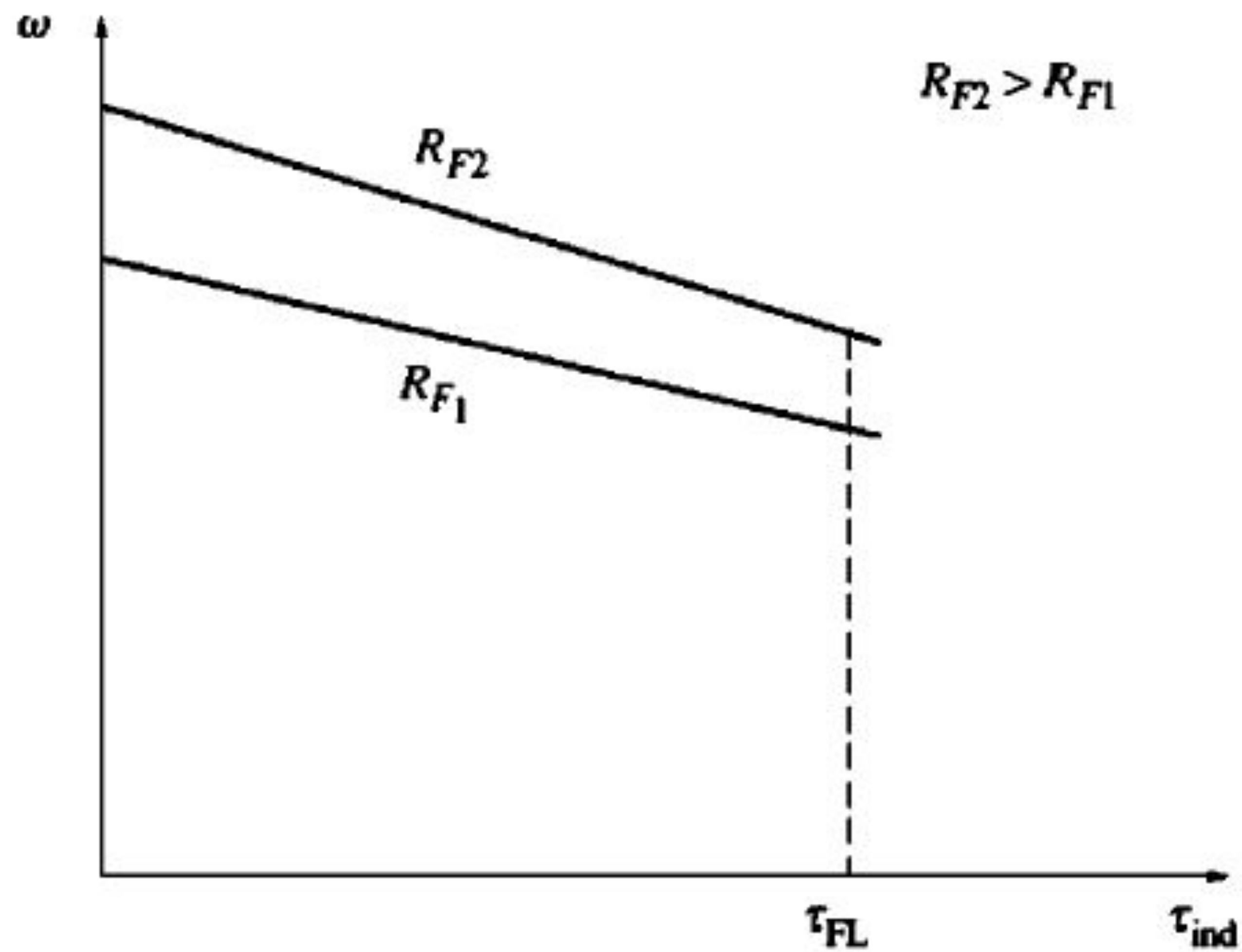
Armature resistance control

Speed control of Shunt motors (2)

Field resistance/field flux control:

1) Increasing R_F decreases I_F.
2) Decreasing I_F decreases φ.
3) Decreases φ lowers E_A.
4) Decreasing E_A increases I_A.
5) Increasing I_A increases induced torque.
6) Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$ and the speed ω increases.
7) Increasing ω increases E_A.
8) Increasing E_A decreases I_A.
9) Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher ω.

Speed control of Shunt motors (3)



Under normal operating range, the increase in field resistance increases the motor speed.

A shunt motor is running at 626 rpm when taking an armature current of 50A from a 440V supply. The armature circuit has a resistance of 0.28Ω. If the flux is suddenly reduced by 5%, find:

(a) The maximum value to which the current increases momentarily and the ratio of the corresponding torque to the initial torque;

(b) The ultimate steady value of the armature current, assuming the torque due to the load to remain unaltered.

(c) The speed at this new condition.

For the initial condition, $E_1 = V_T - I_{A1} R_A = 440 - 50 \times 0.28 = 426$.

(a) $\frac{E_2}{426} = \frac{K \times 0.95 \phi n}{K \phi n} \Rightarrow E_2 = 0.95 \times 426 = 404.7V$

$\therefore I_{A2} R_A = 440 - 404.7 = 35.3 \quad \Rightarrow I_{A2} = 35.3/0.28 = 126A$

$\frac{T_2}{T_1} = \frac{K \times 0.95 \phi \times 126}{K \phi \times 50} = 2.394$

(b) Ultimately at the steady condition, induced torque becomes equal to the load torque T_1 with 0.95ϕ at a new speed. Where, $T_1 = K\phi \times 50$. Therefore,

$K \times 0.95\phi \times I_A = K\phi \times 50 \Rightarrow I_A = 50/0.95 = 52.6A$

(c) Under steady state condition with 0.95ϕ, E is calculated as: $E = V_T - I_A R_A = 440 - 52.6 \times 0.28 = 425.272$

$\frac{K \times 0.95 \phi n}{K \phi \times 626} = \frac{425.272}{426} \Rightarrow n = 657.82$ rpm

Effect of Armature Reaction on Motor

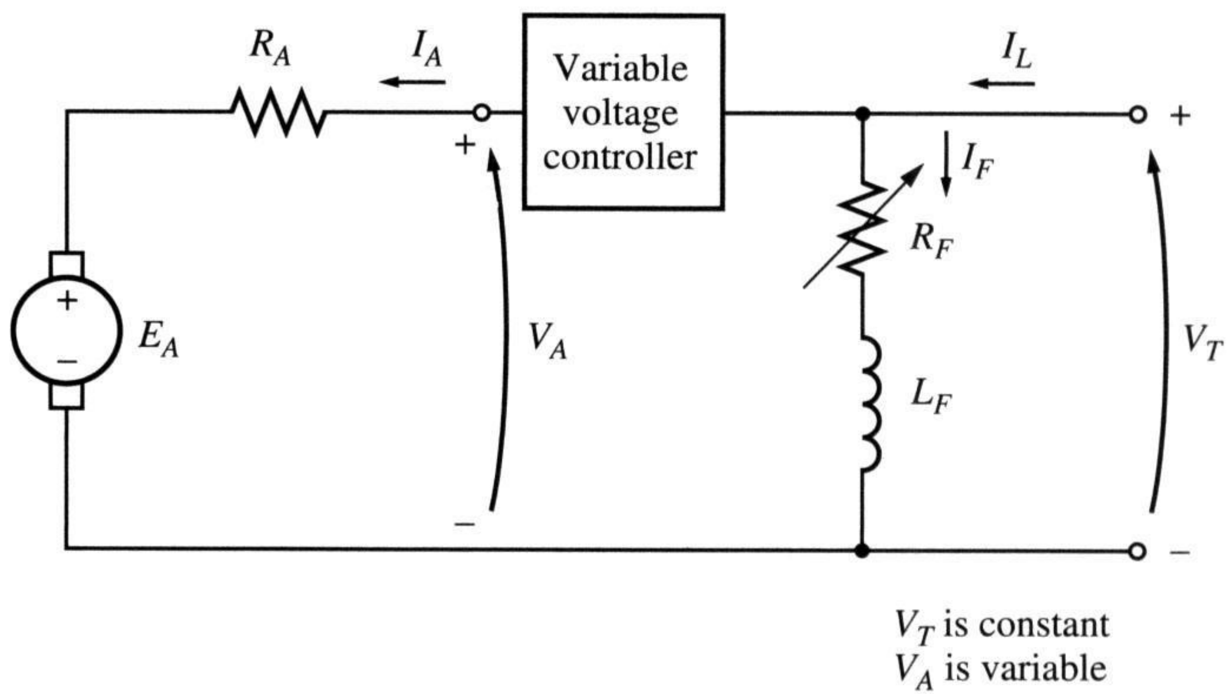
The armature reaction weakens the flux cause the motor’s speed to rise. Over loading a motor without compensating winding may cause runaway.

Shunt motor: Speed control

Armature terminal voltage control:

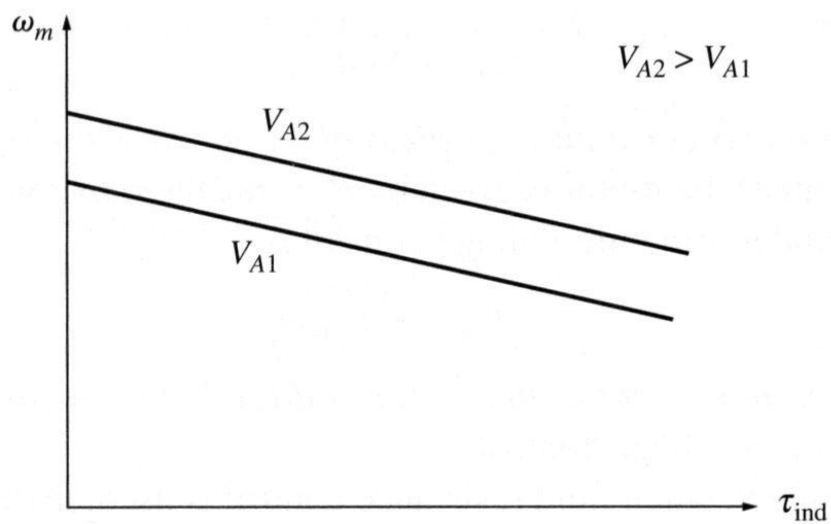
This method implies changing the voltage applied to the armature of the motor without changing the voltage applied to its field. Therefore, the motor must be separately excited to use armature voltage control.

Armature voltage speed control



Shunt motor: Speed control

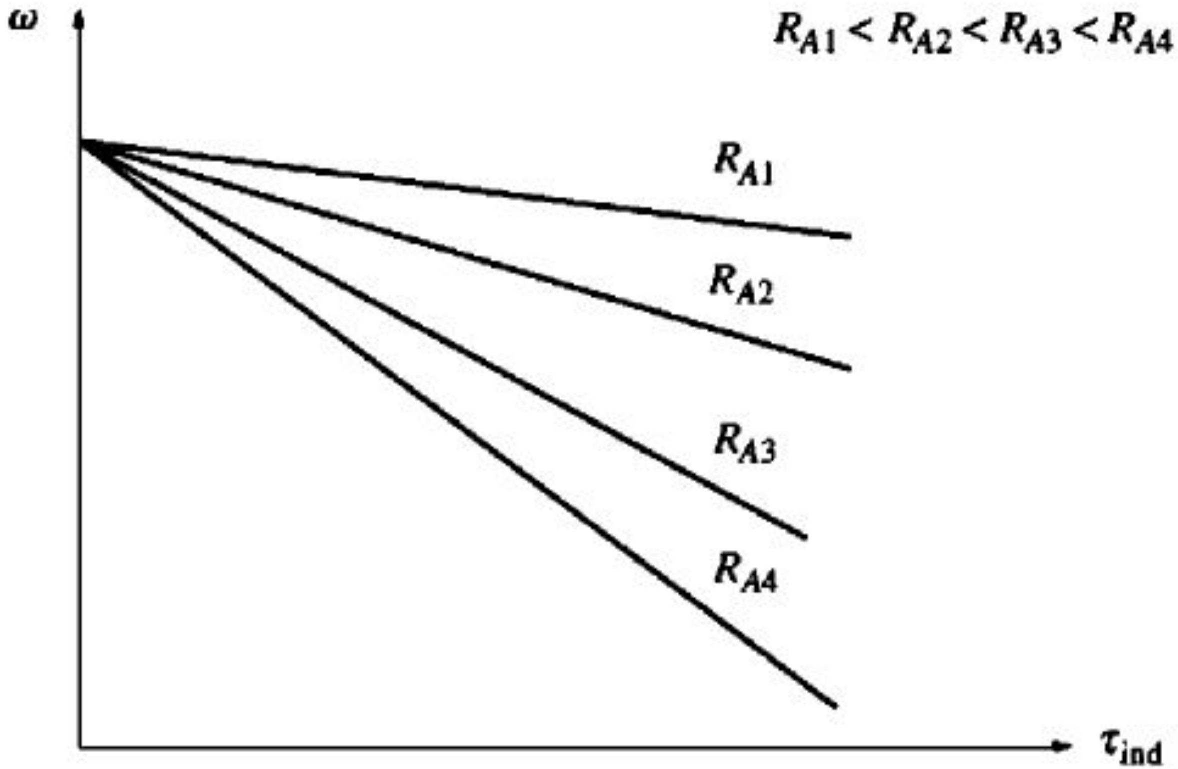
1) Increasing the armature voltage V_A increases the armature current ($I_A = (V_A - E_A)/R_A$);
2) Increasing armature current I_A increases the induced torque τ_{ind} ($\tau_{ind} = K\varphi I_A$);
3) Increased induced torque τ_{ind} is now larger than the load torque τ_{load} and, therefore, the speed ω *increases*;
4) Increasing speed increases the internal generated voltage ($E_A = K\varphi\omega$);
5) Increasing E_A decreases the armature current I_A…
6) Decreasing I_A decreases the induced torque until $\tau_{ind} = \tau_{load}$ at a higher speed ω.



Shunt motor: Speed control

Armature resistance control:

$$V_T = E_A + I_A R_A = K\phi\omega + \frac{\tau_{ind}}{K\phi} R_A \quad \Rightarrow \quad \omega = \frac{V_T}{K\phi} - \frac{\tau_{ind}}{(K\phi)^2} R_A$$



Base speed

If a motor is operated at its rated terminal voltage, power, and field current, it will be running at the rated speed which is also called a base speed.

Shunt motor: Speed control

Field resistance control is mainly used for speeds above the base speed but not below it. Trying to achieve speeds slower than the base speed by the field circuit control, requires large field currents that may damage the field winding.

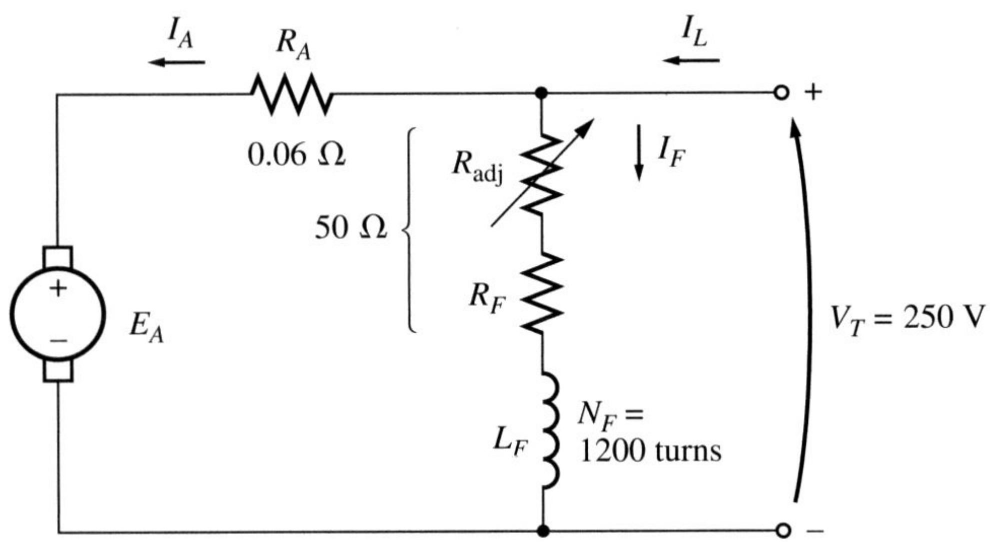
Since the armature terminal voltage is limited to its rated value, no speeds exceeding the base speed can be achieved safely while using this method. It is thus used for speeds below the base speed.

As the insertion of resistor in armature circuit increases the power loss, armature resistance control is very wasteful method of speed control; thus it is rarely used.

Shunt motor: Example

Example 5.1: A 50 hp, 250 V, 1200 rpm DC shunt motor **with** compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50 Ω, which produces a no-load speed of 1200 rpm. The shunt field winding has 1200 turns per pole.

a) Find the motor speed when its input current is 100 A.
b) Find the motor speed when its input current is 200 A.
c) Find the motor speed when its input current is 300 A.
d) Plot the motor torque-speed characteristic.



Shunt motor: Example

The internal generated voltage of a DC machine (with its speed expressed in rad/sec):

$$E_A = K\phi\omega$$

Since the field current is constant (both field resistance and V_T are constant) and since there are no armature reaction (due to compensating windings), we conclude that the flux in the motor is constant. The speed and the internal generated voltages at different loads are related as

$$\frac{E_{A2}}{E_{A1}} = \frac{K\phi\omega_2}{K\phi\omega_1} = \frac{n_2}{n_1}$$

Therefore:

$$n_2 = \frac{E_{A2}}{E_{A1}} n_1$$

At no load, the armature current is zero and therefore $E_{A1} = V_T = 250$ V.

Shunt motor: Example

a) Since the input current is 100 A, the armature current is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F} = 100 - \frac{250}{50} = 95\ A$$

Therefore: $$E_A = V_T - I_A R_A = 250 - 95 \cdot 0.06 = 244.3\,V$$

and the resulting motor speed is:

$$n_2 = \frac{E_{A2}}{E_{A1}} n_1 = \frac{244.3}{250} 1200 = 1173\ rpm$$

b) Similar computations for the input current of 200 A lead to n_2 = 1144 rpm.
c) Similar computations for the input current of 300 A lead to n_2 = 1115 rpm.
d) To plot the output characteristic of the motor, we need to find the torque corresponding to each speed. At no load, the torque is zero.

Shunt motor: Example

Since the induced torque at any load is related to the power converted in a DC motor:

$$P_{conv} = E_A I_A = \tau_{ind}\omega$$

the induced torque is

$$\tau_{ind} = \frac{E_A I_A}{\omega}$$

For the input current of 100 A:

$$\tau_{ind} = \frac{2443 \cdot 95}{2\pi \cdot 1173/60} = 190\ N\text{-}m$$

For the input current of 200 A:

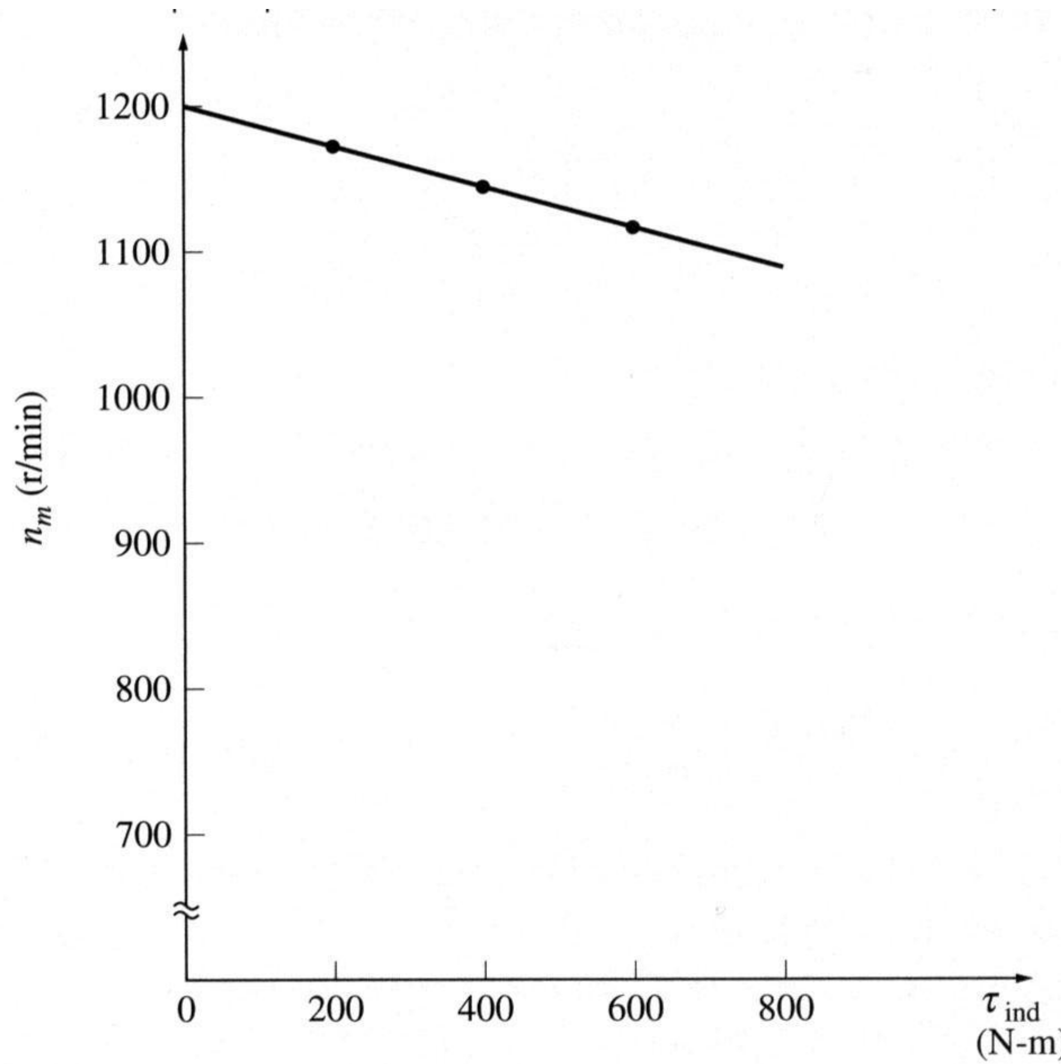
$$\tau_{ind} = \frac{2383 \cdot 195}{2\pi \cdot 1144/60} = 388\ N\text{-}m$$

For the input current of 300 A:

$$\tau_{ind} = \frac{2323 \cdot 295}{2\pi \cdot 1115/60} = 587\ N\text{-}m$$

Shunt motor: Example

The torque-speed characteristic of the motor is:

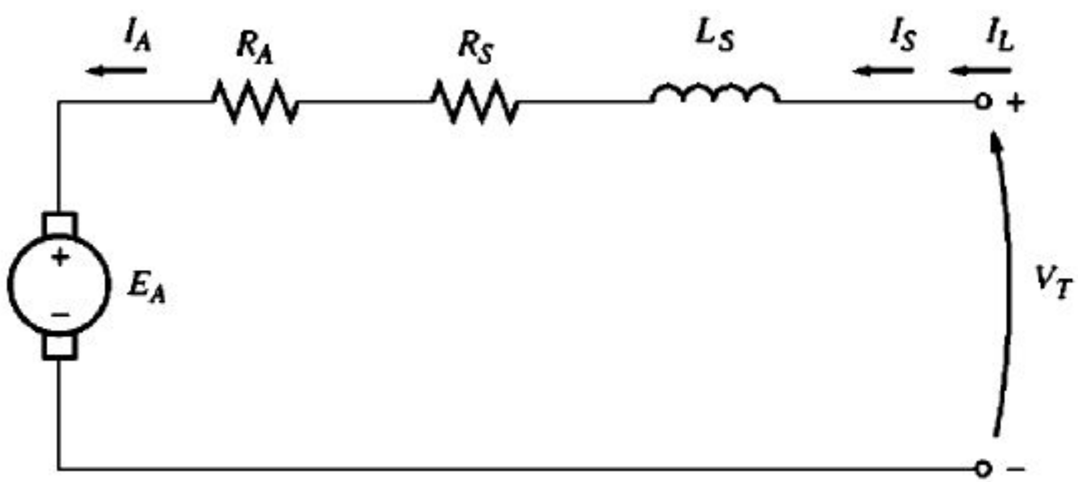


Shunt motor: The effect of an open field circuit

Machine flux drops to residual value, E_A would drop with it. This will increase armature current, induced torque will also increase and becomes higher than that the shaft/load torque. Therefore motor speed starts to rise and just keeps going up.

Series motors: terminal characteristic

$$I_A = I_S = I_L \qquad \therefore V_L = E_A + I_A(R_A + R_s)$$

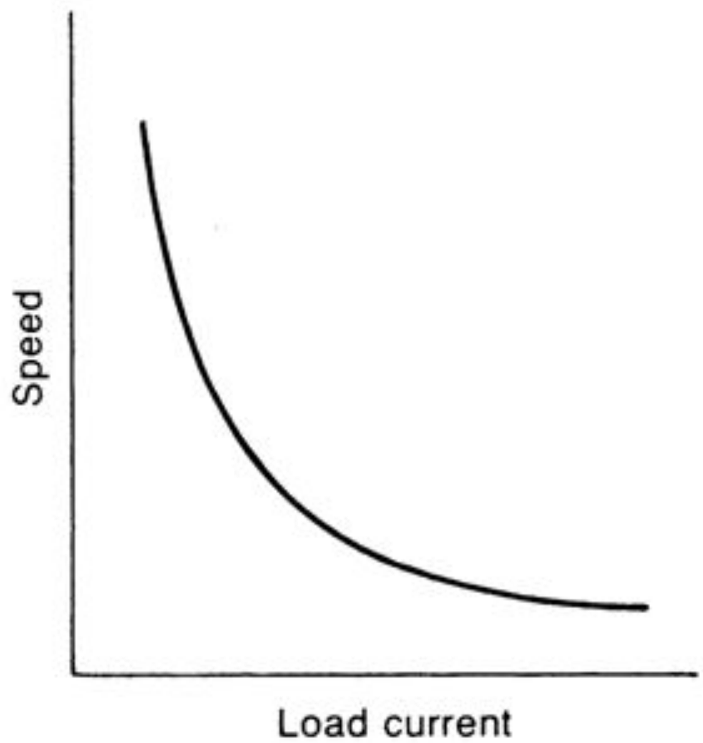


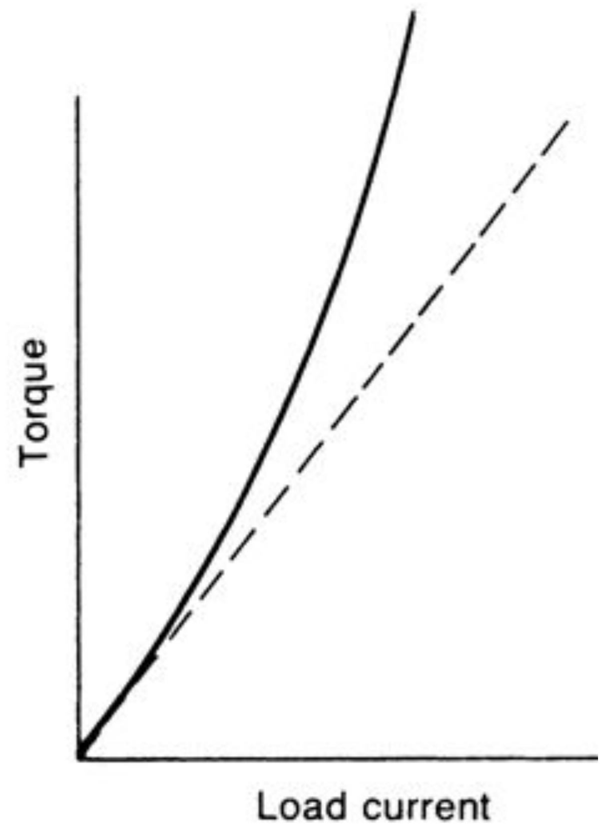
Series motors: terminal characteristic

$$\omega = \frac{V_T - I_A(R_A + R_s)}{K\phi} = \frac{V_T - I_A(R_A + R_s)}{KcI_A}$$

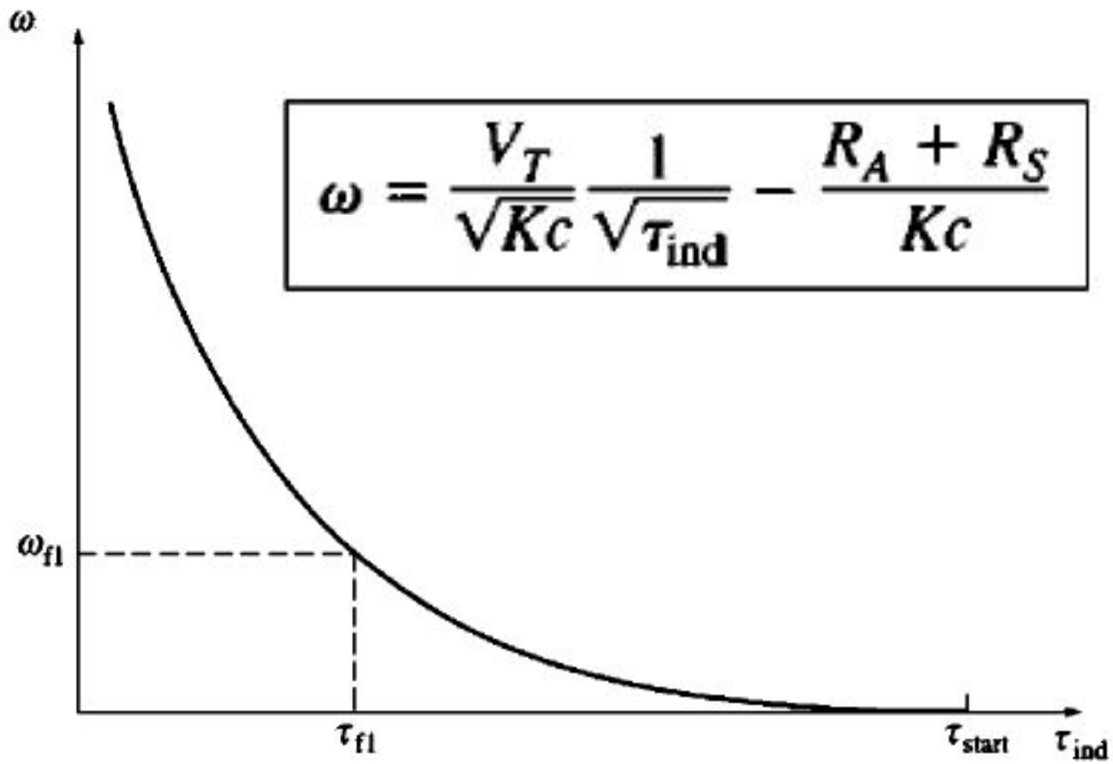
$$\phi = cI_A$$

$$\tau_{ind} = K\phi I_A = KcI_A^2$$





Series motors: terminal characteristic



In absence of sufficient load, the speed of the series motor goes fast; this motor starts always with load.

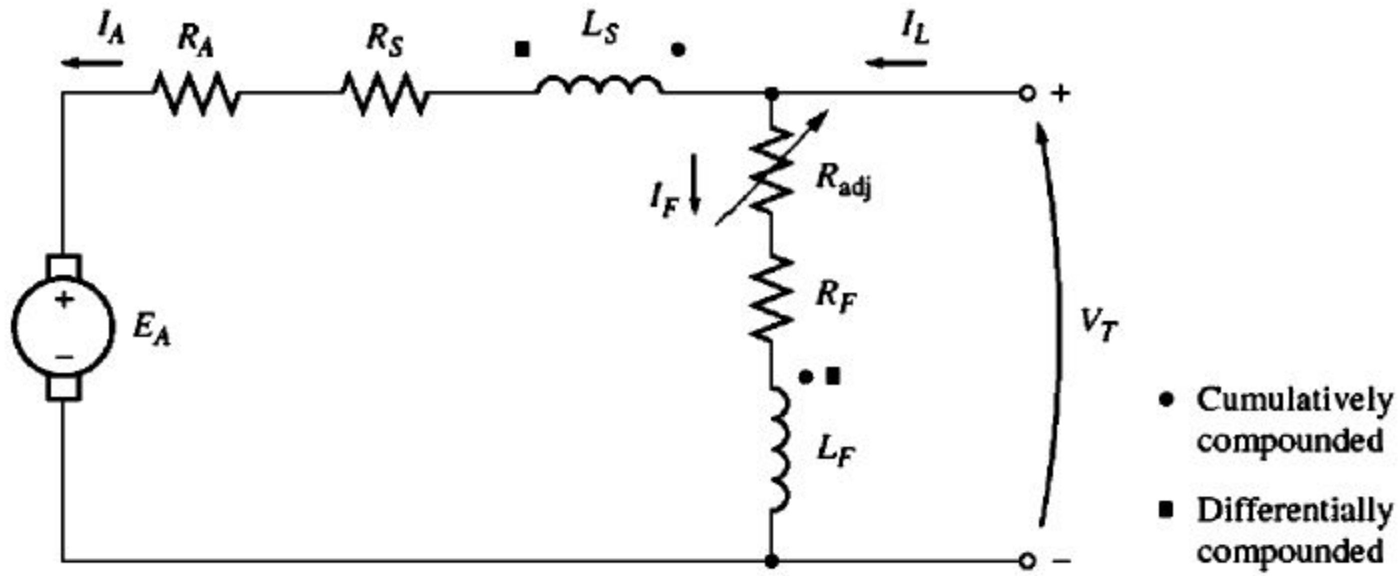
Speed control of Series motors (1)

$$\boxed{\omega = \frac{V_T}{\sqrt{Kc}}\frac{1}{\sqrt{\tau_{\text{ind}}}} - \frac{R_A + R_S}{Kc}}$$

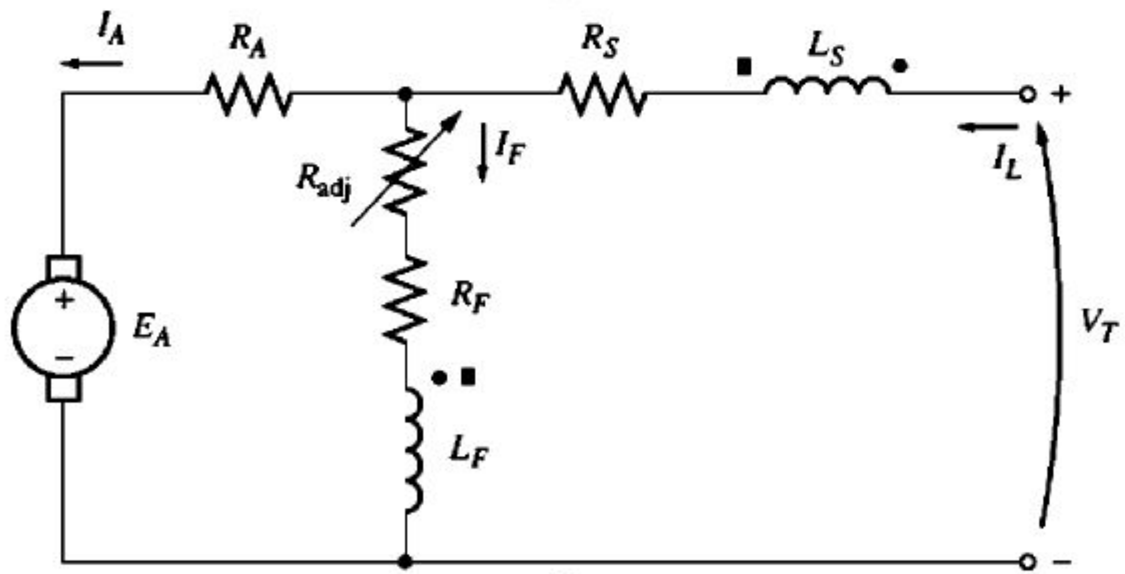
<u>Methods are</u>:

1) Terminal voltage control
2) Armature resistance control (in efficient)

Compound motors (1)



(a)



(b)

Compound motors (2)

$$I_F = \frac{V_T}{R_F}, \; I_L = I_A + I_F$$

$$\therefore V_T = E_A + I_A(R_A + R_s)$$ Long Shunt

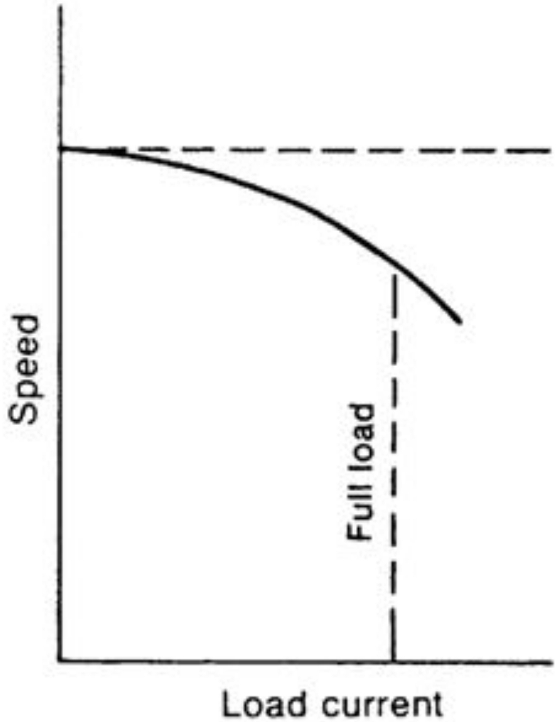
$$or, \quad V_T = E_A + I_A R_A + I_L R_s$$ Short Shunt

$$\omega_{cc} = \frac{V_T}{K\phi_{cc}} - \frac{\tau_{ind}}{(K\phi_{cc})^2}(R_A + R_S)$$

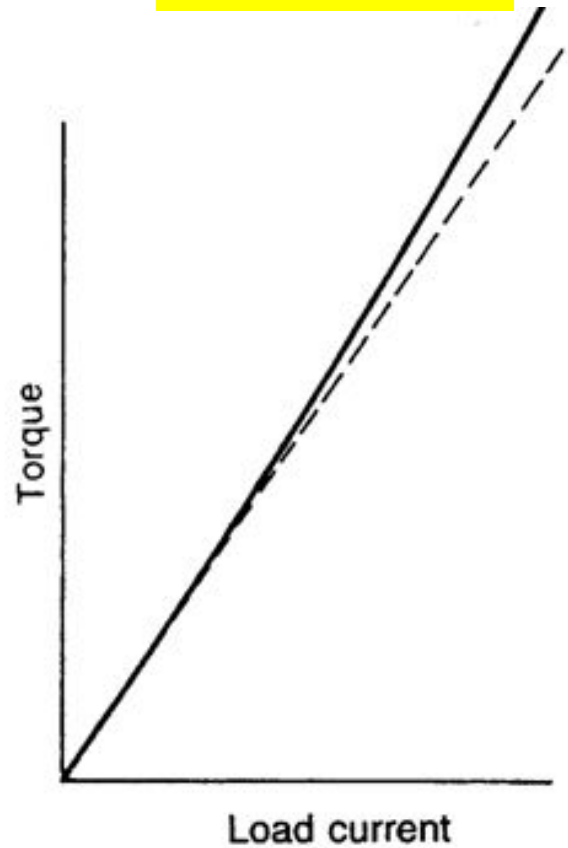
$$\phi_{cc} = \phi_F + \phi_S$$

Cumulative Compound motors (3)

$$\omega = K\frac{V_T - I_A(R_A + R_s)}{\phi}$$



$$\tau_{ind} = K_T \phi I_A$$



Speed drops more than shunt and torque increases more than shunt.

Cumulative Compound motors (4)

At light loads, series field has a very small effect, so the motor behaves as a shunt motor; as the load gets very large, the series flux becomes significant so that its torque-speed curve begins to follow to that of a series motor.

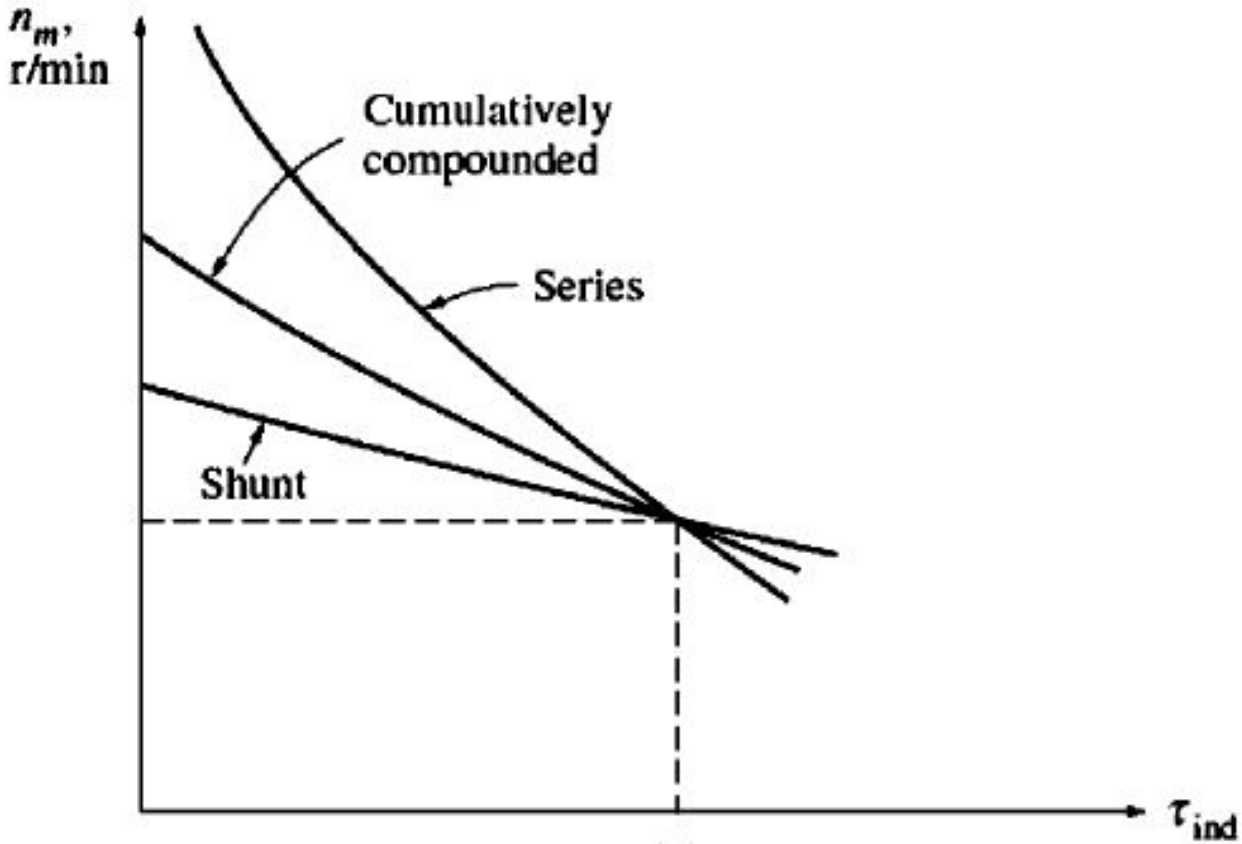
Cumulative compound motor has a higher starting torque than a shunt motor (constant flux) but a lower starting torque than a series motor (flux is proportional to load current).

It combines the best feature of both the shunt and the series motor.

Like series motor, it has extra torque for starting.

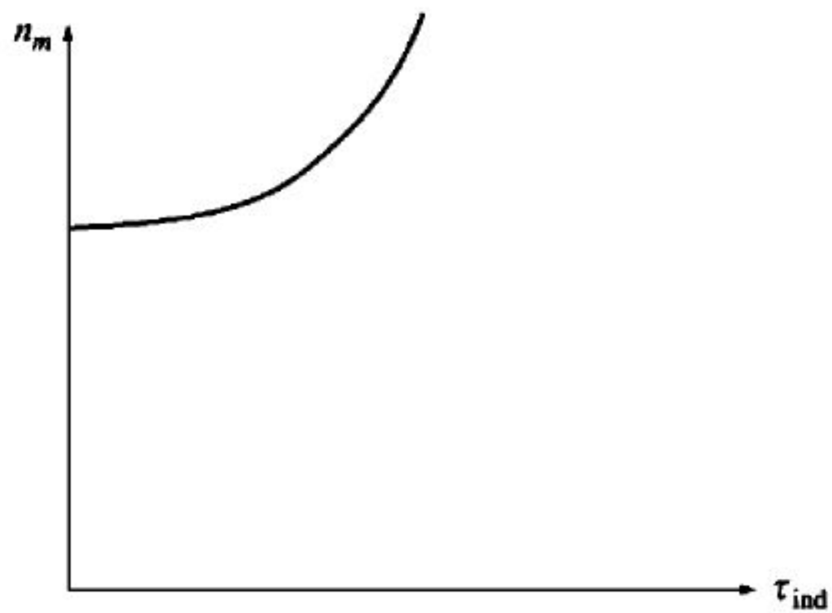
Like shunt motor, it does not over speed at no load.

Cumulative Compound motors (5)



Differential Compound motors

As load increases, I_A increases which decreases flux.
As the flux decreases, speed of the motor increases.
Increased speed which further increases I_A, further decreasing flux.
Increased I_A increases the speed again.
Differentially compound motor becomes unstable and tends to run away.



Speed control of Compound motors

1) Field resistance/field flux control

2) Terminal voltage control

3) Armature resistance control

Speed Regulation of a DC Motor

The speed regulation of a dc motor is defined as the change in speed when the load on the motor is reduced from rated value to zero, expressed as percentage of the rated load speed.

$$\text{Speed Regulation} = \frac{NLspeed - FLspeed}{FLspeed} \times 100\%$$

Mechanical Power Developed in a motor

Out of armature input ($V_T I_A$) some is wasted in armature resistance loss ($I_A{}^2 R_a$) and the rest is co

$$E_A I_A = V_T I_A - I_A^2 R_A = P_m$$

power which is known as developed mechanical power.

This po

Condition for Maximum power:

$$\frac{dP_m}{dI_A} = \frac{d}{dI_A}\left(V_T I_A - I_A^2 R_A\right) = V_T - 2 I_A R_A = 0$$

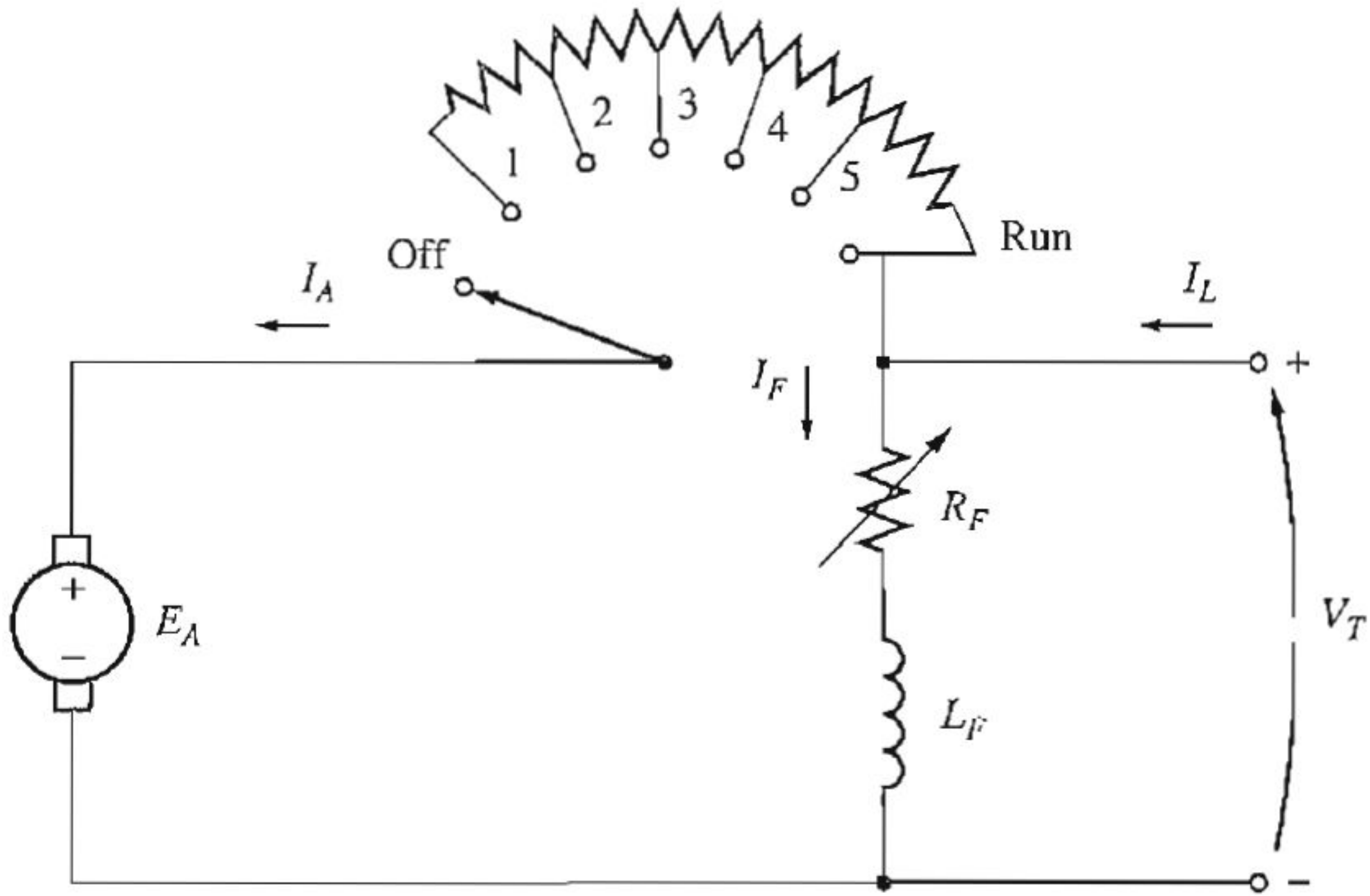
$$I_A R_A = \frac{V_T}{2} \quad \Rightarrow \text{ From } V_T = E_A + I_A R_A$$

$$\text{we have } E_A = \frac{V_T}{2}$$

Starting of dc motor (1)

At the instant of start up, there is no back emf. A high current is thus flows in the armature that may blow the fuse and disconnect the motor from supply. It is therefore necessary to insert some resistance in series with the armature circuit to limit the current flow through the armature winding at the starting time.

As the motor starts to rotate, this resistance is to be taken out in steps because the back emf rise as the motor come up to full speed. This resistance arrangement is called starter.



1
2
3
4
5
Off
Run
I_A
I_L
I_F
+
−
R_F
L_F
E_A
V_T

A starter is required for a 220V shunt motor. The maximum allowable current is 55A and the minimum current is about 35A. Find the number of sections of starter resistance required and the resistance of each section. The armature resistance of the motor is 0.4Ω.

$$n = \frac{\log (R_A / R_{tot})}{\log (I_{min} / I_{max})}$$

$R_{total} = R_1 = \frac{220}{55} = 4\ \Omega$ n=5.1≡6

$R_2 * 55 = R_{total} * 35 \Rightarrow R_2 = (4*35)/55 = 2.55$

Resistance between position 1 and 2, $R_{12} = 4 - 2.55 = 1.45$. $R_3 = R_2 * 35/55 = 1.623$. $R_{23} = R_2 - 1.623 = 0.93$

$$R_{23}=(R_2= 2.55)-(R_3=1.623)=0.93$$

$$R_4= R_3*35/55=1.033.$$

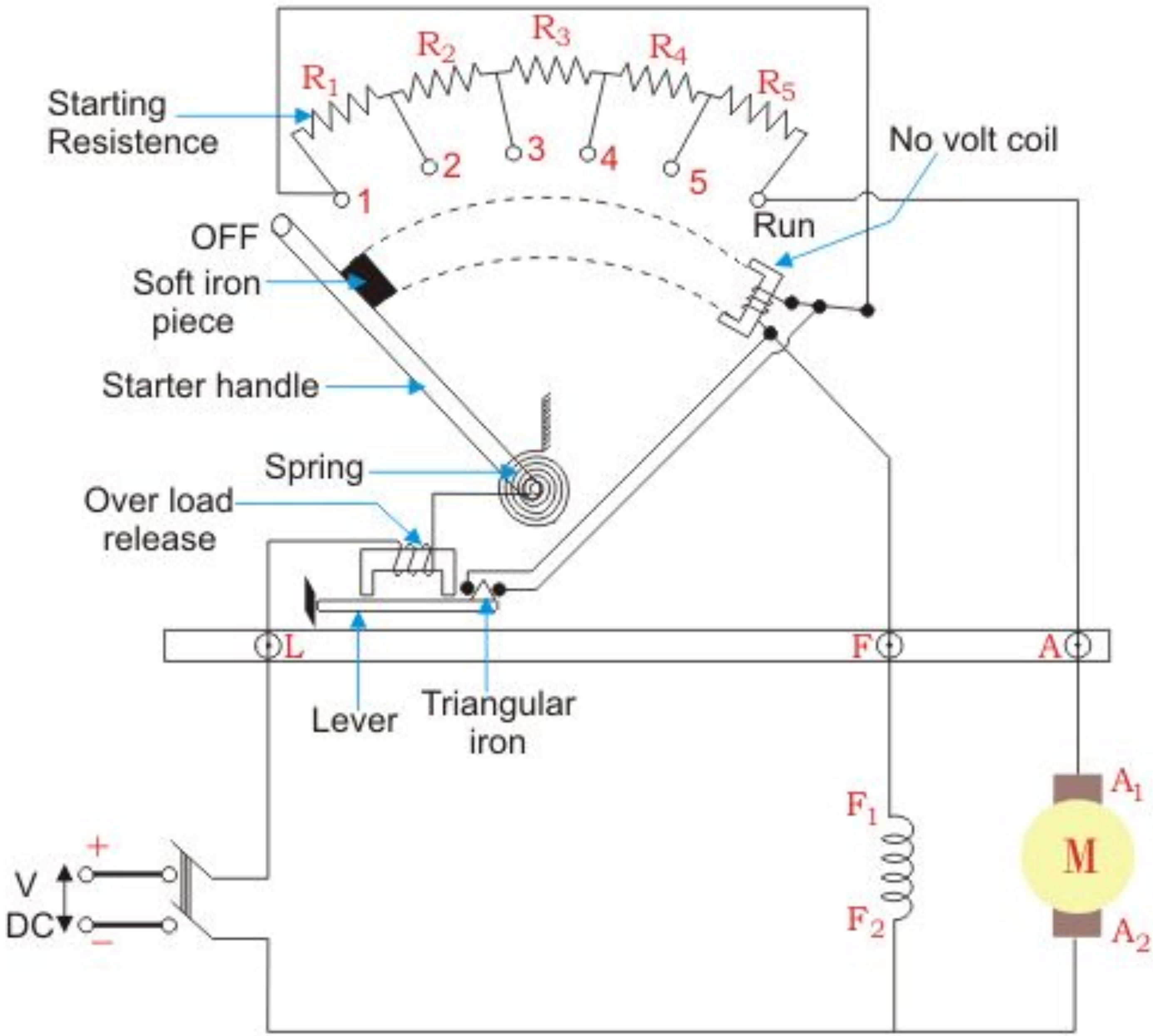
$$R_{34}=1.623-1.033=0.59$$

$$R_5= R_4*35/55=0.6574.$$

$$R_{45}=1.033-0.6574=0.3756$$

$$R_6= R_5*35/55=0.418.$$

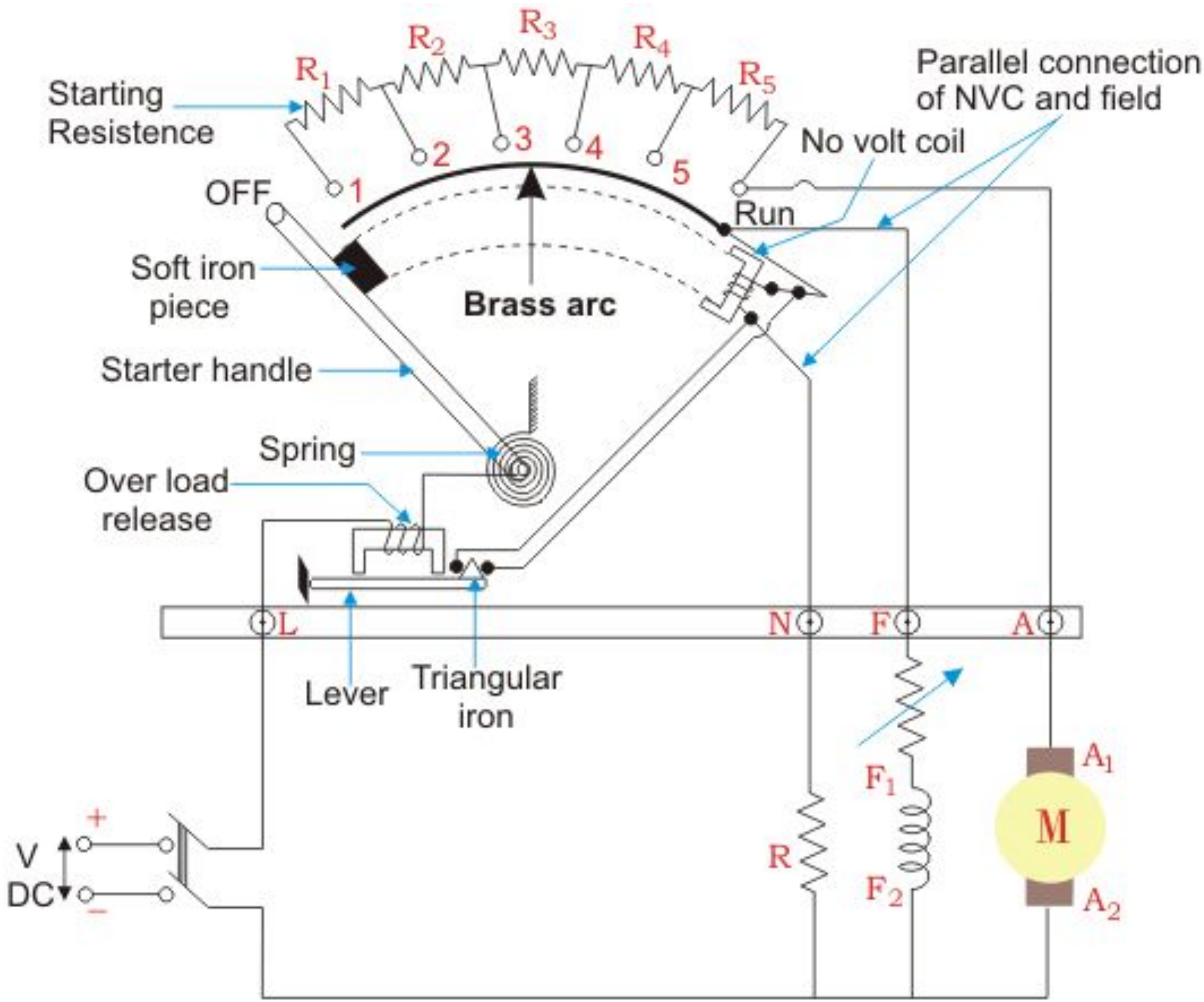
$$R_{56}=0.6574-0.418=0.2394$$



Three Point Starter

Starting of dc motor (2)

Three point starter has one disadvantage. If the machine is run at higher speed by field weakening, the strength of magnet may become so weak that it will fail to hold the handle in the RUN position. Thus we find a false disconnection of the motor takes place even there is neither over load nor any sudden disruption of supply.



Four Point Starter

Braking of dc motor

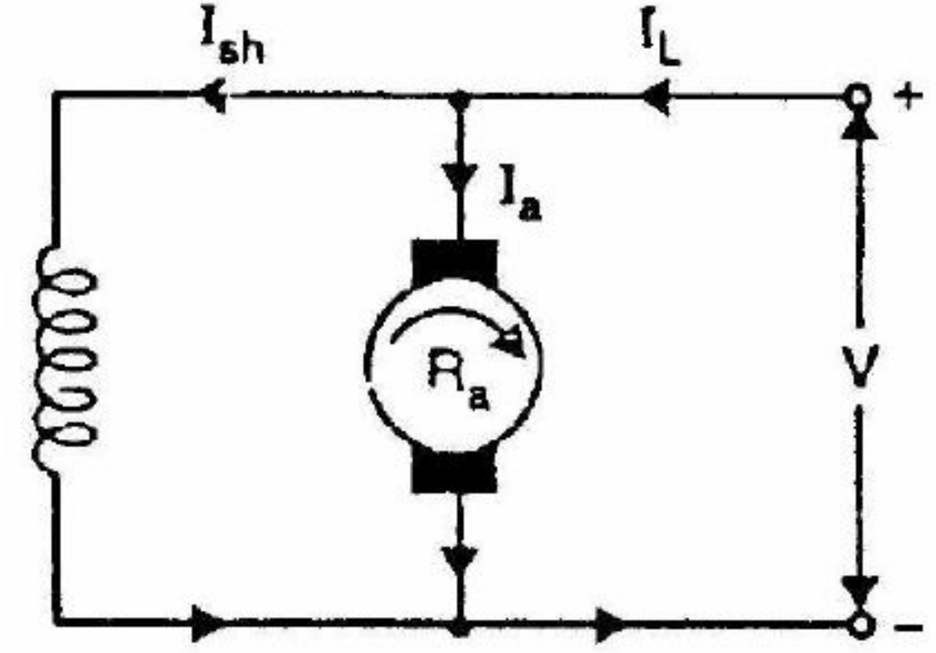
During braking of a motor, the machine is forced to operate as generator as that a torque opposite to the direction of rotation will be imposed on the shaft, helping the machine to come to stop quickly.

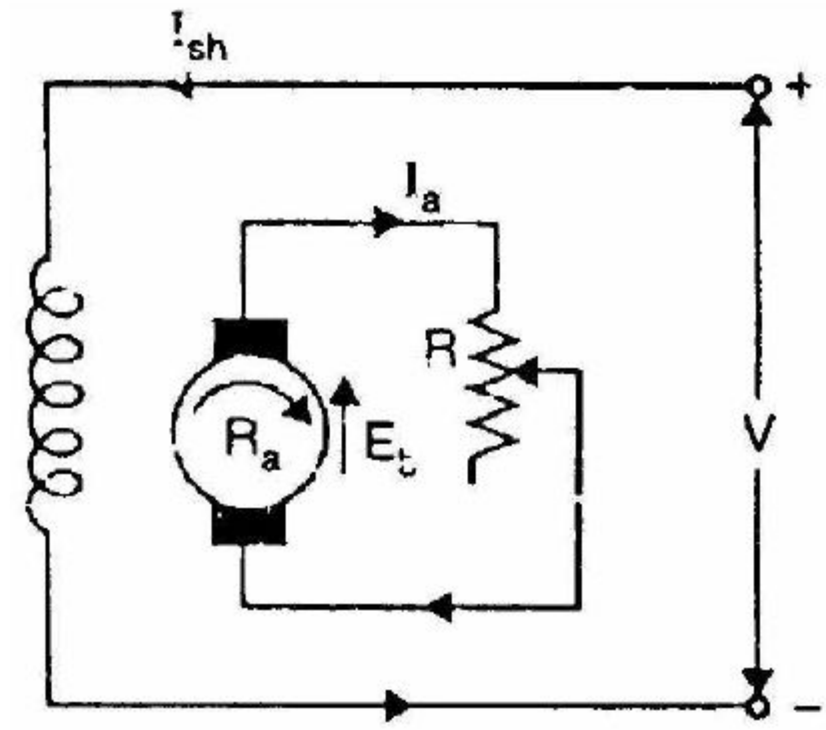
(i) *plugging* or *counter-current,* (ii) *dynamic* or *rheostatic, and* (iii) *regenerative.* These will be discussed here briefly. The dynamics of the braking problem is discussed in Ref. [9].

- Rheostatic or dynamic braking braking
- Plugging
- Regenerative braking

Braking of dc motor

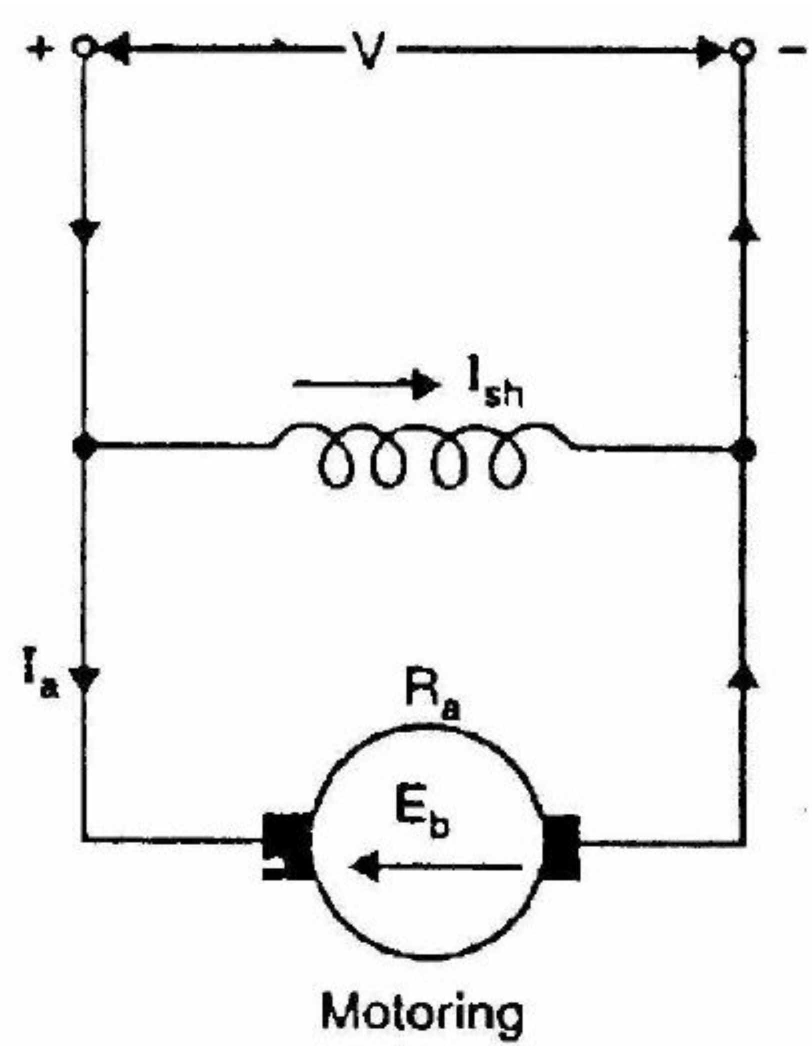
Rheostatic or Dynamic braking

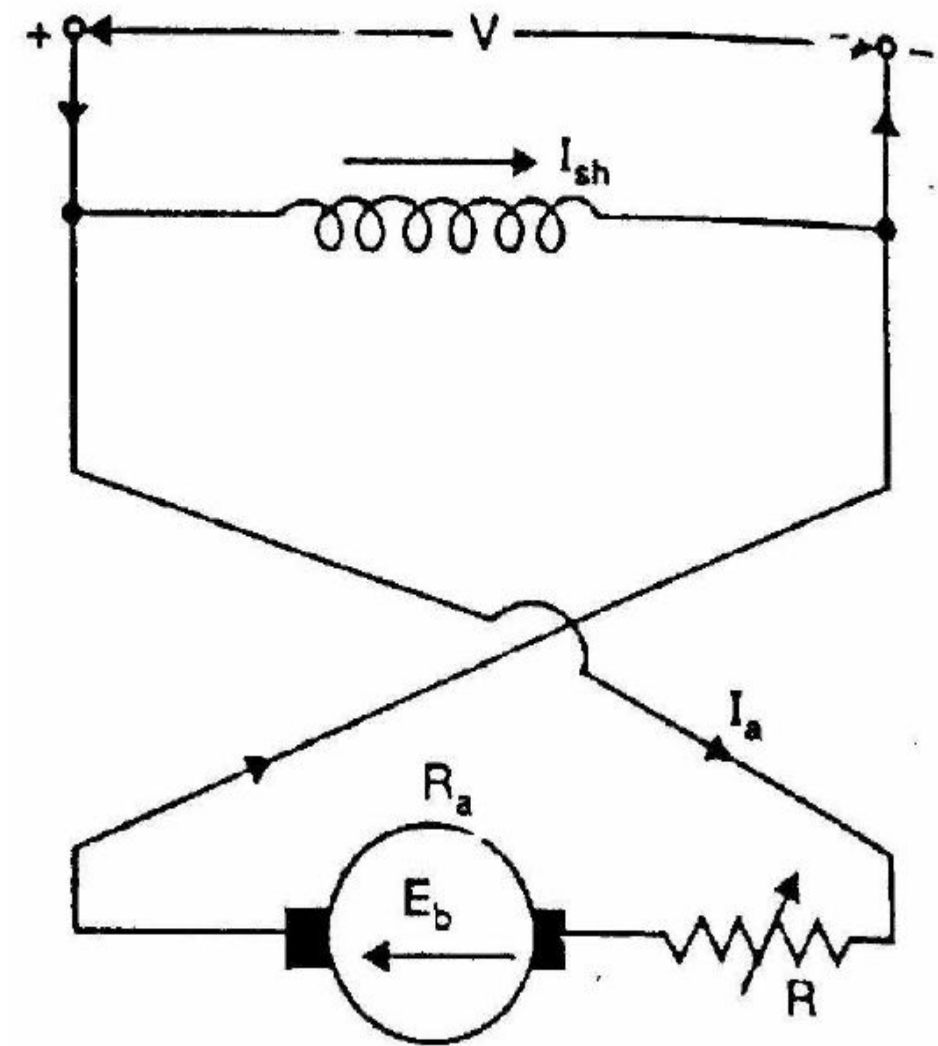




Braking of dc motor

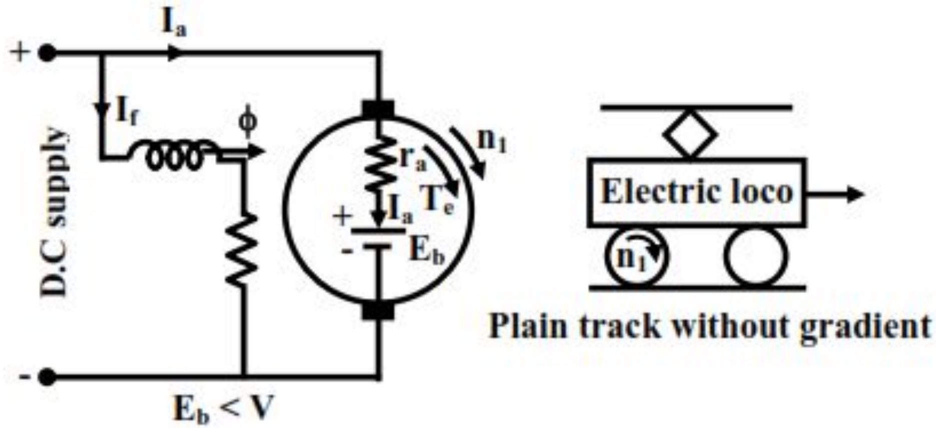
Plugging



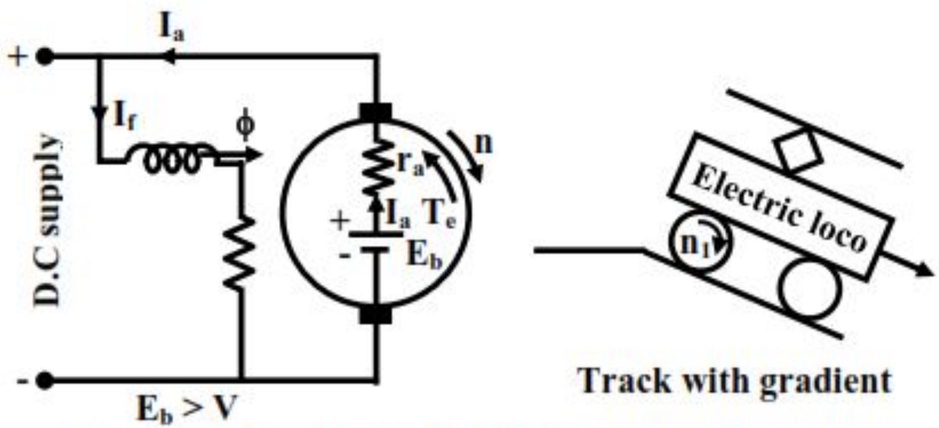


Braking of dc motor

Regenerative braking

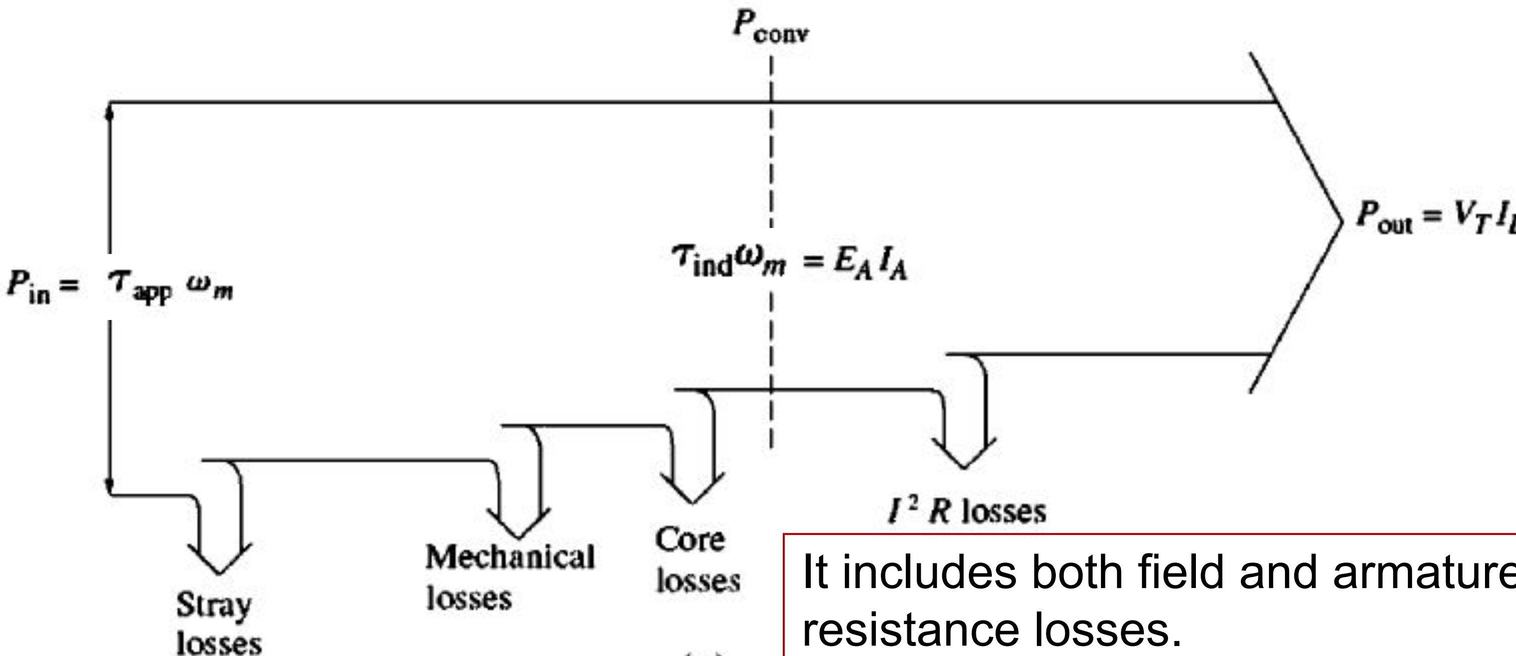


Machine operates as motor



Machine enters regenerative braking mode.

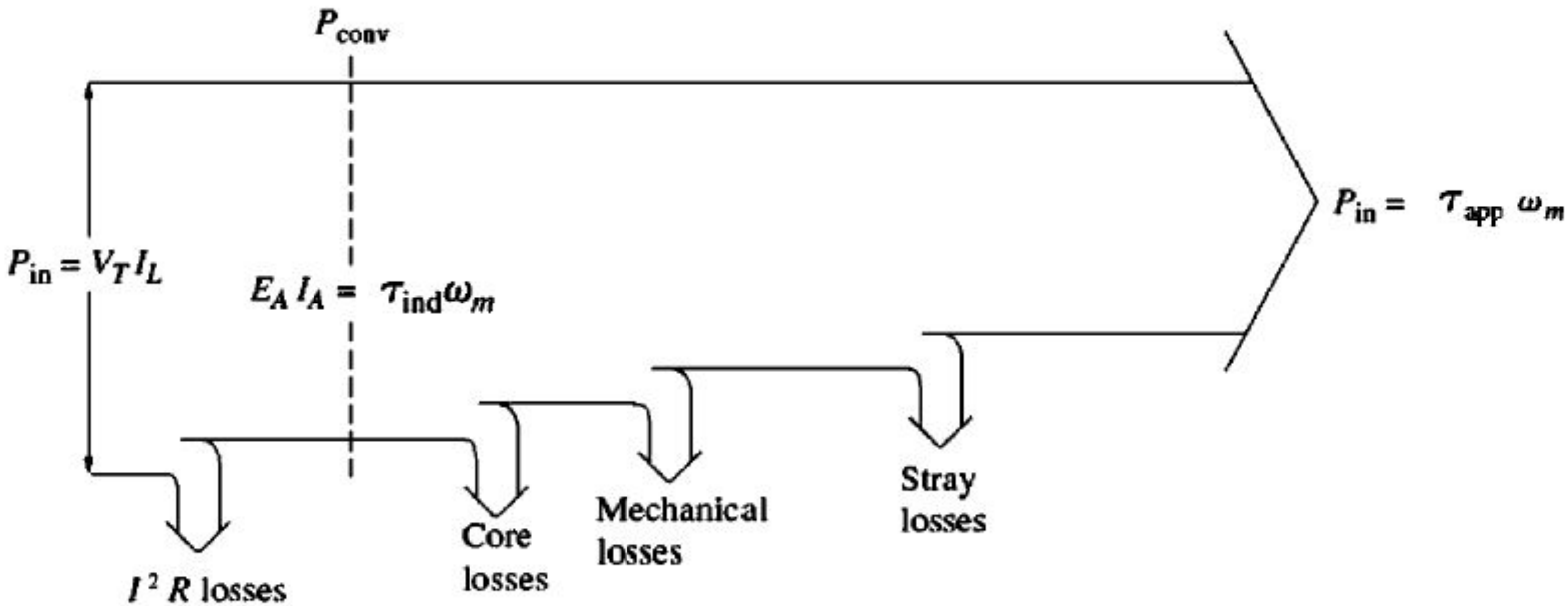
Power flow-diagram of DC Generator



(a)

It includes both field and armature resistance losses.

Power flow-diagram of DC Motor



It includes both field and armature resistance losses.

Note that
Power: HP×746= Watt
Torque: lb-ft×0.7376= N-m

Losses in a DC Machine

1. Copper losses;
2. Brush losses;
3. Mechanical losses;
4. Core losses;
5. Stray load losses.

DC machine efficiency calculations

To find the copper losses, we need to know the currents in the motor and two resistances. In practice, the armature resistance can be found by blocking the rotor and apply a small DC voltage to the armature terminals, such that the armature current will equal to its rated value. The ratio of the applied voltage to the armature current is approximately R_A.

The field resistance is determined by supplying the rated voltage to the field circuit and measuring the resulting field current. The field voltage to field current ratio equals to the field resistance.

DC machine efficiency calculations

Brush drop losses are frequently lumped together with copper losses. If treated separately, brush drop losses are a product of the brush voltage drop V_{BD} and the armature current I_A.

The core and mechanical losses are usually determined together. If the machine is running freely as a motor at no load and at the rated speed, the current I_A is very small and the armature copper losses are negligible. Therefore, if the field copper losses are subtracted from the input power of the motor, the remainder will be the mechanical and core losses. These two losses are also called the no-load rotational losses. As long as the motor's speed remains approximately the same, the no-load rotational losses are a good estimate of mechanical and core losses in the machine under load.

Efficiency of DC Motor

Efficiency of a dc machine is

$$Efficiency,\ \eta = \frac{P_{output}}{P_{input}} \times 100\% = \frac{P_{input} - P_{loss}}{P_{input}} \times 100\% = \frac{P_{output}}{P_{output} + P_{loss}} \times 100\%$$

A 8-pole, 25 kW, 120-V dc generator has a duplex lap Lap-wound armature which has 64 coils with 16 turns Per coil. Its rated speed is 2400 rpm.

(a) How much flux per pole is required to produce the rated voltage at no load conditions?

(b) What is current per path at the rated load?

(c) What is the induced torque at the rated load?

(d) If the resistance of this winding is 0.011 Ω/turn, what is the armature resistance of the machine?

A dc machine has 8-pole and a rated current of 100 A. How much current will flow in each path at rated conditions if the armature is (a) simplex lap wound; (b) duplex lap wound; and (c) simplex wave wound; and (d) duplex wave wound.

Problem 9.22

The magnetization curve for a separately excited dc generator is shown in Figure P9–7. The generator is rated at 6 kW, 120 V, 50 A, and 1800 r/min and is shown in Figure P9–8. Its field circuit is rated at 5A. The following data are known about the machine:

$$R_A = 0.18\ \Omega \qquad V_F = 120\ \text{V}$$

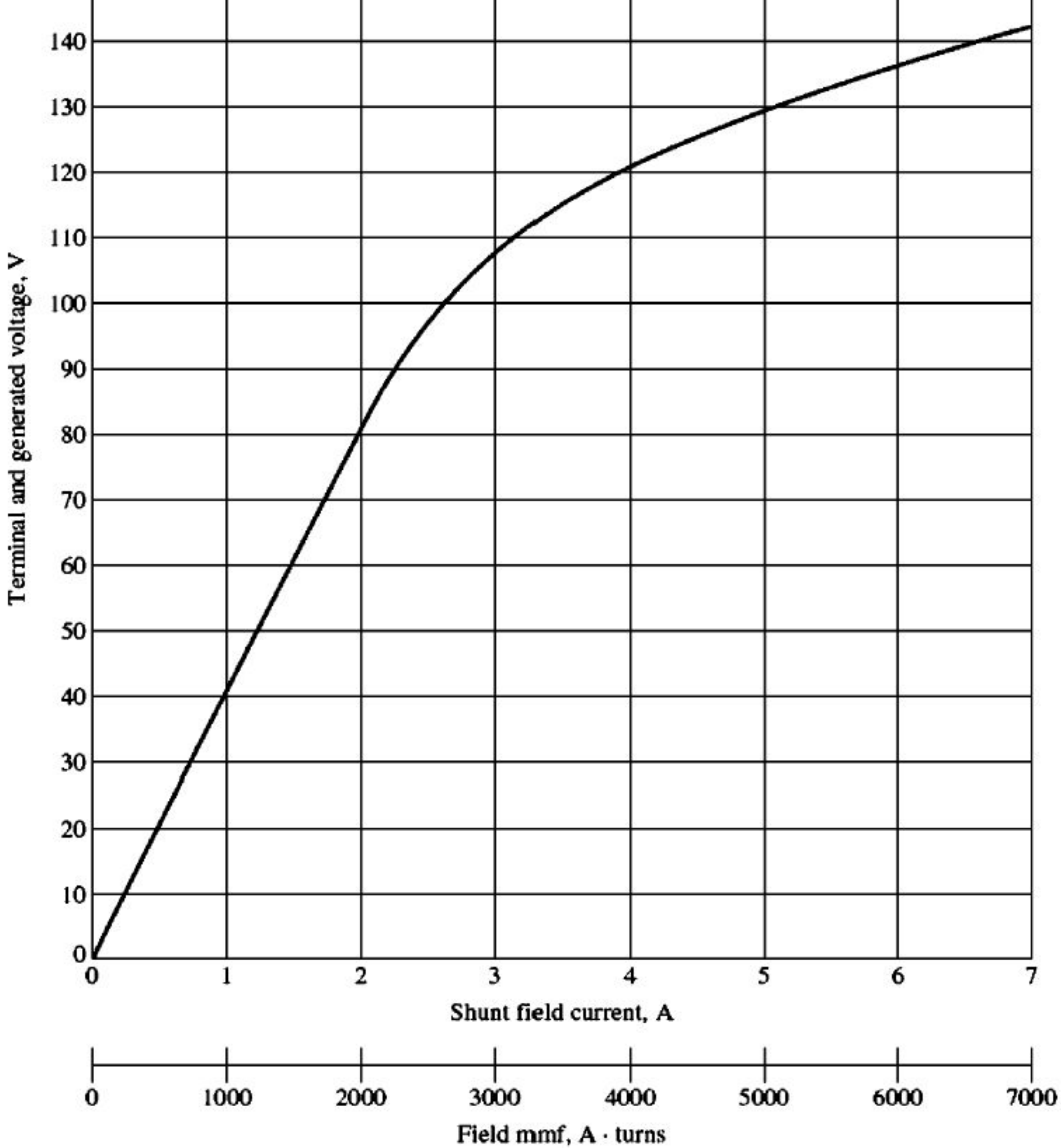
$$R_{\text{adj}} = 0 \text{ to } 30\ \Omega \qquad R_F = 24\ \Omega$$

$$N_F = 1000 \text{ turns per pole}$$

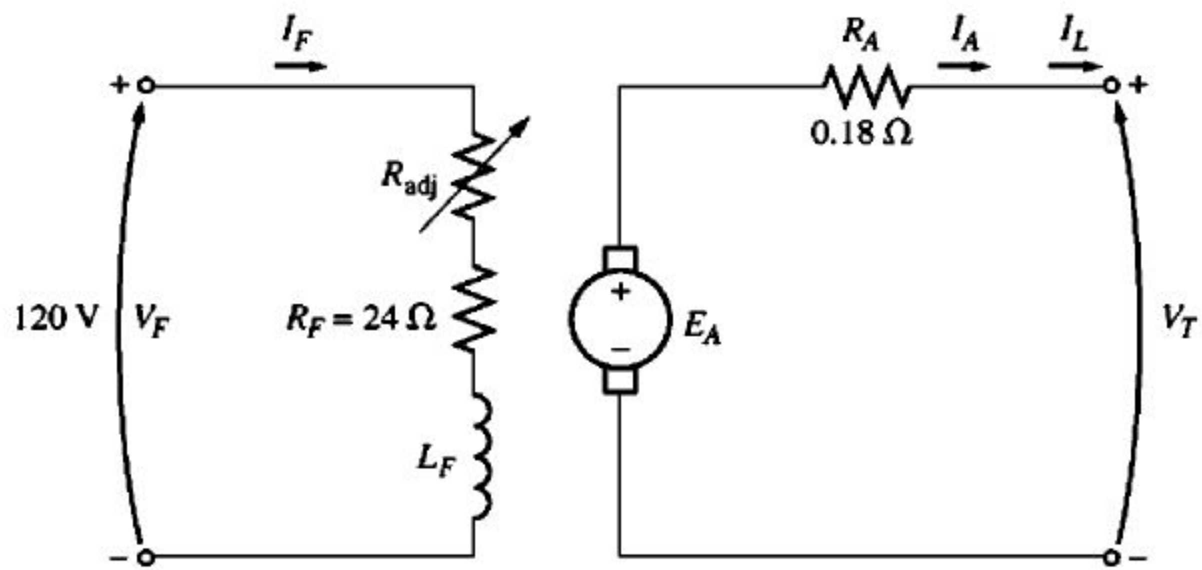
Answer the following questions about this generator, assuming no armature reaction.

(a) If this generator is operating at no load, what is the range of voltage adjustments that can be achieved by changing R_{adj}?

(b) If the field rheostat is allowed to vary from 0 to 30 Ω and the generator's speed is allowed to vary from 1500 to 2000 r/min, what are the maximum and minimum no-load voltages in the generator?



Terminal and generated voltage, V
0
10
20
30
40
50
60
70
80
90
100
110
120
130
140
0 1 2 3 4 5 6 7
Shunt field current, A
0 1000 2000 3000 4000 5000 6000 7000
Field mmf, A · turns


I_F
R_A
I_A
I_L
+
+
0.18 Ω
R_{adj}
120 V
V_F
$R_F = 24\ \Omega$
+
−
E_A
V_T
L_F
−
−

9–23. If the armature current of the generator in Problem 9–22 is 50 A, the speed of the generator is 1700 r/min, and the terminal voltage is 106 V, how much field current must be flowing in the generator?

9–24. Assuming that the generator in Problem 9–22 has an armature reaction at full load equivalent to 400 A • turns of magnetomotive force, what will the terminal voltage of the generator be when $I_F = 5$ A, $n_m = 1700$ r/min, and $I_A = 50$ A?

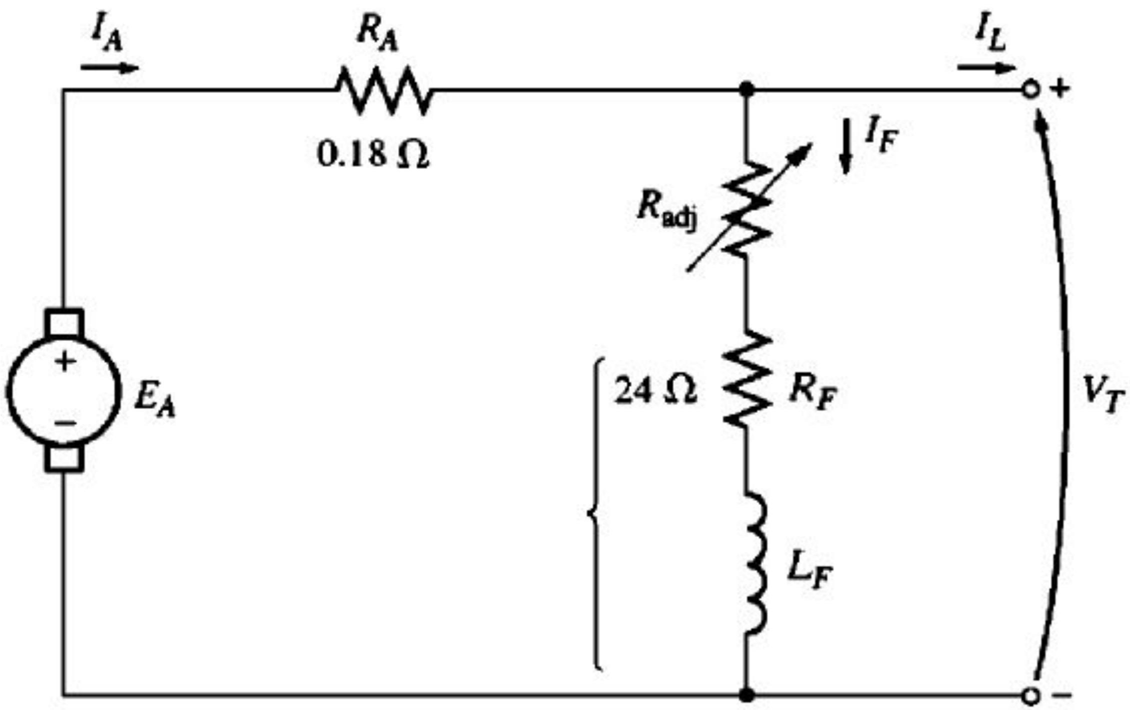
The machine in Problem 9–22 is reconnected as a shunt generator and is shown in Figure P9–9. The shunt field resistor R_{adj} is adjusted to 10 Ω, and the generator's speed is 1800 r/min.

(a) What is the no-load terminal voltage of the generator?

(b) Assuming no armature reaction, what is the terminal voltage of the generator with an armature current of 20 A? 40 A?

(c) Assuming an armature reaction equal to 300 A • turns at full load, what is the terminal voltage of the generator with an armature current of 20 A? 40 A?

(d) Calculate and plot the terminal characteristics of this generator with and without armature reaction.



Example 1

A 230 V, 10 hp d.c. shunt motor delivers power to a load at 1200 r/min. The armature current drawn by the motor is 200 A. The armature circuit resistance of the motor is 0.2 Ω and the field resistance is 115Ω. If the rotational losses are 500W, what is the value of the load torque?

Solution

The back emf induced in the armature is:

$$E_b = V_T - I_a R_a$$

$$= 230 - 0.2 \times 200 = 190 \text{ V}$$

Power developed (in the rotor), $P_{dev} = E_b I_a = 190 \times 200 = 38000$ W

Power delivered to the load, $P_{load} = P_{dev} - P_{rot} = P_{out}$

$$= 38000 - 500 = 37500 \text{ W}$$

Load torque, $T_{load} = P_{load}/\omega_m$

Since speed $\omega_m = 2\pi N/60$

where N is the speed in revolutions per minute (r/min). The torque supplied to the load can be calculated as:

$$T_{load} = \frac{37500}{\left(\frac{2\pi}{60}\right) \times 1200} = 298.4 \ N.m$$

Example 1

Q1) A 240 V, shunt DC motor takes an armature current of 20 A when running at 960 rpm. The armature resistance is 0.2 Ω. Determine the no load speed if the no load armature current is 1 A.

Answer: 975.45 rpm

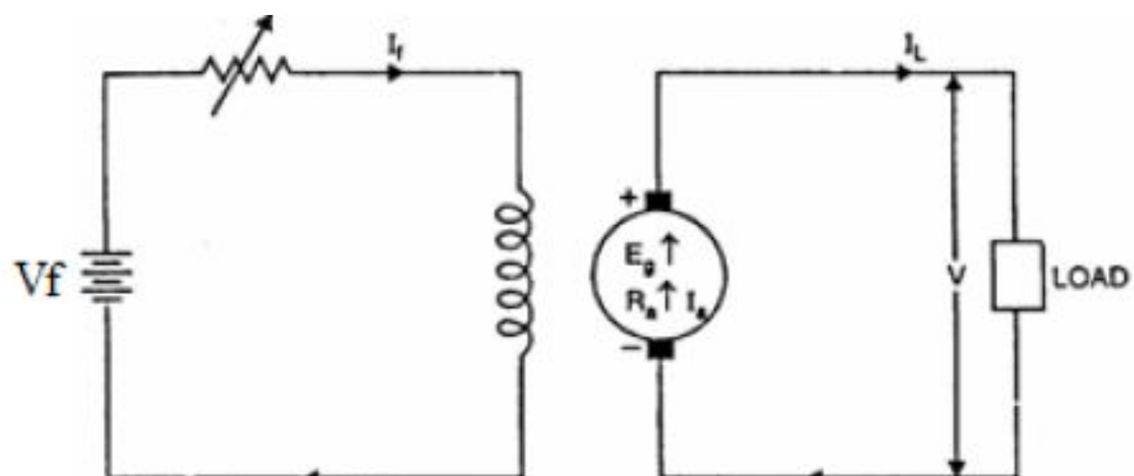
Efficiency of Dc Generator

Example 2: A 12 kW, 240V, 1200 rpm, separately excited DC generator has armature and field winding resistances of 0.20Ω and 200Ω , respectively. At no-load, the terminal voltage is 240V, the field current is 1.25A, and the machine runs at 1200 rpm. When the generator delivers rated current to a load at 240V, calculate

a) The generated voltage E_a
b) The field circuit voltage V_f
c) The developed torque T_e
d) Voltage regulation
e) The efficiency if the total rotational losses amount to 750W.

Solution:

P_L=12kW, V_L=240V,
R_a=0.2Ω, R_f=200Ω



Efficiency of Dc Generator

At no-load, $I_L = I_a = 0$.

$\therefore$ Generated voltage, $E_{a1} = V_t + I_a R_a = V_t = 240V$. This case $I_{f1} = 1.25A$.

$$(a)\ At\, rated\, load:\ I_a = \frac{P_L}{V_t} = \frac{12{,}000}{240} = 50A$$

$$\therefore\ E_{a2} = V_t + I_a R_a = 240 + 50 \times 0.20 = 250V$$

$$(b)\ Now\ we\, have\ \frac{E_{a2}}{E_{a1}} = \frac{K_b I_{f2} n_2}{K_b I_{f1} n_1} = \frac{I_{f2}}{I_{f1}} \qquad \because n_2 = n_1$$

$$\Rightarrow I_{f2} = \frac{E_{a2} I_{f1}}{E_{a1}} = \frac{250 \times 1.25}{240} = 1.30\ A$$

$$\therefore\ V_f = I_f R_f = 1.3 \times 200 = 260V$$

Efficiency of Dc Generator

$$(c)\ T_e = \frac{E_a I_a}{\omega_m} = \frac{250 \times 50}{40\pi} = 99.5\ N-m \qquad \because \omega_m = \frac{2\pi n}{60} = \frac{2\pi(1200)}{60} = 40\pi\ rad/\sec$$

$$(d)\ Voltage\ regulation = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{E_a - V_t}{V_t} \times 100\%$$

$$= \frac{250 - 240}{240} \times 100\% = 4.1667\%$$

$$(e)\ \ P_{out} = V_t I_t = (240)(50) = 12,000W$$

$$The\ losses\ are: P_{rot} = 750W$$

$$I_f^2 R_f = 1.3^2 \times 200 = 338W$$

$$I_a^2 R_a = 50^2 \times 0.20 = 500W$$

$$Total\ power\ loss,\ P_{loss} = 750 + 338 + 500 = 1,588W$$

$$Input\ \ power: P_{in} = P_{out} + P_{loss} = 12,000 + 1,588 = 13,588W$$

$$Efficiency,\ \eta = \frac{P_{out}}{P_{in}} = \frac{12,000}{13,588} \times 100\% = 88.31\%$$

Knowledge Test-1

Q1: What type of magnetic material hard or soft is used for dc machine design?

Q2: What is the purpose of using such magnetic material to design dc machines?

Q3: What law is the basis of dc generator operation?

Q4: How can we vary the generated voltage?

Q5: What factors can vary the motor torque?

Table 26-1 Resistivities of Some Materials at Room Temperature (20°C)

Material	Resistivity, ρ ($\Omega \cdot m$)	Temperature Coefficient of Resistivity, α (K^{-1})
	Typical Metals	
Silver	1.62×10^{-8}	4.1×10^{-3}
Copper	1.69×10^{-8}	4.3×10^{-3}
Gold	2.35×10^{-8}	4.0×10^{-3}
Aluminum	2.75×10^{-8}	4.4×10^{-3}
Manganin[a]	4.82×10^{-8}	0.002×10^{-3}
Tungsten	5.25×10^{-8}	4.5×10^{-3}
Iron	9.68×10^{-8}	6.5×10^{-3}
Platinum	10.6×10^{-8}	3.9×10^{-3}